

# Why **MASTERflex**® Peristaltic Pumps?

1

## Easy

- Quick installation
- Low maintenance
- Easy tubing changeover

2

## Durable

- Trusted for over 60 years
- Broad chemical compatibility

3

## Versatile

- From research to process scale
- Handle just about any fluid
- Multichannel capabilities

4

## Customisable

- Configurable systems
- Many tubing options
- Easy scale-up

5

## Accurate

- Flow ranges from 0.0006- mL/min to 42-L/min
- Rigorous design and testing standards
- Tubing is custom manufactured to precise tolerances



## Selecting a Masterflex® Pump: *As-Easy-As*

### Application Requirements

- What is my required flow rate?
- What chemical am I pumping?
- Do conditions such as viscosity or back pressure play an important role in my decision?

### Base a Pump Series on Your Flow Rate

- Use the table below to find the pump series that best fits your flow rate needs

### Complete Your System

- L/S and I/P pump series are divided into three components—pump head, tubing, and drive
- Select the three components from the same series or choose a complete preconfigured system

| Masterflex pump series | Features                                                                                     | Flow range             | Speed |          | Remote control capability | Multichannel capabilities | Page(s) |
|------------------------|----------------------------------------------------------------------------------------------|------------------------|-------|----------|---------------------------|---------------------------|---------|
|                        |                                                                                              |                        | Fixed | Variable |                           |                           |         |
| <b>C/L</b> ®           | Compact, low-flow pump that easily fits on benchtop or mounts within a panel                 | 0.002 to 43 mL/min     |       | ✓        | Yes                       | Yes                       | 137     |
| <b>L/S</b> ®           | High-accuracy drives for precise flow control and dispensing                                 | 0.00001 to 3400 mL/min | ✓     | ✓        | Yes                       | Yes                       | 121–133 |
| <b>I/P</b> ®           | Powerful systems that provide better protection from possible hazards such as dust and water | 0.0006 to 26 LPM       | ✓     | ✓        | Yes                       | Yes                       | 134–135 |
| <b>B/T</b> ®           | Highest flow rates—ideal for large batch fluid transfer with excellent hazard protection     | 0.71 to 42 LPM         | ✓     | ✓        | Yes                       | Yes                       | 136     |

## Custom **OEM** Pump Systems

**When you need more than an off-the-shelf solution, you can look to Cole-Parmer as a trusted partner**

With more than 100 years of combined experience, our design engineers will work directly with you to create the exact pump you need. Contact our Application Specialists to get started!

# MASTERflexLIVE™

## Masterflex® Cloud-Enabled Drives Featuring MasterflexLive™

Real-time remote pump control  
and monitoring

- Secure, cloud-enabled platform for control and monitoring of select L/S® and I/P® pumps
- Get real-time control of all pump parameters, including speed, flow rate, dispense volume, and more
- Push notifications alert you to operating conditions and error messages
- Monitor and adjust critical processes running 24/7—no need to be onsite
- “Hands-free” operation lets you control pumps operating inside cleanrooms, glove boxes or isolation chambers
- Access from your PC, tablet or smartphone (iOS™ and Android™ compatible)
- Internet connection via standard Ethernet (RJ45) jack on rear of drive



**EU Annex 11  
compliant**



# Monitor and control your pumps via the cloud.

## 24/7 remote access from anywhere

Create a free account

Manage & control pump parameters

Get alerts & notifications

| rpm                                                                      | Flow range<br>(mL/min) | Speed control<br>(resolution) | Open-head<br>sensor | Pump heads<br>accepted | Motor<br>size   | IP<br>rating | Power<br>(50/60 Hz) | Catalogue<br>number |
|--------------------------------------------------------------------------|------------------------|-------------------------------|---------------------|------------------------|-----------------|--------------|---------------------|---------------------|
| L/S cloud-enabled digital drive                                          |                        |                               |                     |                        |                 |              |                     |                     |
| 0.02 to 100                                                              | 0.001 to 588           | ±0.1%                         | Yes                 | 2                      | ¼ hp<br>(75 W)  | IP33         | 90 to 260 VAC       | MY-07551-60         |
| 0.1 to 600                                                               | 0.006 to 3400          |                               |                     |                        |                 |              |                     | MY-07551-50         |
| L/S cloud-enabled digital process drive with 316 stainless steel housing |                        |                               |                     |                        |                 |              |                     |                     |
| 0.1 to 600                                                               | 0.006 to 3400          | ±0.1%                         | Yes                 | 2                      | ¼ hp<br>(75 W)  | IP66         | 90 to 260 VAC       | MY-07575-70         |
| L/S cloud-enabled digital process drive with powder-coat steel housing   |                        |                               |                     |                        |                 |              |                     |                     |
| 0.1 to 600                                                               | 0.006 to 3400          | ±0.1%                         | Yes                 | 2                      | ¼ hp<br>(75 W)  | IP66         | 90 to 260 VAC       | MY-07575-72         |
| I/P cloud-enabled digital process drive with 316 stainless steel housing |                        |                               |                     |                        |                 |              |                     |                     |
| 0.1 to 650                                                               | 0.6 to 26,000          | ±0.1%                         | Yes                 | 2                      | ¾ hp<br>(280 W) | IP66         | 90 to 260 VAC       | MY-77420-50         |
| I/P cloud-enabled digital process drive with powder-coat steel housing   |                        |                               |                     |                        |                 |              |                     |                     |
| 0.1 to 650                                                               | 0.6 to 26,000          | ±0.1%                         | Yes                 | 2                      | ¾ hp<br>(280 W) | IP66         | 90 to 260 VAC       | MY-77420-60         |

## Masterflex® L/S® Variable-Speed Console Drives

- Compact, cost-effective for general fluid transfer applications
- Flow range: 0.42 to 2900 mL/min
- Simple operation with single-turn speed control
- Separate power switch enables you to maintain speed setting when turning drive on/off; front-panel LED indicates power on
- Reversible motor lets you easily prime/purge tubing and pump in either direction
- Remote control of start/stop via DB9 connector on back of drive

Drives accept nine different pump head types: Standard, Easy-Load®, Easy-Load II, Easy-Load 3, Multichannel, High-Performance, Cartridge, PTFE tubing, and PTFE-diaphragm. Drive 07555-05 is not recommended for use with the PTFE-tubing and PTFE-diaphragm pump heads.

**What's included:** 1.8-m (6-ft) line cord with IEC 320/CEE22 socket. Shipped with country-specific cord/plug set; please specify ultimate destination when ordering.



### FEATURES

- ✓ Compact and cost effective
- ✓ Reversible motor accepts up to two pump heads
- ✓ Remote start/stop

Shown with L/S Easy-Load II pump head 77200-60

**Flow Rates in mL/min** (flow rates in parentheses can only be obtained with the High-Performance pump head—see page 131)

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| rpm       | Precision pump tubing                                           |            |            |            |            |            | High-performance precision pump tubing                                            |                          |                          |                           |
|-----------|-----------------------------------------------------------------|------------|------------|------------|------------|------------|-----------------------------------------------------------------------------------|--------------------------|--------------------------|---------------------------|
|           | L/S 13                                                          | L/S 14     | L/S 16     | L/S 25     | L/S 17     | L/S 18     | L/S 15                                                                            | L/S 24                   | L/S 35                   | L/S 36                    |
|           | Standard <sup>†</sup> , Easy-Load®, Easy-Load® II, Easy-Load® 3 |            |            |            |            |            | Multichannel <sup>‡</sup> , Standard, Easy-Load II, Easy-Load 3, High-Performance |                          |                          |                           |
|           | Cartridge, Multichannel                                         |            |            | Cartridge  |            |            | Easy-Load                                                                         |                          |                          |                           |
| 20 to 600 | 1.2 to 36                                                       | 4.2 to 130 | 16 to 480  | 34 to 1000 | 56 to 1700 | 76 to 2300 | 34 to 1000                                                                        | 56 to 1700               | 76 to 2300               | 96 to 2900                |
| 7 to 200  | 0.42 to 12                                                      | 1.4 to 43  | 5.6 to 160 | 12 to 330  | 20 to 570  | 27 to 770  | 12 to 330<br>(13 to 370)                                                          | 20 to 570<br>(21 to 600) | 27 to 770<br>(30 to 870) | 34 to 970<br>(41 to 1130) |

<sup>†</sup>The Standard pump head is not available for L/S 25. <sup>‡</sup>Multichannel head is not available for L/S 36.



| rpm       | Speed control (resolution) | Pump heads accepted | Motor size     | IP rating | Dimensions (L x W x H)                      | Power (50/60 Hz) |      | Catalogue number |
|-----------|----------------------------|---------------------|----------------|-----------|---------------------------------------------|------------------|------|------------------|
|           |                            |                     |                |           |                                             | VAC              | Amps |                  |
| 20 to 600 | ±5%                        | 1                   | 1/20 hp (37 W) | IP22      | 23 x 18.3 x 13.4 cm<br>(9.1" x 7.2" x 5.3") | 180 to 260       | 0.8  | MY-07555-05      |
| 7 to 200  |                            | 2                   |                |           |                                             | 180 to 260       | 0.8  | MY-07555-15      |

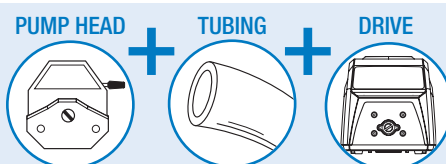
### Accessories

**MY-77595-35** Foot switch, DB9 male, 1.8-m (6-ft) cable

**MY-17050-01** ILAC MRA, NIST-traceable calibration for peristaltic pump drive

**MY-07528-80** Handheld remote controller; for control of start/stop, route tubing through handle for dispensing into containers

Choose a convenient  
Complete System  
One catalogue number includes all 3!



| System | Flow range        | What's included        |                                                  |                                      | Power             | Catalogue number |
|--------|-------------------|------------------------|--------------------------------------------------|--------------------------------------|-------------------|------------------|
|        |                   | Pump head              | Tubing                                           | Drive                                |                   |                  |
|        | 16 to 480 mL/min  | Easy-Load® II 77200-60 | C-Flex® ULTRA L/S 16 tubing 06434-16 3 m (10 ft) | Console drive 07555-05 20 to 600 rpm | 230 VAC, 50/60 Hz | MY-77910-45      |
|        | 56 to 1700 mL/min | Easy-Load II 77200-62  | C-Flex ULTRA L/S 24 tubing 06434-24 3 m (10 ft)  | Console drive 07555-05 20 to 600 rpm | 230 VAC, 50/60 Hz | MY-77910-55      |



# Masterflex® L/S® Computer-Compatible Digital Drives

## Automate your process by interfacing drives with a computer

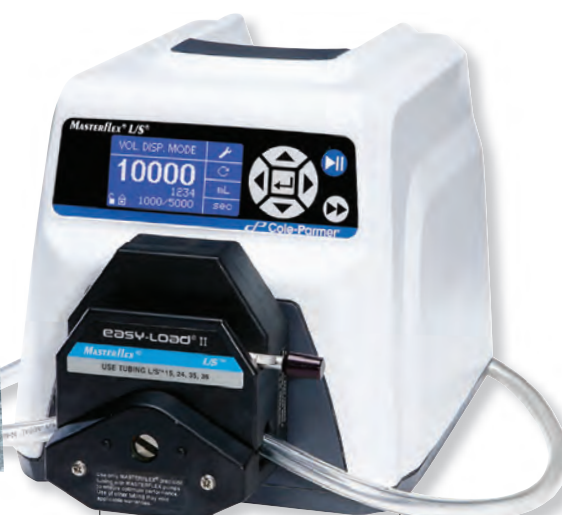
- Flow rate: 0.001 to 3400-mL/min with L/S tubing
- Interface drive to your PC via DB9 serial or USB connector
- Graphical LCD shows four operating modes: continuous run, time dispense, copy dispense, and volume dispense
- View pump performance continually
- Antidrip function ensures dispensing accuracy
- Remote control capabilities via DB25 female connector on drive

Drives offer analogue remote control options and simple programming—ideal for automated process applications. Precise PWM speed control ( $\pm 0.1\%$ ) maximises accuracy and efficiency while the brushless motor gives you quiet, maintenance-free reliability.

**What's included:** 1.8-m (6-ft) line cord with IEC 320/EE22 socket. Shipped with country-specific plug/cord set; please specify ultimate destination when ordering.

### FEATURES

- ✓ Flow rates: 0.001 to 3400 mL/min
- ✓ Variable speed from 0.02 to 600 rpm
- ✓ Programmable, reversible motor accepts up to four pump heads
- ✓ Remote functionality



Drive 07551-20 shown with L/S Easy-Load II pump head 77200-62

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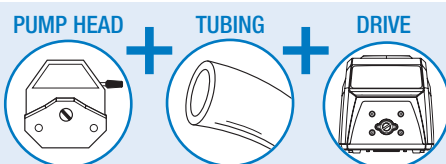


| rpm         | Speed control (resolution)                                  | Open-head sensor | Pump heads accepted | Motor size                  | IP rating | Dimensions (L x W x H)                     | Power (50/60 Hz) |                | Catalogue number |
|-------------|-------------------------------------------------------------|------------------|---------------------|-----------------------------|-----------|--------------------------------------------|------------------|----------------|------------------|
|             |                                                             |                  |                     |                             |           |                                            | VAC              | Amps           |                  |
| 0.1 to 600  | $\pm 0.1\%$<br>(0.1 rpm at 600 rpm;<br>0.01 rpm at 100 rpm) | No               | 2                   | $\frac{1}{10}$ hp<br>(75 W) | IP33      | 25.4 x 21.6 x 21.6 cm<br>(10" x 8½" x 8½") | 90 to 260 VAC    | 1.1 at 230 VAC | MY-07551-20      |
| 0.02 to 100 |                                                             | No               | 4                   |                             |           |                                            |                  |                | MY-07551-30      |

### Accessories

**MY-07523-92 Foot switch**, momentary start/stop; 1.8-m (6-ft) cable  
**MY-07523-94 DB25 male connector**. Use to create your own cable  
**MY-07523-95 Cable assembly**, DB25 male connector and 7.9-m (25-ft) cable with stripped wire ends for remote control

**MY-07523-98 Tilt bail**, use to securely elevate front of drive  
**MY-07523-97 Handheld remote controller**, DB9; for on/off control; route tubing through handle for filling and dispensing applications  
**MY-17050-01 ILAC MRA, NIST-traceable calibration** with data for peristaltic pump drive



Choose a convenient  
**Complete System**  
One catalogue number includes all 3!

| System | Flow range          | What's included          |                                                           |                                                         | Power                    | Catalogue number |
|--------|---------------------|--------------------------|-----------------------------------------------------------|---------------------------------------------------------|--------------------------|------------------|
|        |                     | Pump head                | Tubing                                                    | Drive                                                   |                          |                  |
|        | 0.08 to 480-mL/min  | Easy-Load® 3<br>77800-60 | C-Flex® ULTRA<br>L/S 16 tubing<br>06434-16<br>3-m (10 ft) | Computer-compatible<br>drive 07551-20<br>0.1 to 600 rpm | 115/230 VAC,<br>50/60 Hz | MY-77924-55      |
|        | 0.28 to 1700-mL/min | Easy-Load II<br>77200-62 | C-Flex® ULTRA<br>L/S 24 tubing<br>06434-24<br>3-m (10 ft) |                                                         |                          | MY-77924-65      |

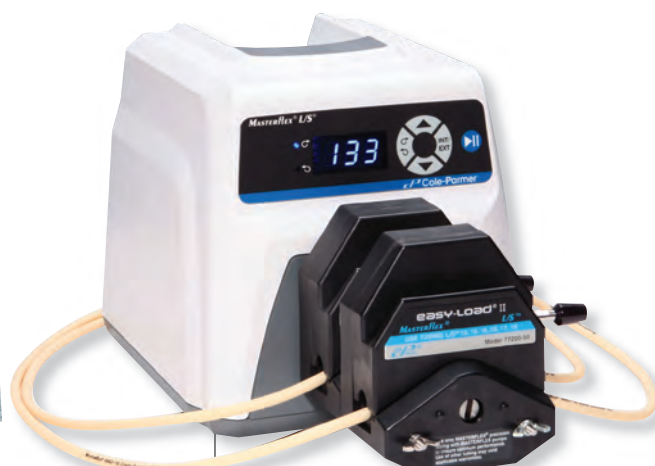
# Masterflex® L/S® Precision Variable-Speed Console Drives

Don't lose control of your metering applications—digital display gives you repeatable control

- Accurate and repeatable settings with 3-digit LED display of rpm
- Pump fluid in either direction with reversible motor—perfect for purging tubing
- Save space and increase productivity by stacking 2 or 4 pump heads on the same drive to create a multichannel pump system
- Maximise processes in a minimal footprint by stacking splash-resistant, IP33-rated drives
- Two ways to control speed, direction, and start/stop—use either the membrane keypad or remote control capabilities
- Customise for your application—drives accept nine pump head types: Standard, Easy-Load®, Easy-Load II, Easy-Load 3, Multichannel, High-Performance, Cartridge, PTFE-Tubing, and PTFE-Diaphragm

## FEATURES

- ✓ Flow rates: 0.06 to 3400 mL/min
- ✓ Variable speed from 1 to 600 rpm
- ✓ Reversible motor accepts up to four pump heads



Stack up to 4 heads to maximise productivity in less space

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**Flow Rates in mL/min** (flow rates in parentheses can only be obtained with the High-Performance pump head—see page 131)

| rpm      | Precision pump tubing                           |            |            |            |            |            | High-performance precision pump tubing                                |                         |                         |                         |
|----------|-------------------------------------------------|------------|------------|------------|------------|------------|-----------------------------------------------------------------------|-------------------------|-------------------------|-------------------------|
|          | L/S 13                                          | L/S 14     | L/S 16     | L/S 25     | L/S 17     | L/S 18     | L/S 15                                                                | L/S 24                  | L/S 35                  | L/S 36                  |
|          | Standard†, Easy-Load, Easy-Load II, Easy-Load 3 |            |            |            |            |            | Multichannel†, Standard†, Easy-Load II, Easy-Load 3, High-Performance |                         |                         |                         |
|          | Cartridge, Multichannel                         |            |            | Cartridge  |            |            | Easy-Load                                                             |                         |                         |                         |
| 6 to 600 | 0.36 to 36                                      | 1.3 to 130 | 4.8 to 480 | 10 to 1000 | 17 to 1700 | 23 to 2300 | 10 to 1000 (11 to 1100)                                               | 17 to 1700 (18 to 1800) | 23 to 2300 (26 to 2600) | 29 to 2900 (34 to 3400) |
| 1 to 100 | 0.06 to 6                                       | 0.21 to 21 | 0.8 to 80  | 1.7 to 170 | 2.8 to 280 | 3.8 to 380 | 1.7 to 170 (1.8 to 180)                                               | 2.8 to 280 (3.0 to 300) | 3.8 to 380 (4.3 to 430) | 4.8 to 480 (5.8 to 580) |

†The Standard pump head is not available for L/S 25. ‡Multichannel head is not available for L/S 36. Multichannel and Cartridge heads are available for microbore tubing.

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| rpm      | Speed control (resolution)                       | Pump heads accepted | Motor size     | IP rating | Dimensions (L x W x H)<br>(10" x 8.5" x 8.5") | Power (50/60 Hz) |                | Catalogue number |
|----------|--------------------------------------------------|---------------------|----------------|-----------|-----------------------------------------------|------------------|----------------|------------------|
|          |                                                  |                     |                |           |                                               | VAC              | Amps           |                  |
| 6 to 600 | ±0.25% (1 rpm at 600 rpm;<br>0.1 rpm at 100 rpm) | 2                   | 1/10 hp (75 W) | IP33      | 25.4 x 21.6 x 21.6 cm<br>(10" x 8.5" x 8.5")  | 90 to 260        | 1.2 at 230 VAC | MY-07528-10      |
| 3 to 300 |                                                  | 2                   |                |           |                                               |                  |                | MY-07528-20      |
| 1 to 100 |                                                  | 4                   |                |           |                                               |                  |                | MY-07528-30      |

## Accessories

**MY-07595-45 Connector**, DB9 male. Use for 4 to 20 mA remote control

**MY-07595-47 Cable assembly**, DB9 male connector and 7.9-m (25-ft) cable with stripped wire terminal ends. Use for remote control

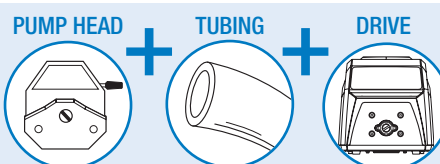
**MY-77595-35 Foot switch**, DB9 male

**MY-07528-80 Handheld remote controller**; for remote control of start/stop, route tubing through handle for dispensing into containers

**MY-17050-01 ILAC MRA, NIST-traceable calibration** with data for peristaltic pump drive

## Choose a convenient Complete System

One catalogue number includes all 3!



| System | Flow range         | What's included          |                                                           |                                                     | Power                      | Catalogue number |
|--------|--------------------|--------------------------|-----------------------------------------------------------|-----------------------------------------------------|----------------------------|------------------|
|        |                    | Pump head                | Tubing                                                    | Drive                                               |                            |                  |
|        | 4.8- to 480-mL/min | Easy-Load® 3<br>77800-60 | C-Flex® ULTRA<br>L/S 16 tubing<br>06434-16<br>3 m (10 ft) | Precision console<br>drive 07528-10<br>6 to 600 rpm | 90 to 260 VAC,<br>50/60 Hz | MY-77916-10      |
|        | 17- to 1700-mL/min | Easy-Load II<br>77200-62 | C-Flex® ULTRA<br>L/S 24 tubing<br>06434-24<br>3 m (10 ft) |                                                     |                            | MY-77916-20      |



# Masterflex® L/S® Variable-Speed Digital Drives

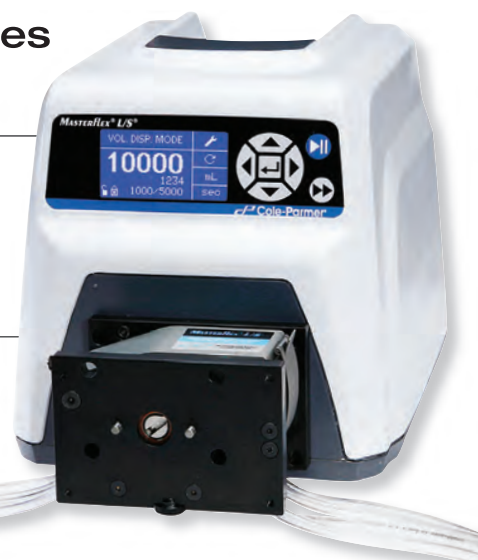
## Versatile, worry-free operation

- Brushless, maintenance-free motor provides  $\pm 0.1\%$  speed control accuracy with 6000:1 turndown ratio
- Bright graphical LCD offers easy programming, four operating modes, and seven languages
- Get more accurate dispensing with preprogrammed calibration
- Drives accept all L/S pump heads for more pumping versatility
- Models equipped with open-head sensor stop drive motor when pump head is opened—pair with an open-head sensor equipped pump head (see pages 130–131)
- Operate remotely via DB25 female connector

### FEATURES

- ✓ Flow rates: 0.001 to 3400 mL/min
- ✓ Variable speed from 0.02 to 600 rpm
- ✓ Programmable, reversible motor accepts up to four pump heads
- ✓ Ensure dispensing accuracy with antidrip function
- ✓ Hands-free dispensing via time interval dispense cycles

Four operating modes for more functionality: continuous run, time dispense, copy dispense, and volume dispense



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| rpm         | Speed control (resolution)                                  | Open-head sensor | Pump heads accepted | Motor size        | IP rating | Dimensions (L x W x H)                       | Power                      | Catalogue number |
|-------------|-------------------------------------------------------------|------------------|---------------------|-------------------|-----------|----------------------------------------------|----------------------------|------------------|
| 0.1 to 600  | $\pm 0.1\%$<br>(0.1 rpm at 600 rpm;<br>0.01 rpm at 100 rpm) | No               | 2                   | 1/10 hp<br>(75 W) | IP33      | 25.4 x 21.6 x 21.6 cm<br>(10" x 8.5" x 8.5") | 90 to 260 VAC,<br>50/60 Hz | MY-07522-20      |
|             |                                                             | Yes              |                     |                   |           |                                              |                            | MY-07522-28      |
| 0.02 to 100 | $\pm 0.1\%$<br>(0.1 rpm at 600 rpm;<br>0.01 rpm at 100 rpm) | No               | 4                   | 1/10 hp<br>(75 W) | IP33      | 25.4 x 21.6 x 21.6 cm<br>(10" x 8.5" x 8.5") | 90 to 260 VAC,<br>50/60 Hz | MY-07522-30      |
|             |                                                             | Yes              |                     |                   |           |                                              |                            | MY-07522-38      |

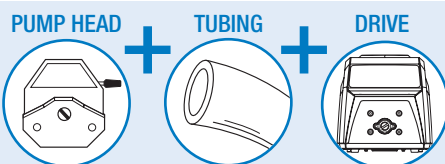
**MY-07523-92 Foot switch**, momentary start/stop; 1.8-m (6-ft) cable

**MY-07523-94 DB25 male connector**. Use to create your own cable

**MY-07523-95 Cable assembly** for remote control; DB25 male connector and 7.9-m (25-ft) cable with stripped wire ends

**MY-07523-97 Handheld remote controller**; route tubing through handle for dispensing into containers

**MY-17050-01 ILAC MRA, NIST-traceable calibration** with data for peristaltic pump drive



Choose a convenient  
**Complete System**

One catalogue number includes all 3!

| System | Flow range          | What's included          |                                                           |                                             | Power                      | Catalogue number |
|--------|---------------------|--------------------------|-----------------------------------------------------------|---------------------------------------------|----------------------------|------------------|
|        |                     | Pump head                | Tubing                                                    | Drive                                       |                            |                  |
|        | 0.08 to 480-mL/min  | Easy-Load® 3<br>77800-60 | C-Flex® ULTRA<br>L/S 16 tubing<br>06434-16<br>3-m (10 ft) | Digital drive<br>07522-20<br>0.1 to 600 rpm | 90 to 260 VAC,<br>50/60 Hz | MY-77921-65      |
|        | 0.28 to 1700-mL/min | Easy-Load II<br>77200-62 | C-Flex ULTRA<br>L/S 24 tubing<br>06434-24<br>3-m (10 ft)  |                                             |                            | MY-77921-75      |

### Application Solution



### Challenge

A manufacturer of safety glass and laminated glass was looking for a pump that could dispense coating resins for plate glass to provide protection and build shatter resistance. The resins used in the process could be polyurethane, epoxy, or other coatings, and in the past the resins ate through dispensing tubing. As a result, the manufacturer needed to find a pump system with tubing that had the appropriate chemical compatibility. Likewise, the pump system needed to offer quick maintenance for tube changes.

### Solution

The customer tested silicone tubing for chemical resistance and found it to be suitable for the resins being dispensed. The customer then chose the L/S variable-speed digital drive 07522-20 for its precision pumping and speed control. The manufacturer also selected an Easy-Load II pump head for quick maintenance and reduced downtime during tubing changes.



## Masterflex® L/S® Digital Modular Dispensing Drives

### Three dispensing modes and precision low-maintenance motor

- Graphical LCD shows four operating modes: continuous run, timed dispense, copy dispense, volume dispense
- Dispense by mL—volume in mL of each dispense, 0.001 to 9999 mL; Copy—1 to 9999 dispense cycles; or SEC—time interval between each dispense cycle (1 to 9999 secs)
- Antidrip function ensures dispense accuracy
- Brushless, maintenance-free motor offers  $\pm 0.1\%$  speed control accuracy and a 6000 to 1 turndown ratio

**A. Modular Drive with Benchtop Controller** features a stackable, splash-resistant ABS plastic controller housing and epoxy finished steel drive housing.

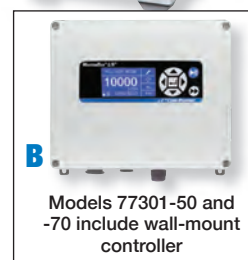
**B. Washdown Modular Drive with Wall-Mount Controller** has a powder-coated aluminum controller housing and powder-coated steel drive housing.

**What's included:** 1.8-m (6-ft) cable to connect drive and controller and 1.8-m (6-ft) line cord with IEC 320/CEE22 socket. Shipped with country-specific plug/cord set; please specify destination when ordering.



#### FEATURES

- ✓ Flow rates: 0.001 to 3400 mL/min depending on drive rpm and tubing size
- ✓ Variable speed
- ✓ Programmable, reversible motor accepts up to two pump heads and offers remote functionality



Models 77301-50 and -70 include wall-mount controller

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| rpm                                                                    | Speed control | Pump heads accepted | Motor size     | IP rating        | Dimensions (L x W x H)                                                                              | Power (50/60 Hz) |                | Catalogue number |
|------------------------------------------------------------------------|---------------|---------------------|----------------|------------------|-----------------------------------------------------------------------------------------------------|------------------|----------------|------------------|
|                                                                        |               |                     |                |                  |                                                                                                     | VAC              | Amps           |                  |
| A. Digital modular drive with benchtop controller                      |               |                     |                |                  |                                                                                                     |                  |                |                  |
| 0.1 to 600                                                             | ±0.1%         | 2                   | 1/10 hp (75 W) | IP33             | Controller: 26.7 x 21 x 12.7 cm (10½" x 8¼" x 5")<br>Drive: 26.7 x 9.7 x 11.7 cm (10½" x 3⅜" x 4⅝") | 90 to 260        | 1.1 at 230 VAC | MY-77301-40      |
| 0.02 to 100                                                            |               | 4                   |                |                  |                                                                                                     |                  |                | MY-77301-60      |
| B. Digital washdown modular drive with washdown, wall-mount controller |               |                     |                |                  |                                                                                                     |                  |                |                  |
| 0.1 to 600                                                             | ±0.1%         | 2                   | 1/10 hp (75 W) | IP66/<br>NEMA 4X | Controller: 23 x 30 x 11.4 cm (9" x 11¼" x 4½")<br>Drive: 39.5 x 17.8 x 15.2 cm (15½" x 7" x 6")    | 90 to 260        | 1.1 at 230 VAC | MY-77301-50      |
| 0.02 to 100                                                            |               | 4                   |                |                  |                                                                                                     |                  |                | MY-77301-70      |

## Masterflex® L/S® Digital Console Process Drives

### Transfer, dispense, and meter in lab, process and washdown environments

- IP66 and NEMA 4X rated. Sealed housing and sealed keypad—simply hose down to clean
- Flow rate: 0.001 to 3400-mL/min with L/S tubing depending on drive rpm and tubing size
- Graphical LCD shows six different parameters: motor rpm, flow rate, dispense volume, cumulative volume, dispense interval time, and copy number.
- Dispense by: volume, copy, or time with programmable delay interval
- Remote control capabilities via waterproof connection
- Models equipped with open-head sensor stop drive motor when pump head is opened—pair with an open-head sensor equipped pump head

**What's included:** 1.8-m (6-ft) line cord—features IEC 320/CEE 22 connector coupler; watertight cord connection. Please specify destination country when ordering; drive is shipped with country-specific plug/cord set.

#### FEATURES

- ✓ Flow rates: 0.001 to 3400 mL/min
- ✓ IP66 and NEMA 4X rated
- ✓ Programmable, reversible motor accepts two or four pump heads and offers remote functionality



Drive 07575-38 shown with L/S Easy-Load II pump head 77200-32

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| rpm                                                           | Speed control (resolution) | Open-head sensor | Pump heads accepted | Motor size     | IP rating | Dimensions (L x W x H)                     | Power                   | Catalogue number |
|---------------------------------------------------------------|----------------------------|------------------|---------------------|----------------|-----------|--------------------------------------------|-------------------------|------------------|
| <b>Digital process drive with 316 stainless steel housing</b> |                            |                  |                     |                |           |                                            |                         |                  |
| 0.1 to 600                                                    | $\pm 0.1\%$                | No               | 2                   | 1/10 hp (75 W) | IP66      | 30.5 x 22.9 x 23.6 cm (12" x 9" x 9 9/16") | 90 to 260 VAC, 50/60 Hz | MY-07575-30      |
| 0.02 to 100                                                   |                            | Yes              |                     |                |           |                                            |                         | MY-07575-38      |
|                                                               |                            | No               | 4                   |                |           |                                            |                         | MY-07575-35      |
| <b>Digital process drive with powder-coat steel housing</b>   |                            |                  |                     |                |           |                                            |                         |                  |
| 0.1 to 600                                                    | $\pm 0.1\%$                | No               | 2                   | 1/10 hp (75 W) | IP66      | 30.5 x 22.9 x 23.6 cm (12" x 9" x 9 9/16") | 90 to 260 VAC, 50/60 Hz | MY-07575-40      |
|                                                               |                            | Yes              |                     |                |           |                                            |                         | MY-07575-48      |
| 0.02 to 100                                                   |                            | No               | 4                   |                |           |                                            |                         | MY-07575-45      |

**MY-07575-84 Washdown foot switch**, momentary start/stop; 1.8-m (6-ft) cable  
**MY-07575-80 Cable assembly** for remote control; 31-pin round connector and 7.9-m (25-ft) cable with stripped wire ends

**MY-17050-01 ILAC MRA, NIST-traceable calibration** with data for peristaltic pump drive

## Masterflex® L/S® Precision Variable-Speed Modular Drives

### Ideal for limited spaces, hoods, or isolation chambers

- Three-digit LED display shows motor rpm—confirm speed setting at a glance
- Remote control capabilities: speed, direction, and start/stop
- ±0.25% speed control for precise flow regulation—ideal for metering

#### A. Modular Drives with Benchtop Controller

feature a stackable, splash-resistant ABS plastic controller housing and epoxy-finished steel drive housing. Remote control via a DB9 female connector on back of controller.

#### B. Washdown Modular Drives with Wall-Mount Controller

have a powder-coated aluminum controller housing and powder-coated steel drive housing. Remote control via weather-resistant connection on controller.

#### FEATURES

- ✓ Flow rates: 0.06 to 3400 mL/min
- ✓ Easy up-front control of speed, direction, and start/stop
- ✓ Reversible motor—purge tubing before or after pumping; pump fluid in either direction



**B**  
Models 07557-60 and -70 include wall-mount controller

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| rpm                                                                      | Speed control<br>(resolution)                    | Pump<br>heads accepted | Motor<br>size     | IP<br>rating     | Dimensions<br>(L x W x H)                                 | Power (50/60 Hz) |                | Catalogue<br>number |
|--------------------------------------------------------------------------|--------------------------------------------------|------------------------|-------------------|------------------|-----------------------------------------------------------|------------------|----------------|---------------------|
|                                                                          |                                                  |                        |                   |                  |                                                           | VAC              | Amps           |                     |
| A. Precision modular drives with benchtop controller                     |                                                  |                        |                   |                  |                                                           |                  |                |                     |
| 6 to 600                                                                 | ±0.25% (1 rpm at 600 rpm;<br>0.1 rpm at 100 rpm) | 2                      | 1/10 hp<br>(75 W) | IP33             | Controller: 3.2 x 2.5 x 1.5 m<br>Drive: 3.3 x 1.3 x 1.4 m | 90 to 260        | 1.1 at 230 VAC | MY-07557-00         |
| 1 to 100                                                                 |                                                  | 4                      |                   |                  |                                                           |                  |                | MY-07557-10         |
| B. Precision washdown modular drives with washdown wall-mount controller |                                                  |                        |                   |                  |                                                           |                  |                |                     |
| 6 to 600                                                                 | ±0.25% (1 rpm at 600 rpm;<br>0.1 rpm at 100 rpm) | 2                      | 1/10 hp<br>(75 W) | IP66/<br>NEMA 4X | Controller: 2.7 x 3.6 x 1.4 m<br>Drive: 4.7 x 2.1 x 1.8 m | 90 to 260        | 1.1 at 230 VAC | MY-07557-60         |
| 1 to 100                                                                 |                                                  | 4                      |                   |                  |                                                           |                  |                | MY-07557-70         |

**MY-07557-51 Extension cable**, 2.7-m (9-ft); extend distance between motor and controller

**MY-07595-47 Cable assembly**, for benchtop modular drives; DB9 male connector and 7.9-m (25-ft) cable with stripped wire terminal ends. Use for remote control

**MY-77300-32 Cable assembly**, for washdown modular drives; 18-pin round connector and 7.9-m (25-ft) cable with stripped wire terminal ends. Use for remote control

**MY-17050-01 ILAC MRA, NIST-traceable calibration** with data for peristaltic pump drive

## Masterflex® L/S® Precision Console Process Drives

### Stand up to process and washdown applications

- IP66 and NEMA 4X rated; sealed housing and sealed membrane keypad
- Three-digit LED display lets you confirm speed setting at a glance
- Open-head sensor stops drive motor when pump head is opened—pair with an open-head sensor equipped pump head
- Remote control capabilities via a watertight connector on back of drive

Accurate console drives have powerful motors, speed display, and ±0.25% speed control for precise flow regulation—ideal for metering applications. Separate speed and power (on/off) control lets you maintain speed setting while turning pump on/off. Remote speed control requires cable assembly 77300-32.

**What's included:** 1.8-m (6-ft) line cord with IEC 320/CEE22 socket. Shipped with country-specific plug/cord set; please specify ultimate destination when ordering.

#### FEATURES

- ✓ Flow rates: 0.06 to 3400 mL/min
- ✓ Variable speed from 1 to 600 rpm
- ✓ Stack up to four pump heads



Drive 07528-40 shown with L/S Easy-Load® II pump head 77200-32

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| rpm                                                       | Speed control<br>(resolution) | Open-head<br>sensor | Pump heads<br>accepted | Motor size     | IP rating        | Dimensions<br>(L x W x H)                     | Power                      | Catalogue number |
|-----------------------------------------------------------|-------------------------------|---------------------|------------------------|----------------|------------------|-----------------------------------------------|----------------------------|------------------|
| Precision process drives with 316 stainless steel housing |                               |                     |                        |                |                  |                                               |                            |                  |
| 6 to 600                                                  | ±0.25%                        | Yes                 | 2                      | 1/10 hp (75 W) | IP66/<br>NEMA 4X | 30.5 x 22.9 x 23.6 cm<br>(12" x 9" x 9 5/16") | 90 to 260 VAC,<br>50/60 Hz | MY-07528-40      |
| 1 to 100                                                  |                               |                     | 4                      |                |                  |                                               |                            | MY-07528-45      |
| Precision process drives with powder-coat steel housing   |                               |                     |                        |                |                  |                                               |                            |                  |
| 6 to 600                                                  | ±0.25%                        | Yes                 | 2                      | 1/10 hp (75 W) | IP66/<br>NEMA 4X | 30.5 x 22.9 x 23.6 cm<br>(12" x 9" x 9 5/16") | 90 to 260 VAC,<br>50/60 Hz | MY-07528-50      |
| 1 to 100                                                  |                               |                     | 4                      |                |                  |                                               |                            | MY-07528-55      |

#### Accessories

**MY-77300-32 Cable assembly**, for precision process drives; 18-pin round connector and 7.9-m (25-ft) cable with stripped wire ends

**MY-17050-01 ILAC MRA, NIST-traceable calibration** for peristaltic pump drive



## Masterflex® L/S® Variable-Speed Economy Modular Drives

### Versatile with a wide flow range and simple controls

- Separate motor and controller are connected via a 1.9-m (6-ft) cable—place components where convenient or where space allows
- Single-turn speed control and on/off switch with green power indicator—maintain speed setting while turning pump on and off
- Painted steel housing is IP33 rated

A perfect upgrade from a fixed-speed drive, these variable-speed drives offer a wider rpm range,  $\pm 2\%$  speed regulation, and ETL and CE listings. For added versatility, drives accept all L/S pump heads.

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#### FEATURES

- ✓ Flow rates: 0.06 to 3400 mL/min
- ✓ Variable speed—1 to 100 or 6 to 600 rpm
- ✓ Convenient modular design



**Flow Rates in mL/min** (flow rates in parentheses can only be obtained with the High-Performance pump head—see page 131)

| rpm      | Precision pump tubing                            |            |            |            |            |            | High-performance precision pump tubing                                |                         |                         |                         |
|----------|--------------------------------------------------|------------|------------|------------|------------|------------|-----------------------------------------------------------------------|-------------------------|-------------------------|-------------------------|
|          | L/S 13                                           | L/S 14     | L/S 16     | L/S 25     | L/S 17     | L/S 18     | L/S 15                                                                | L/S 24                  | L/S 35                  | L/S 36                  |
|          | Standard†, Easy-Load®, Easy-Load II, Easy-Load 3 |            |            |            |            |            | Multichannel‡, Standard†, Easy-Load II, Easy-Load 3, High-Performance |                         |                         |                         |
|          | Cartridge, Multichannel                          |            |            | Cartridge  |            |            | Easy-Load                                                             |                         |                         |                         |
| 6 to 600 | 0.36 to 36                                       | 1.3 to 130 | 4.8 to 480 | 10 to 1000 | 17 to 1700 | 23 to 2300 | 10 to 1000 (11 to 1100)                                               | 17 to 1700 (18 to 1800) | 23 to 2300 (26 to 2600) | 29 to 2900 (34 to 3400) |
| 1 to 100 | 0.06 to 6                                        | 0.21 to 21 | 0.8 to 80  | 1.7 to 170 | 2.8 to 280 | 3.8 to 380 | 1.7 to 170 (1.8 to 180)                                               | 2.8 to 280 (3.0 to 300) | 3.8 to 380 (4.3 to 430) | 4.8 to 480 (5.8 to 580) |

†The Standard pump head is not available for L/S 25.

‡Multichannel head is not available for L/S 36. Multichannel and Cartridge heads are available for microbore tubing.

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| rpm      | Speed control (resolution) | Pump heads accepted | Motor size     | IP rating                       | Dimensions (L x W x H)                                                                                  | Power (50/60 Hz) |                | Catalogue number |
|----------|----------------------------|---------------------|----------------|---------------------------------|---------------------------------------------------------------------------------------------------------|------------------|----------------|------------------|
|          |                            |                     |                |                                 |                                                                                                         | VAC              | Amps           |                  |
| 6 to 600 | $\pm 2\%$                  | 2                   | 1/10 hp (75 W) | Controller: IP33<br>Motor: IP21 | Controller: 20.4 x 16.4 x 8.7 cm (8" x 6.4" x 3.4")<br>Motor: 19.8 x 9.7 x 11.5 cm (7.8" x 3.8" x 4.5") | 190 to 260       | 1.2 at 230 VAC | MY-07559-05      |
| 1 to 100 | $\pm 2\%$                  | 4                   | 1/10 hp (75 W) | Controller: IP33<br>Motor: IP21 | Controller: 20.4 x 16.4 x 8.7 cm (8" x 6.4" x 3.4")<br>Motor: 19.8 x 9.7 x 11.5 cm (7.8" x 3.8" x 4.5") | 190 to 260       | 1.2 at 230 VAC | MY-07559-15      |

MY-17050-01 ILAC MRA, NIST-traceable calibration with data for peristaltic pump

## Masterflex® L/S® High-Pressure Pumps

### Pump at pressures up to 125 psi (8.5 bar)

- Three-digit LED display shows motor rpm—confirm speed setting at a glance
- High-performance pump head offers automatic occlusion—optimised when pump head is closed and latched—load tubing in 15 seconds or less
- Pump at pressures up to 125 psi (8.5 bar) with included L/S 16HP tubing—ideal for feed, filtration, and metering applications
- Models equipped with open-head sensor stop the drive motor when pump head is opened



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77914-25,  
-27

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| System | Flow range       | What's included                              |                                                                 |                                                                              | Power                   | Catalogue number |
|--------|------------------|----------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------|------------------|
|        |                  | Pump head                                    | Tubing                                                          | Drive                                                                        |                         |                  |
|        | 0.9 to 90 mL/min | High-Performance 77252-72                    | PharMed BPT L/S 16HP high-pressure tubing 05664-16 2.4-m (8 ft) | Precision console drive with ABS plastic housing 07528-30 1 to 100 rpm       | 90 to 260 VAC, 50/60 Hz | MY-77914-20      |
|        | 0.9 to 90 mL/min | High-Performance (open-head sensor) 77252-62 |                                                                 | Precision process drive with powder-coat steel housing 07528-55 1 to 100 rpm |                         | MY-77914-25      |
|        | 0.9 to 90 mL/min | High-Performance (open-head sensor) 77252-62 |                                                                 | Precision process drive with stainless steel housing 07528-45 1 to 100 rpm   |                         | MY-77914-27      |



## Masterflex® L/S® Fixed-Speed Drives

### Get consistent, repeatable flow rates

- Stackable drives are ideal for field or lab use
- Lighted power switch on front of drive indicates that pump is running
- Accept five different pump head types: Standard, Easy-Load®, Easy-Load II, Easy-Load 3, and Cartridge

**What's included:** 1.8-m (6-ft) line cord, European plug.

#### Flow Rates in mL/min<sup>††</sup>

| Flow rates in mL/min |                                                 |        |        |        |        |        |
|----------------------|-------------------------------------------------|--------|--------|--------|--------|--------|
| rpm                  | Precision pump tubing                           |        |        |        |        |        |
|                      | L/S 13                                          | L/S 14 | L/S 16 | L/S 25 | L/S 17 | L/S 18 |
|                      | Standard†, Easy-Load, Easy-Load II, Easy-Load 3 |        |        |        |        |        |
|                      | Cartridge‡                                      |        |        |        |        |        |
| 1                    | 0.06                                            | 0.21   | 0.8    | 1.7    | 2.8    | 3.8    |
| 2                    | 0.12                                            | 0.42   | 1.6    | 3.4    | 5.6    | 7.6    |
| 6                    | 0.36                                            | 1.3    | 4.8    | 10     | 17     | 23     |
| 12                   | 0.72                                            | 2.5    | 9.6    | 20     | 34     | 46     |
| 20                   | 1.2                                             | 4.2    | 16     | 34     | 56     | 76     |
| 30                   | 1.8                                             | 6.3    | 24     | 50     | 84     | 110    |
| 60                   | 3.6                                             | 13     | 48     | 100    | 170    | 230    |

| rpm | High-Performance Precision pump tubing |        |        |        |
|-----|----------------------------------------|--------|--------|--------|
|     | L/S 15                                 | L/S 24 | L/S 35 | L/S 36 |
|     | Standard, Easy-Load II, Easy-Load 3    |        |        |        |
|     | Easy-Load                              |        |        |        |
| 1   | 1.7                                    | 2.8    | 3.8    | 4.8    |
| 2   | 3.4                                    | 5.6    | 7.6    | 9.6    |
| 6   | 10                                     | 17     | 23     | 29     |
| 12  | 20                                     | 34     | 46     | 58     |
| 20  | 34                                     | 56     | 76     | 96     |
| 30  | 50                                     | 84     | 110    | 140    |
| 60  | 100                                    | 170    | 230    | 290    |

<sup>†</sup>Standard pump head is not available for L/S 25. <sup>‡</sup>Cartridge heads are available for microbore tubing. <sup>††</sup>Motor rpm and flow rates for the 230 VAC, 50 Hz economy drives are approximately 5% of values shown.



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#### FEATURES

- ✓ Flow rates: 0.06 to 290 mL/min
- ✓ General, intermittent-duty fluid transfer applications

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| rpm                  | Speed control      | Pump heads accepted | Motor size                               | IP rating | Dimensions (L x W x H)                       | Catalogue number |
|----------------------|--------------------|---------------------|------------------------------------------|-----------|----------------------------------------------|------------------|
| 230 VAC/50 Hz models |                    |                     |                                          |           |                                              |                  |
| 1                    | None <sup>††</sup> | 2                   | Sub-fractional (<1/25 hp) <sup>†††</sup> | IP22      | 15.2 x 13.4 x 11.9 cm (6" x 5 1/4" x 4 3/8") | MY-07542-01      |
| 2                    |                    |                     |                                          |           |                                              | MY-07542-02      |
| 5                    |                    |                     |                                          |           |                                              | MY-07542-06      |
| 10                   |                    |                     |                                          |           |                                              | MY-07542-12      |
| 17                   | None <sup>††</sup> | 2                   | Sub-fractional (<1/25 hp) <sup>†††</sup> | IP22      | 15.2 x 13.4 x 11.9 cm (6" x 5 1/4" x 4 3/8") | MY-07542-20      |
| 25                   |                    |                     |                                          |           |                                              | MY-07542-30      |
| 50                   |                    |                     |                                          |           |                                              | MY-07542-60      |

<sup>††</sup>Drive speed may vary  $\pm 15\%$ , subject to fluctuations in line voltage and frequency.

<sup>†††</sup>Maximum power required is 70 W or less.

## Masterflex® L/S® Digital Miniflex® Pumps

### Small footprint, yet delivers accurate flow over a wide range

- Flow range of 0.005 to 381 mL/min (depending on tubing size)
- Tubing calibration function ensures accuracy
- Open-head sensor stops drive when pump head is opened
- Remote control via 25-pin connector on back of drive
- Automatic tubing retention makes tube loading quick and easy

Compact, stackable pump with durable ABS housing is easy to wipe down. The Miniflex pump head features a wide occlusion angle that prevents fluid backflow up to the rated pressure of the tubing. Self-lubricating thermoplastic polyester rollers are low friction and long wearing for smooth operation and long life.

#### Flow Rates (mL/min) at 0.1 to 300 rpm

| Pump model | Precision pump tubing |             |              |              |
|------------|-----------------------|-------------|--------------|--------------|
|            | L/S 13                | L/S 14      | L/S 16       | L/S 25       |
| 07525-20   | 0.005 to 14           | 0.018 to 55 | 0.061 to 184 | 0.127 to 381 |
| 07525-40   | 0.005 to 14           | 0.018 to 55 | 0.061 to 184 | —            |

| Number of channels | rpm        | Speed control (repeatability) | Reversible motor | IP rating | Dimensions (L x W x H)                     | Power (50/60 Hz) |                | Catalogue number |
|--------------------|------------|-------------------------------|------------------|-----------|--------------------------------------------|------------------|----------------|------------------|
|                    |            |                               |                  |           |                                            | VAC              | Amps           |                  |
| 1                  | 0.1 to 300 | $\pm 0.1\%$                   | Yes              | IP33      | 23.9 x 14.0 x 15.7 cm (9.4" x 5.5" x 6.2") | 90 to 260        | 0.2 at 230 VAC | MY-07525-20      |
| 2                  |            |                               |                  |           |                                            |                  |                | MY-07525-40      |

**MY-07523-92 Foot switch**, momentary start/stop; DB25 male; 1.8-m (6-ft) cable

**MY-07523-95 Cable assembly**, DB25 male connector and 7.9-m (25-ft) cable with stripped wire ends



#### FEATURES

- ✓ Flow rates: 0.005 to 381 mL/min
- ✓ Intuitive icon-driven programming
- ✓ Tubing calibration for accuracy
- ✓ Single- or dual-channel models

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Power supplies are:

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**MY-07523-97 Handheld remote controller**, DB25 male, route tubing through handle for dispensing applications; 1.8-m (6-ft) cable  
**MY-17050-01 ILAC MRA, NIST-traceable calibration** with data for peristaltic pump drive



# Masterflex® Drives Powering Quattroflow™ Pumps

## Move your delicate biological fluids

- Diaphragms enable soft, gentle pumping through the strokes generated by the eccentric shaft
- No mechanical seal or rotating parts that can fail—only the diaphragm and body come in contact with the fluid

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**Quattroflow Pumps** offer a unique four diaphragm (chamber) design, providing the constant low-pulsation, low-shear pumping needed for your biological fluids. In addition, the quaternary diaphragm design enables risk-free dry-running, self-priming, and a high turn-down ratio. Both the single-use and multiple-use models safely and reliably pump plasma products, proteins, monoclonal antibodies, vaccines, and more.

Single-use pump heads have a disposable wetted chamber constructed of solid polypropylene (PP) that can be replaced as a complete unit—eliminating the cleaning validation, sterilisation process. You can quickly turn around equipment from one batch to the next, maximising production operations. Come pre-sterilised and double bagged.

Multiple-use pump systems ensure total product containment without abrasion and minimal particle creation. Stainless steel construction handles higher pressure applications including cross-flow filtration and chromatography columns.

**Special Masterflex Drives** were developed to best take advantage of the operating parameters of the Quattroflow pumps. Drives feature an intuitive interface, remote control functionality, and a brushless, maintenance-free motor with  $\pm 0.1\%$  speed control accuracy. The graphical LCD shows six different parameters: motor rpm, flow rate, dispense volume, cumulative volume, dispense interval time, and copy number.

Plastic-enclosed drive has an IP33 rating for benchtop lab applications and has been combined with the lower flow QF150 series pump head.

Stainless steel drives have IP66 and NEMA 4X ratings and are designed for both the QF150 and QF1200 series. Sealed stainless steel housing and sealed keypad are protected from dust, water, and chemicals in the washdown environment—simply hose down to clean.



### Applications

- Cross-flow systems
- Tangential flow filtration
- Chromatography systems
- Reaction dosing
- Virus or sterile filtration
- Buffer mixing systems
- Virus cultures
- Bacterial and viral vaccines
- Cell cultures
- Cell culture supernatants
- Enzyme solutions
- Antibodies
- Virus inactivation

| Systems                |                                   |                  | Replacement pump heads |
|------------------------|-----------------------------------|------------------|------------------------|
| Pump head model        | Drive type                        | Catalogue number | Catalogue number       |
| QF150SU (single use)   | ABS plastic, IP33                 | MY-74000-35      | MY-76000-00            |
|                        | Stainless steel, IP66 and NEMA 4X | MY-74000-37      | MY-76000-00            |
| QF150S (multiple use)  | ABS plastic, IP33                 | MY-74000-31      | MY-76000-06            |
|                        | Stainless steel, IP66 and NEMA 4X | MY-74000-33      | MY-76000-06            |
| QF1200SU (single use)  | Stainless steel, IP66 and NEMA 4X | MY-74000-47      | MY-76000-02            |
| QF1200S (multiple use) | Stainless steel, IP66 and NEMA 4X | MY-74000-43      | MY-76000-08            |

### Specifications for Quattroflow Pumps

| Model                       | QF150SU (single use)                                                     | QF150S (multiple use)                                                         | QF1200SU (single use)                                                    | QF1200S (multiple use)                                                        |
|-----------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Flow range                  | 1 to 180 L/h<br>(0.26 to 48 GPH)                                         | 1 to 180 L/h<br>(0.26 to 48 GPH)                                              | 20 to 960 L/h<br>(5.3 to 253 GPH)                                        | 20 to 960 L/h<br>(5.3 to 253 GPH)                                             |
| Max pressure                | For fluid temp <40°C (104°F)<br>4 bar (58 psi)                           | 4 bar (58 psi)                                                                | 4 bar (58 psi)                                                           | 6 bar (87 psi)                                                                |
|                             | For fluid temp >40°C (104°F)<br>4 bar (58 psi)                           | 4 bar (58 psi)                                                                | 4 bar (58 psi)                                                           | 4 bar (58 psi)                                                                |
| Max fluid temperature       | 60°C (140°F)                                                             | 80°C (176°F)                                                                  | 60°C (140°F)                                                             | 80°C (176°F)                                                                  |
| Suction lift at 300 rpm     | 2 to 3 m (6.6 to 9.8 ft)                                                 | 2 to 3 m (6.6 to 9.8 ft)                                                      | 2 to 3 m (6.6 to 9.8 ft)                                                 | 2 to 3 m (6.6 to 9.8 ft)                                                      |
| Volume per revolution       | At free output<br>1.2 mL                                                 | 1.2 mL                                                                        | 9.6 mL                                                                   | 9.6 mL                                                                        |
|                             | Without connector<br>15 mL                                               | 15 mL                                                                         | 75 mL                                                                    | 75 mL                                                                         |
| Connection type             | 6.4 mm (¼") Tri-Clamp®                                                   | 6.4 mm (¼") Tri-Clamp                                                         | 19 mm (¾") Tri-Clamp                                                     | 19 mm (¾") Tri-Clamp                                                          |
| Position of connectors      | In-line                                                                  | In-line                                                                       | In-line                                                                  | In-line                                                                       |
| Connector direction options | 4                                                                        | 4                                                                             | 4                                                                        | 4                                                                             |
| Wetted materials            | PP (housing, valve plate),<br>TPE (diaphragm), EPDM<br>(valves, O-rings) | 316L SS (housing, valve plate),<br>TPE (diaphragm), EPDM<br>(valves, O-rings) | PP (housing, valve plate),<br>TPE (diaphragm), EPDM<br>(valves, O-rings) | 316L SS (housing, valve plate),<br>TPE (diaphragm), EPDM<br>(valves, O-rings) |

### Specifications for Masterflex Drives

| Description                 | ABS plastic drive for QF150                                                                                                                                                                                                                           | Stainless steel drive for QF150 | Stainless steel drive for QF1200 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|
| Speed                       | 3000 rpm                                                                                                                                                                                                                                              | 3000 rpm                        | 1700 rpm                         |
| Speed control               | $\pm 0.1\%$                                                                                                                                                                                                                                           | $\pm 0.1\%$                     | $\pm 0.1\%$                      |
| Motor size                  | 1/16 hp (0.05 kW)                                                                                                                                                                                                                                     | 1/16 hp (0.05 kW)               | 3/8 hp (0.26 kW)                 |
| IP rating                   | IP33                                                                                                                                                                                                                                                  | IP66 and NEMA 4X                | IP66 and NEMA 4X                 |
| Dispense modes              | Motor rpm, flow rate, dispense volume, cumulative volume, dispense interval time, and copy number                                                                                                                                                     |                                 |                                  |
| Input/output communications | Speed control: scalable 0 to 20 mA, 4 to 20 mA, 0 to 10 V; Pumping direction: requires open collector or contact closure; Start/stop/purge: requires open collector or contact closure; Tachometer output: scalable 0 to 20 mA, 4 to 20 mA, 0 to 10 V |                                 |                                  |
| Power                       | 90 to 260 VAC                                                                                                                                                                                                                                         |                                 |                                  |

# Masterflex® Drives Enhance Worker Protection

**You take your safety seriously, and so do we.**

You review every step of the process to take care of your staff and to benefit your business. We've added a new option to our Masterflex product line to help enhance protection in your workplace.

## Open-Head Interlock Sensor Technology

Available integrated into select L/S® and I/P® drives or as a retrofit system, the open-head interlock sensors add an extra level of protection for operators who maintain pumps. When paired with an open-head sensor equipped pump head, the drive automatically shuts down when the pump head is opened, eliminating exposure to the moving rollers connected to the drive shaft.

Read the article *How Masterflex Pumps with Open Head Sensors Enhance Worker Protection*  
[tech-article/enhanced-worker-safety-open-head-interlock-sensor](#)

## Masterflex products with integrated open-head sensors

L/S Computer-Compatible Digital Drives [page 121](#)

L/S Variable-Speed Digital Drives [page 122](#)

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I/P Digital Console Process Drives [page 134](#)

I/P Easy-Load Pump Head [page 135](#)

## Upgrade your Existing Drives

Use the retrofit system controller to incorporate open-head sensors into pump systems you already have. The system includes a control box that easily connects to your drive and an open-head sensor equipped pump head. If pump head is opened during operation, the interlock control box instantly shuts down the drive.

The retrofit system is compatible with select L/S and I/P drives. Controller 77097-00 includes power cables and a single extension cable to connect one sensor-equipped pump head. Order optional extension cable 77097-20 to connect a second pump head.

MY-77097-00 Open-head interlock sensor control box

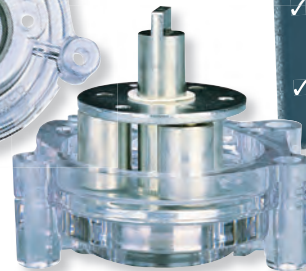
MY-77097-20 Optional extension cable, 1.8-m (6-ft)



## Masterflex® L/S® Standard Pump Heads

### Ideal for accurate and repeatable fluid transfer applications

- PC housing allows you to see pump in operation;
- PPS housing and SS rotors have excellent chemical resistance

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#### FEATURES

- ✓ Best overall pump head performance
- ✓ Designed and optimised for one tubing size
- ✓ Stack up to four heads

| Pump tubing size                                                         | mL per rev | Flow rates in mL/min |              | Max system pressure bar (psi) | Catalogue number† |             |             |
|--------------------------------------------------------------------------|------------|----------------------|--------------|-------------------------------|-------------------|-------------|-------------|
|                                                                          |            | 0.02 to 100 rpm      | 6 to 600 rpm |                               | PC housing        | PPS housing | SS rotor    |
|                                                                          |            |                      |              |                               | CRS rotor         | SS rotor    |             |
| <b>A. Standard pump heads for Precision pump tubing</b>                  |            |                      |              |                               |                   |             |             |
| L/S 13                                                                   | 0.06       | 0.001 to 6           | 0.36 to 36   | 2.7 (40)                      | MY-07013-20       | MY-07013-21 | MY-07013-52 |
| L/S 14                                                                   | 0.21       | 0.004 to 21          | 1.3 to 130   | 2.7 (40)                      | MY-07014-20       | MY-07014-21 | MY-07014-52 |
| L/S 16                                                                   | 0.8        | 0.016 to 80          | 4.8 to 480   | 2.7 (40)                      | MY-07016-20       | MY-07016-21 | MY-07016-52 |
| L/S 17                                                                   | 2.8        | 0.059 to 280         | 17 to 1700   | 1.4 (20)                      | MY-07017-20       | MY-07017-21 | MY-07017-52 |
| L/S 18                                                                   | 3.8        | 0.077 to 380         | 23 to 2300   | 1.0 (15)                      | MY-07018-20       | MY-07018-21 | MY-07018-52 |
| <b>B. Standard pump heads for High-Performance Precision pump tubing</b> |            |                      |              |                               |                   |             |             |
| L/S 15                                                                   | 1.7        | 0.033 to 170         | 10 to 1000   | 2.7 (40)                      | MY-07015-20       | MY-07015-21 | MY-07015-52 |
| L/S 24                                                                   | 2.8        | 0.057 to 280         | 17 to 1700   | 2.7 (40)                      | MY-07024-20       | MY-07024-21 | MY-07024-52 |
| L/S 35                                                                   | 3.8        | 0.077 to 380         | 23 to 2300   | 2.4 (35)                      | MY-07035-20       | MY-07035-21 | —           |
| L/S 36                                                                   | 4.8        | 0.096 to 480         | 29 to 2900   | 1.4 (20)                      | MY-07036-30       | MY-07036-31 | —           |

†PC = polycarbonate PPS = polyphenylene sulfide CRS = cold-rolled steel SS = stainless steel

#### Mounting Hardware

Stack up to four L/S Standard pump heads on one drive. Mounting hardware is stainless steel.

| For tubing size  | Heads to be mounted | Catalogue number |
|------------------|---------------------|------------------|
| L/S 13 to L/S 35 | One                 | MY-07013-04      |
|                  | Two                 | MY-07013-05      |
|                  | Three               | MY-07013-08      |
|                  | Four                | MY-07013-09      |
| L/S 36           | One                 | MY-07036-01      |
|                  | Two                 | MY-07036-02      |

## Masterflex® L/S® Easy-Load® II Pump Heads

### Automatic retention holds tubing securely in place

- Reduced pulsation and improved pressure performance and mechanical stability with standard four-roller rotor
- Choice of rotors—economical CRS rotors or SS rotors for better chemical resistance and durability
- Don't let pressure get to you. Choose models with adjustable occlusion—tighten for higher pressure, loosen for longer tubing life
- Models with open-head interlock sensor stop drive motor when pump head is open (when used with sensor-equipped drives)

#### FEATURES

- ✓ Load tubing easily
- ✓ Accepts several tubing sizes for a wide flow range
- ✓ Stack up to four heads
- ✓ Durable PPS housing



**What's included:** single-channel mounting hardware and a 38-cm (15") length of silicone pump tubing. Pump head comes ready to mount to drive.

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| Pump tubing size                                                      | mL per rev | Flow rates in mL/min |              | Max system pressure† bar (psi) | Fixed occlusion |             |                                | Adjustable occlusion |
|-----------------------------------------------------------------------|------------|----------------------|--------------|--------------------------------|-----------------|-------------|--------------------------------|----------------------|
|                                                                       |            | 1 to 100 rpm         | 6 to 600 rpm |                                | CRS‡ rotor      | SS‡ rotor   | SS‡ rotor and open-head sensor | SS‡ rotor            |
|                                                                       |            |                      |              |                                | Cat. no.        | Cat. no.    | Cat. no.                       |                      |
| A. Easy-Load II pump heads for Precision pump tubing (not shown)      |            |                      |              |                                |                 |             |                                |                      |
| L/S 13                                                                | 0.06       | 0.06 to 6            | 0.36 to 36   | 2.7 (40)                       | MY-77200-50     | MY-77200-60 | MY-77200-30                    | MY-77201-60          |
| L/S 14                                                                | 0.21       | 0.21 to 21           | 1.3 to 130   | 2.7 (40)                       |                 |             |                                |                      |
| L/S 16                                                                | 0.8        | 0.8 to 80            | 4.8 to 480   | 2.7 (40)                       |                 |             |                                |                      |
| L/S 25                                                                | 1.7        | 1.7 to 170           | 10 to 1000   | 2.4 (35)                       |                 |             |                                |                      |
| L/S 17                                                                | 2.8        | 2.8 to 280           | 17 to 1700   | 1.4 (20)                       |                 |             |                                |                      |
| L/S 18                                                                | 3.8        | 3.8 to 380           | 23 to 2300   | 1.0 (15)                       |                 |             |                                |                      |
| B. Easy-Load II pump heads for High-Performance Precision pump tubing |            |                      |              |                                |                 |             |                                |                      |
| L/S 15                                                                | 1.7        | 1.7 to 170           | 10 to 1000   | 2.7 (40)                       | MY-77200-52     | MY-77200-62 | MY-77200-32                    | MY-77201-62          |
| L/S 24                                                                | 2.8        | 2.8 to 280           | 17 to 1700   | 2.7 (40)                       |                 |             |                                |                      |
| L/S 35                                                                | 3.8        | 3.8 to 380           | 23 to 2300   | 2.4 (35)                       |                 |             |                                |                      |
| L/S 36                                                                | 4.8        | 4.8 to 480           | 29 to 2900   | 1.4 (20)                       |                 |             |                                |                      |
| C. Two-channel Easy-Load II pump heads for Precision pump tubing      |            |                      |              |                                |                 |             |                                |                      |
| L/S 13                                                                | 0.06       | 0.06 to 6            | 0.36 to 36   | 2.7 (40)                       | MY-77202-50     | MY-77202-60 | —                              | —                    |
| L/S 14                                                                | 0.21       | 0.21 to 21           | 1.3 to 130   | 2.7 (40)                       |                 |             |                                |                      |
| L/S 16                                                                | 0.8        | 0.8 to 80            | 4.8 to 480   | 2.7 (40)                       |                 |             |                                |                      |
| L/S 25                                                                | 1.7        | 1.7 to 170           | 10 to 1000   | 2.4 (35)                       |                 |             |                                |                      |
|                                                                       |            |                      |              |                                |                 |             |                                |                      |

†Actual performance varies depending on pump tubing formulation—values shown are for firm pump tubing.

‡CRS = cold-rolled steel SS = stainless steel

**Mounting Hardware.** Mount up to four L/S Easy-Load II pump heads on a single drive. Mounting hardware is stainless steel.

| Heads to be mounted | One         | Two         | Three       | Four        |
|---------------------|-------------|-------------|-------------|-------------|
| Catalogue number    | MY-77200-01 | MY-77200-02 | MY-77200-03 | MY-77200-04 |

## Masterflex® L/S® High-Performance Pump Heads

### All-new design offers faster tubing changes and improved ergonomics

- Simply close the pump head and tubing occlusion is optimal and automatic
- Flow rates up to 3400-mL/min with L/S high-performance precision tubing
- Reinforced nylon housing and occlusion bed with stainless steel rotor/roller assembly
- Models with open-head interlock sensor stop drive motor when pump head is open (when used with sensor-equipped drives)
- Compatible with all L/S drives that accept two or more pump heads

#### FEATURES

- ✓ Pressures up to 150 psi with L/S high-pressure (HP) pump tubing
- ✓ Automatic tubing occlusion—no need to adjust
- ✓ ATEX Zone 2 rated models available



**What's included:** mounting hardware and 38 cm (15") of pump tubing.

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**ATEX**  
Zone 2  
77252-80, 82



| Pump tubing size                                                       | mL per rev | Flow rates in mL/min |                 | Max system pressure bar (psi) | Catalogue number                                 |                            |                                          |
|------------------------------------------------------------------------|------------|----------------------|-----------------|-------------------------------|--------------------------------------------------|----------------------------|------------------------------------------|
|                                                                        |            | 0.02 to 100 rpm      | 0.1 to 600 rpm  |                               | High-Performance pump head with open-head sensor | High-Performance pump head | ATEX-approved High-Performance pump head |
| High-performance pump heads for Precision pump tubing                  |            |                      |                 |                               |                                                  |                            |                                          |
| L/S 16                                                                 | 0.9        | 0.018 to 90          | 0.09 to 614     | 2.7 (40)                      | MY-77252-60                                      | MY-77252-70                | MY-77252-80                              |
| L/S 25                                                                 | 1.9        | 0.038 to 187         | 0.19 to 1278    | 2.7 (40)                      |                                                  |                            |                                          |
| L/S 17                                                                 | 3.0        | 0.06 to 307          | 0.30 to 1883    | 1.4 (20)                      |                                                  |                            |                                          |
| L/S 18                                                                 | 3.8        | 0.076 to 377         | 0.38 to 2812    | 2.7 (40)                      |                                                  |                            |                                          |
| High-performance pump heads for High-Performance Precision pump tubing |            |                      |                 |                               |                                                  |                            |                                          |
| L/S 14HP†                                                              | 0.3        | 0.006 to 30          | Not recommended | 10.3 (150)                    | MY-77252-62                                      | MY-77252-72                | MY-77252-82                              |
| L/S 16HP†                                                              | 0.9        | 0.018 to 90          | Not recommended | 8.5 (125)                     |                                                  |                            |                                          |
| L/S 15HP†                                                              | 1.7        | 0.034 to 170         | Not recommended | 6.8 (100)                     |                                                  |                            |                                          |
| L/S 24HP†                                                              | 2.4        | 0.048 to 240         | Not recommended | 5.5 (80)                      |                                                  |                            |                                          |
| L/S 15                                                                 | 1.8        | 0.036 to 180         | 0.18 to 1100    | 2.7 (40)                      |                                                  |                            |                                          |
| L/S 24                                                                 | 3.0        | 0.06 to 300          | 0.30 to 1800    | 2.7 (40)                      |                                                  |                            |                                          |
| L/S 35                                                                 | 4.3        | 0.086 to 430         | 0.43 to 2600    | 2.4 (35)                      |                                                  |                            |                                          |
| L/S 36                                                                 | 5.8        | 0.116 to 580         | 0.58 to 3400    | 1.4 (20)                      |                                                  |                            |                                          |

†L/S 14HP, L/S 15HP, L/S 16HP and L/S 24HP pump tubing is for use only with the L/S High-Performance pump head mounted on any L/S 100 rpm drive capable of running two or more pump heads. **Note:** High-Performance pump heads are not stackable.

## Masterflex® L/S® Easy-Load® 3 Pump Heads

### No tools or hardware needed

- Mount and stack heads in seconds with twist-lock mounting feature
- CRS rotor is more economical while SS offers greater durability and chemical resistance
- Included mounting plate adapts Easy-Load 3 pump heads to L/S drives—mount up to four pump heads without tools or hardware

#### FEATURES

- ✓ Same side tubing entry/exit for space-limited applications
- ✓ Automatic tubing retention
- ✓ Stack up to four heads



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| Pump tubing sizes                                                 | mL per rev | Flow rates in mL/min |              | Max system pressure† bar (psi) | Catalogue number‡ |             |
|-------------------------------------------------------------------|------------|----------------------|--------------|--------------------------------|-------------------|-------------|
|                                                                   |            | 1 to 100 rpm         | 6 to 600 rpm |                                | CRS rotor         | SS rotor    |
| Easy-Load 3 pump heads for Precision pump tubing                  |            |                      |              |                                |                   |             |
| L/S 13                                                            | 0.06       | 0.06 to 6            | 0.36 to 36   | 2.7 (40)                       | MY-77800-50       | MY-77800-60 |
| L/S 14                                                            | 0.22       | 0.22 to 22           | 1.3 to 130   | 2.7 (40)                       |                   |             |
| L/S 16                                                            | 0.8        | 0.8 to 80            | 4.8 to 480   | 2.7 (40)                       |                   |             |
| L/S 25                                                            | 1.7        | 1.7 to 170           | 10 to 1000   | 2.4 (35)                       |                   |             |
| L/S 17                                                            | 2.8        | 2.8 to 280           | 17 to 1700   | 1.4 (20)                       |                   |             |
| L/S 18                                                            | 3.8        | 3.8 to 380           | 23 to 2300   | 1.0 (15)                       |                   |             |
| Easy-Load 3 pump heads for High-Performance Precision pump tubing |            |                      |              |                                |                   |             |
| L/S 15                                                            | 1.7        | 1.7 to 170           | 10 to 1000   | 2.0 (30)                       | MY-77800-52       | MY-77800-62 |
| L/S 24                                                            | 2.8        | 2.8 to 280           | 17 to 1700   | 2.0 (30)                       |                   |             |
| L/S 35                                                            | 3.8        | 3.8 to 380           | 23 to 2300   | 1.7 (25)                       |                   |             |
| L/S 36                                                            | 4.8        | 4.8 to 480           | 29 to 2900   | 1.4 (20)                       |                   |             |

†Actual performance varies depending on pump tubing formulation—values shown are for firm pump tubing.

‡CRS = cold-rolled steel SS = stainless steel

**MY-77800-00 Replacement mounting kit**, for mounting Easy-Load 3 pump head to Masterflex L/S pump drives. Includes mounting plate and four screws

## Masterflex® L/S® Easy-Load® Pump Heads

### New design, with smaller footprint, reduced pulsation and even faster tubing changes!

- No need to remove head to change tubing
- SS rotor offers durability and chemical resistance
- Polyarylamide (PARA) housing offers strength and durability
- Get four separate pumping channels by mounting up to four heads on one drive

**What's included:** single-channel mounting hardware.

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| Pump tubing sizes                                               | mL per rev | Flow rates in mL/min |                | Max system pressure bar (psi) | Catalogue number† |                                    |
|-----------------------------------------------------------------|------------|----------------------|----------------|-------------------------------|-------------------|------------------------------------|
|                                                                 |            |                      |                |                               | PARA housing      | PARA housing with open head sensor |
|                                                                 |            | 0.02 to 100 rpm      | 0.1 to 600 rpm |                               | SS rotor          | SS rotor                           |
| Easy-Load pump heads for Precision pump tubing                  |            |                      |                |                               |                   |                                    |
| L/S 13                                                          | 0.06       | 0.001 to 6           | 0.006 to 36    | 2.7 (40)                      | MY-07514-10       | MY-07514-30                        |
| L/S 14                                                          | 0.21       | 0.004 to 21          | 0.02 to 130    | 2.7 (40)                      |                   |                                    |
| L/S 16                                                          | 0.8        | 0.016 to 80          | 0.08 to 480    | 2.7 (40)                      |                   |                                    |
| L/S 25                                                          | 1.7        | 0.034 to 170         | 0.17 to 1000   | 2.4 (35)                      |                   |                                    |
| L/S 17                                                          | 2.8        | 0.056 to 280         | 0.28 to 1700   | 1.4 (20)                      |                   |                                    |
| L/S 18                                                          | 3.8        | 0.076 to 380         | 0.38 to 2300   | 1.0 (15)                      |                   |                                    |
| Easy-Load pump heads for High-Performance Precision pump tubing |            |                      |                |                               |                   |                                    |
| L/S 15                                                          | 1.7        | 0.034 to 170         | 0.17 to 1000   | 2.7 (40)                      | MY-07514-12       | MY-07514-32                        |
| L/S 24                                                          | 2.8        | 0.056 to 280         | 0.28 to 1700   | 2.7 (40)                      |                   |                                    |

†PARA = Polyarylamide SS = stainless steel



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**Mounting Hardware.** Stack up to four L/S Easy-Load pump heads on one drive. Hardware is stainless steel.

| Heads to be mounted | Catalogue number |
|---------------------|------------------|
| One                 | MY-07514-04      |
| Two                 | MY-07013-05      |
| Three               | MY-07013-08      |
| Four                | MY-07013-09      |

## Masterflex® L/S® PTFE-Tubing Pump Head

### Ideal for high-purity fluids and aggressive organic solvents

- PTFE tubing is the only wetted part; anodised aluminum and acetal body; stainless steel rotor assembly
- Compatible with L/S drives that accept two or more pump heads (1/10 hp min); For best results, run at 1 to 300 rpm

**Note:** PTFE-tubing pump head accepts **only** the Masterflex L/S PTFE tubing sets sold below; ordinary PTFE tubing cannot be substituted.

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| Flow rates (mL/min at 1 to 300 rpm) |                | Max continuous pressure | Max suction lift | Avg tubing life (100 rpm, 0 psi) | Number of rollers | Catalogue number |
|-------------------------------------|----------------|-------------------------|------------------|----------------------------------|-------------------|------------------|
| 4-mm OD tubing                      | 6-mm OD tubing |                         |                  |                                  |                   |                  |
| 0.06 to 17                          | 0.25 to 65     | 6.9 bar (100 psi)       | 2.4 m (8 ft)     | 500 hours                        | Six               | MY-77390-00      |

**PTFE-Tubing Sets** include two 38 cm (15") lengths, cannot be substituted with ordinary PTFE tubing.

**MY-77390-50 PTFE-tubing set.** 2-mm ID, 4-mm OD. Set of 2

**MY-77390-60 PTFE-tubing set.** 4-mm ID, 6-mm OD. Set of 2

**Compression Fittings** adapt tubing sets to your system.

For 4-mm OD PTFE-Tubing Sets

**MY-31321-61 Straight connector,** 2.8 bar (40 psi) max

**MY-06605-53 PTFE extension tubing,** 4-mm OD. Pack of 7.6 m (25 ft)

For 6-mm OD PTFE-Tubing Sets

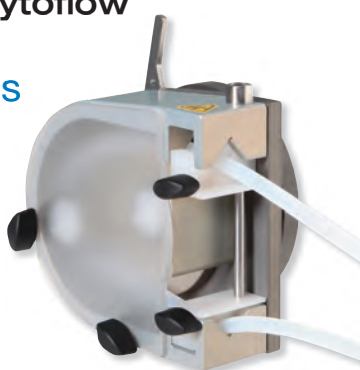
**MY-31321-64 Straight connector,** 9.2 bar (135 psi) max

**MY-06605-54 PTFE extension tubing,** 6-mm OD. Pack of 7.6 m (25 ft)

## Masterflex® L/S® Cytoflow™ Pump Heads

### Gentler on live cells

- Low-speed pumping lets you maintain the consistency and characteristics of shear-sensitive fluids
- Choose 2 rollers for maximum flow rate, 3 rollers for less pulsation
- Compatible with all Masterflex L/S drives that accept two or more pump heads



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| Pump tubing size                                               | mL per rev | Flow rates in mL/min |                | Max system pressure bar (psi) | Catalogue number |
|----------------------------------------------------------------|------------|----------------------|----------------|-------------------------------|------------------|
|                                                                |            | 0.02 to 100 rpm      | 0.1 to 600 rpm |                               |                  |
| 2-roller pump heads for High-Performance Precision pump tubing |            |                      |                |                               |                  |
| L/S 15                                                         | 4.0        | 0.08 to 390          | 0.40 to 2300   | 1.5 (22)                      | MY-77253-02      |
| L/S 24                                                         | 6.2        | 0.12 to 600          | 0.62 to 3700   | 1.5 (22)                      |                  |
| L/S 35                                                         | 9.5        | 0.19 to 930          | 0.95 to 5600   | 1.5 (22)                      |                  |
| L/S 36                                                         | 12.8       | 0.26 to 1270         | 1.3 to 7600    | 1.5 (22)                      |                  |
| 3-roller pump heads for High-Performance Precision pump tubing |            |                      |                |                               |                  |
| L/S 15                                                         | 3.5        | 0.07 to 340          | 0.35 to 2000   | 1.5 (22)                      | MY-77253-12      |
| L/S 24                                                         | 5.7        | 0.11 to 570          | 0.57 to 3400   | 1.5 (22)                      |                  |
| L/S 35                                                         | 7.8        | 0.16 to 780          | 0.78 to 4600   | 1.5 (22)                      |                  |
| L/S 36                                                         | 9.6        | 0.19 to 950          | 0.96 to 5700   | 1.5 (22)                      |                  |



# Masterflex® L/S® Multichannel Pump Heads

## Get low-pulsation synchronous flow in multiple channels without cartridges

- Load the tubing, latch the occlusion bed, and run the pump—two-stop tubing gives you optimal tension with no adjustment
- Stackable two, four, or eight-channel heads for up to 24 channels
- Six roller heads deliver low-pulsation flow; three-roller heads offer higher flow rates
- Heads for L/S Precision pump tubing accept two-stop tubing set sizes L/S 13, L/S 14, and L/S 16; heads for L/S High-Performance Precision pump tubing accept two-stop tubing set sizes L/S 15, L/S 24, and L/S 35
- Compatible with L/S drives that accept two or more pump heads; 1/10 hp min

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### FEATURES

- ✓ Stackable for up to 24 synchronous flow channels
- ✓ Models that accept L/S two-stop tubing sets available
- ✓ Durable stainless steel and anodised aluminum construction



**A**



Stack heads for even more channels!

**What's included:** mounting hardware and hex key tool.

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| Number of channels                                                                                      | Number of rollers | Max channels with stacked heads† |               | Catalogue number |
|---------------------------------------------------------------------------------------------------------|-------------------|----------------------------------|---------------|------------------|
|                                                                                                         |                   | 100 rpm drive                    | 600 rpm drive |                  |
| A. L/S Multichannel pump heads for Precision pump tubing, sizes L/S 13, L/S 14, L/S 16                  |                   |                                  |               |                  |
| Four                                                                                                    | Six               | 24                               | 12            | MY-07535-04      |
| Eight                                                                                                   |                   |                                  |               | MY-07535-08      |
| B. L/S Multichannel pump heads for High-Performance Precision pump tubing, sizes L/S 15, L/S 24, L/S 35 |                   |                                  |               |                  |
| Two                                                                                                     | Three             | 12                               | Six           | MY-07536-02      |
| Four                                                                                                    |                   |                                  |               | MY-07536-04      |

†Maximum channel capacity depends on drive torque capability and tubing formulation used.

### L/S Two-Stop Precision Tubing Sets Ordering Information for 07535- and 07536-series pump heads

| Pump tubing formulation                 |  | L/S Precision pump tubing |                        |                        | L/S High-Performance Precision pump tubing |                        |                        |
|-----------------------------------------|--|---------------------------|------------------------|------------------------|--------------------------------------------|------------------------|------------------------|
|                                         |  | L/S 13                    | L/S 14                 | L/S 16                 | L/S 15                                     | L/S 24                 | L/S 35                 |
| Tygon® E-LFL                            |  | MY-06447-13<br>Pk of 8    | MY-06447-14<br>Pk of 8 | MY-06447-16<br>Pk of 8 | MY-06447-15<br>Pk of 4                     | MY-06447-24<br>Pk of 4 | MY-06447-35<br>Pk of 4 |
| Silicone (platinum-cured)               |  | MY-06421-13<br>Pk of 8    | MY-06421-14<br>Pk of 8 | MY-06421-16<br>Pk of 8 | MY-06421-15<br>Pk of 4                     | MY-06421-24<br>Pk of 4 | MY-06421-35<br>Pk of 4 |
| BioPharm Plus Silicone (platinum-cured) |  | MY-96116-13<br>Pk of 8    | MY-96116-14<br>Pk of 8 | MY-96116-16<br>Pk of 8 | MY-96116-15<br>Pk of 4                     | MY-96116-24<br>Pk of 4 | MY-96116-35<br>Pk of 4 |
| PharMed® BPT                            |  | MY-96114-13<br>Pk of 8    | MY-96114-14<br>Pk of 8 | MY-96114-16<br>Pk of 8 | MY-96114-15<br>Pk of 4                     | MY-96114-24<br>Pk of 4 | MY-96114-35<br>Pk of 4 |
| Chem-Durance® Bio                       |  | MY-96117-13<br>Pk of 8    | MY-96117-14<br>Pk of 8 | MY-96117-16<br>Pk of 8 | MY-96117-15<br>Pk of 4                     | MY-96117-24<br>Pk of 4 | MY-96117-35<br>Pk of 4 |
| Viton®                                  |  | MY-96428-13<br>Pk of 8    | MY-96428-14<br>Pk of 8 | MY-96428-16<br>Pk of 8 | MY-96428-15<br>Pk of 4                     | MY-96428-24<br>Pk of 4 | MY-96428-35<br>Pk of 4 |

### Flow Rates per Channel in mL/min with L/S Two-Stop Precision Pump Tubing Sets above

| Drive rpm | Precision pump tubing |            |            | High-Performance Precision pump tubing |            |            |
|-----------|-----------------------|------------|------------|----------------------------------------|------------|------------|
|           | L/S 13                | L/S 14     | L/S 16     | L/S 15                                 | L/S 24     | L/S 35     |
| 1 to 100  | 0.045 to 4.5          | 0.16 to 16 | 0.47 to 47 | 1.6 to 160                             | 2.8 to 280 | 3.8 to 380 |
| 6 to 600  | 0.27 to 27            | 0.96 to 96 | 2.8 to 280 | 10 to 1000                             | 17 to 1700 | 23 to 2300 |

### Application Solution



#### Challenge

A pharmaceutical manufacturer needed to dispense volumes of 300-mL in 15 seconds or less repeating three times per minute; and 20-mL in 2 seconds or less repeating 20 times per minute. The desired accuracy target for both applications was  $\pm 1.5\%$ . The efficiency of simultaneously pumping multiple channels was critical.

#### Solution

The manufacturer now uses a Masterflex L/S digital drive with an L/S four-channel pump head 07536-04. Size L/S 15 two-stop silicone tubing (06421-15) met the smaller flow requirement; while size L/S 35 two-stop silicone tubing (06421-35) met the larger flow requirement. Careful calibration of the drive, and the synchronous multichannel flow capability of the pump head enabled the manufacturer to meet the accuracy target.

## Masterflex® I/P® Digital Console Process Drives

### Maintenance-free drives built for process applications

- Graphical LCD shows four operating modes: continuous run, timed dispense, copy dispense, volume dispense
- Dispense by volume, copy, or time with programmable delay interval
- Models equipped with open-head sensor stop drive motor when pump head is opened—pair with an open-head sensor equipped pump head (see page 135)

**Models 77420-10 and -18** feature a 316 stainless steel housing with an anodised aluminum base. Ideal for pharmaceutical, dairy, biotech, and food and beverage applications.

**Models 77420-20 and -28** feature a powder-coated steel housing with anodised aluminum base. Use in process, manufacturing, and water-treatment applications where stainless steel is not a critical requirement.

**What's included:** 1.8-m (6-ft) line cord with IEC 320/CEE22 socket. Shipped with country-specific plug/cord set; please specify ultimate destination when ordering.



77420-10 shown with I/P High-Performance pump head 77600-62

#### FEATURES

- ✓ Flow rates: 0.0006 to 19 LPM
- ✓ Accuracy of  $\pm 0.1\%$  speed control
- ✓ Four-line graphical display
- ✓ Reversible motor
- ✓ IP66-rated for easy washdown
- ✓ Accept up to 2 pump heads

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| rpm                                                    | Speed control<br>(resolution) | Open-head<br>sensor | Pump heads<br>accepted | Motor<br>size   | IP<br>rating | Dimensions<br>(L x W x H)                                                                                                         | Power                      | Catalogue<br>number |
|--------------------------------------------------------|-------------------------------|---------------------|------------------------|-----------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------|
| Digital process drive with 316 stainless steel housing |                               |                     |                        |                 |              |                                                                                                                                   |                            |                     |
| 0.1 to 650                                             | ±0.1%                         | No                  | Two                    | ¾ hp<br>(280 W) | IP66         | 42.4 x 28.3 x 7.3 cm<br>(16 <sup>11</sup> / <sub>16</sub> " x 11 <sup>1</sup> / <sub>8</sub> " x 12 <sup>7</sup> / <sub>8</sub> " | 90 to 260 VAC,<br>50/60 Hz | MY-77420-10         |
|                                                        |                               | Yes                 |                        |                 |              |                                                                                                                                   |                            | MY-77420-18         |
| Digital process drive with powder-coat steel housing   |                               |                     |                        |                 |              |                                                                                                                                   |                            |                     |
| 0.1 to 650                                             | ±0.1%                         | No                  | Two                    | ¾ hp<br>(280 W) | IP66         | 42.4 x 28.3 x 7.3 cm<br>(16 <sup>11</sup> / <sub>16</sub> " x 11 <sup>1</sup> / <sub>8</sub> " x 12 <sup>7</sup> / <sub>8</sub> " | 90 to 260 VAC,<br>50/60 Hz | MY-77420-20         |
|                                                        |                               | Yes                 |                        |                 |              |                                                                                                                                   |                            | MY-77420-28         |

**MY-07575-84 Washdown foot switch**, momentary start/stop, 1.8-m (6-ft) cable  
**MY-07575-80 Cable assembly**, 31-pin round connector and 7.9-m (26-ft) cable with stripped wire ends for remote control

**MY-17050-01 ILAC MRA, NIST-traceable calibration** for peristaltic pump drive

## Masterflex® I/P® Precision Process Drives

### Powerful, low-maintenance drives keep you pumping

- Don't worry about moisture, chemicals, or dust—IP55-rated nonchip epoxy-coated steel housing
- Goes where you need it—compact and light enough to carry with one hand, powerful enough to drive two pump heads
- Get up to speed quickly—easy to adjust speed knob and LED display for improved repeatability

#### FEATURES

- ✓ Flow rates: 0.2 to 19 LPM
- ✓ Accuracy of  $\pm 0.25\%$  speed control
- ✓ LED display
- ✓ Reversible motor and remote control capabilities
- ✓ Accept up to 2 pump heads



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| Motor speed<br>(rpm)                                            | Speed control<br>(repeatability) | Pump heads<br>accepted | Motor<br>size     | Dimensions                                   | Power (50/60 Hz) |                | Catalogue<br>number |
|-----------------------------------------------------------------|----------------------------------|------------------------|-------------------|----------------------------------------------|------------------|----------------|---------------------|
|                                                                 |                                  |                        |                   |                                              | VAC              | Amps           |                     |
| Variable-speed brushless process drive                          |                                  |                        |                   |                                              |                  |                |                     |
| 33 to 650                                                       | ±0.25%<br>(2% of max rpm)        | Two                    | ⅓ hp<br>(0.25 kW) | 35.6 x 25.4 x 22.9 cm<br>(14"L x 10"W x 9"H) | 90 to 260        | 2.6 at 230 VAC | MY-77410-10         |
| Variable-speed brushless process drive with remote input/output |                                  |                        |                   |                                              |                  |                |                     |
| 33 to 650                                                       | ±0.25%<br>(2% of max rpm)        | Two                    | ⅓ hp<br>(0.25 kW) | 35.6 x 25.4 x 22.9 cm<br>(14"L x 10"W x 9"H) | 90 to 260        | 2.6 at 230 VAC | MY-77411-00         |

**MY-07595-43 Washdown foot switch** for model 77411-00

**MY-77300-32 Remote cable kit.** Required for model 77411-00 to operate the remote inputs and outputs



## Masterflex® I/P® High-Performance Pump Heads

### Get higher flow rates and pressure

- Easily transfer viscous fluids
- Pump up to 19-LPM for higher volume processing
- Durable design for longer life—made of stainless steel rollers, bearings, rotor plates, and rotor shaft. Occlusion bed is made of thermoset polyester
- Compatible with I/P drives that accept two or more pump heads; pump head is not stackable



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ATEX  
Zone 2  
77600-82



| Pump tubing sizes accepted | mL per rev | Flow rates (LPM) at 20 to 650 rpm | Maximum system bar (psi) | Catalogue number†                  |                                                   |
|----------------------------|------------|-----------------------------------|--------------------------|------------------------------------|---------------------------------------------------|
|                            |            |                                   |                          | Polyester and SS housing, SS rotor | ATEX-approved; polyester and SS housing, SS rotor |
| I/P 70                     | 12.3       | 0.26 to 8                         | 2.7 (40)                 | MY-77600-62                        | MY-77600-82                                       |
| I/P 88                     | 26.2       | 0.4 to 17                         | 2.4 (35)                 |                                    |                                                   |
| I/P 89                     | 29.2       | 0.52 to 19                        | 1.4 (20)                 |                                    |                                                   |

†SS = stainless steel

#### FEATURES

- ✓ Easy to load tubing
- ✓ Adjustable tubing retention
- ✓ “C”-shaped design—tubing enters/exits the same side
- ✓ ATEX Zone 2 rated model available

## Masterflex® I/P® Easy-Load® Pump Heads

### Minimise downtime with faster tubing changes

- Optimise pressure capabilities and tubing life with adjustable occlusion
- Less maintenance and quieter operation
- Adjust flow rate to your needs—can pump from 0.01 to 13-LPM
- PPS housing with CRS or SS rotor available. CRS rotors have sealed bearings; SS rotors have PTFE-sealed bearings
- Model with open-head interlock sensor stops drive motor when pump head is open (when used with sensor-equipped drives)

#### FEATURES

- ✓ Load tubing without removing pump head from drive
- ✓ Automatic tubing retention
- ✓ Stack up to two pump heads
- ✓ ATEX Zone 2 rated model available



**MASTERFLEX®**

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ATEX  
Zone 2  
77602-80



| Pump tubing sizes accepted | mL per rev | Flow rates (LPM) at indicated rpm |      |           | Maximum system bar (psi) | Catalogue number† |             |                               |                   |                            |
|----------------------------|------------|-----------------------------------|------|-----------|--------------------------|-------------------|-------------|-------------------------------|-------------------|----------------------------|
|                            |            |                                   |      |           |                          | White PPS housing |             |                               | Black PPS housing | ATEX-approved; PPS housing |
|                            |            | 100                               | 540  | 20 to 650 |                          | CRS rotor         | SS rotor    | SS rotor and open-head sensor | SS rotor          | SS rotor                   |
| I/P 26                     | 6.15       | 0.6                               | 3.2  | 0.12 to 4 | 2.7 (40)                 | MY-77602-00       | MY-77602-10 | MY-77602-30                   | MY-77601-60       | MY-77602-80                |
| I/P 73                     | 12.3       | 1.2                               | 6.8  | 0.2 to 8  |                          |                   |             |                               |                   |                            |
| I/P 82                     | 20.0       | 2.0                               | 11.0 | 0.4 to 13 |                          |                   |             |                               |                   |                            |

†PPS = polyphenylene sulfide CRS = cold-rolled steel SS = stainless steel

**MY-77601-96 Dual-channel mounting hardware** for two pump heads, stainless steel

## Masterflex® B/T® Fixed- and Variable-Speed Pumps

Achieve the highest flow rates of any Masterflex pump—up to 42-LPM

### Fixed-Speed Pumps

- Economical and easy to operate
- Reversible motor: purge before or after pumping, pump fluid in either direction
- IP55-rated motor—wash off dirt and debris

### Air-Powered Pump

- Use in locations where electricity is unsafe or impractical
- Includes pressure regulator—control your compressor air to match pump requirements
- 24-cfm (0.68 m<sup>3</sup>), 6.9 bar (100 psi) minimum air required, supply your own compressor to power pump
- IP56-rated motor protects against dust and strong water jets
- Rated for ATEX Zone 2 EEx II 3 G c IIC T6; NEC Class 1, Division 2, Groups A, B, C, D, T6

### Variable-Speed Modular Pump

- Place washdown controller wherever it's convenient
- Controller with 4.9-m (16-ft) cord mounts on pump or wall
- IP56-rated motor protects against dust and strong water jets
- Reversible motor: purge before or after pumping, pump fluid in either direction

### Variable-Speed Modular Digital Pump

- High-volume, high-accuracy dispensing
- Dispense by volume or time, or pump continuously
- Separate motor and controller by up to 1.8-meters (6-feet) for convenient operation
- IP56-rated controller and motor protect against dust and water

**What's included:** a Rapid-Load® pump head, 1 m (3 ft) of Tygon® E-LFL B/T 91 pump tubing, and a 1.8-m (6-ft) power cord 230 VAC: European plug.

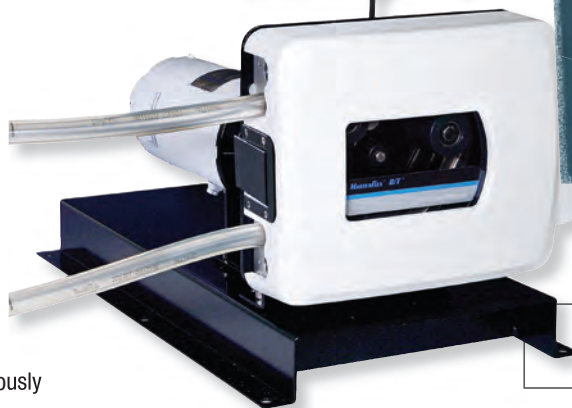


Fixed-speed pump  
77111-60



### FEATURES

- ✓ Flow rates up to 42 LPM
- ✓ Interlock switch shuts off power when pump head is opened
- ✓ Washdown pumps handle the toughest process and production applications, then clean up with ease
- ✓ Gentle peristaltic action is ideal for pumping shear-sensitive or viscous fluids (up to 10,000 cps)
- ✓ Rotor and occlusion bed design allow easier tube loading and prevent "tubing roll" during operation



Variable-speed  
modular digital pump  
77111-70

### Flow Rates in LPM (GPM)

| rpm       | PerfectPosition® pump tubing |                         |
|-----------|------------------------------|-------------------------|
|           | B/T 87                       | B/T 91                  |
| 321       | 18.9 (5.0)                   | 42 (11.1)               |
| 271       | 16.0 (4.2)                   | 35 (9.4)                |
| 35 to 321 | 1.9 to 17.7 (0.5 to 4.7)     | 4.0 to 37 (1.06 to 9.8) |
| 12 to 321 | 0.71 to 18.9 (0.19 to 5.0)   | 1.4 to 42 (0.4 to 11.1) |

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77111-37,  
-67, -77, -80

ATEX  
Zone 2  
77111-80

IP 55  
77111-30,  
-37

IP 56  
Except  
77111-30,  
-37



| Motor type                     | Motor size     | rpm       | IP rating                       | Mounting type | Dimensions (L x W x H)                                                                                 | Power                                                            |      | Catalogue number |
|--------------------------------|----------------|-----------|---------------------------------|---------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|------|------------------|
|                                |                |           |                                 |               |                                                                                                        | VAC (Hz)                                                         | Amps |                  |
| Fixed-speed                    | 1 hp (0.75 kW) | 271       | IP55                            | Direct-drive  | 68.6 x 40 x 38.4 cm (27" x 15.8" x 15.1")                                                              | 220 (50)                                                         | 6.3  | MY-77111-37      |
| Variable-speed, air powered    | ¾ hp (0.56 kW) | 35 to 321 | IP56                            | Direct-drive  | 68.6 x 40 x 38.4 cm (27" x 15.8" x 15.1")                                                              | Requires 24 cfm (0.68 m <sup>3</sup> ) 100 psi (6.9 bar) minimum |      | MY-77111-80      |
| Variable-speed (±5%)           | ½ hp (0.37 kW) | 12 to 321 | Controller: IP56<br>Drive: IP56 | Direct-drive  | 68.6 x 40 x 38.4 cm (27" x 15.8" x 15.1")                                                              | 230 (50/60)                                                      | 3.3  | MY-77111-67      |
| Variable-speed digital (±0.3%) | ¾ hp (0.56 kW) | 12 to 321 |                                 | Direct-drive  | Controller: 22.9 x 27.9 x 11.4 cm (9" x 11" x 4½")<br>Drive: 68.6 x 40 x 38.4 cm (27" x 15.8" x 15.1") | 190 to 260 (50/60)                                               | 5.7  | MY-77111-77      |

**MY-77300-32 Remote cable**, required for remote control operation of models 77111-70 and -77; 7.9-m (25-ft)

**MY-77111-90 Bracket for digital controller** on models 77111-70 and -77

## Masterflex® C/L® Variable-Speed Compact Pumps

### Pump in even the tightest spaces!

- Pump fluids in either direction—easily switchable reversing motor
- Easy tubing changes—open cover and release occlusion bed in one motion

#### A. C/L Single-Channel Pumps

- Stop/start remotely via contact closure
- MAX button quickly primes or purges tubing
- Pump from 0.002- to 43-mL/min

#### B. C/L Digital Microflex® Pump

- Icon-driven menu allows quick programming and setup
- Flow range of 0.00013- to 43-mL/min
- Copy dispense function for filling and dispensing applications
- Remote control of speed, direction, and start/stop/prime via 25-pin connector on back of drive
- Microflex pump head accepts continuous lengths of tubing for clean flow paths
- Tubing calibration ensures accuracy



#### FEATURES

- ✓ Flow range: 0.00013 to 43 mL/min
- ✓ Reversible pumping
- ✓ Remote control capabilities

MASTERFLEX®

Power supplies are:

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| rpm                                 | Speed control<br>(repeatability) | Number of<br>channels | Number of<br>rollers | Reversible<br>motor | IP<br>rating | Dimensions—¼ DIN<br>(L x W x H)          | Power                       |                                    | Catalogue<br>number        |
|-------------------------------------|----------------------------------|-----------------------|----------------------|---------------------|--------------|------------------------------------------|-----------------------------|------------------------------------|----------------------------|
|                                     |                                  |                       |                      |                     |              |                                          | VAC, Hz                     | Amps                               |                            |
| A. C/L single-channel compact pumps |                                  |                       |                      |                     |              |                                          |                             |                                    |                            |
| 1.7 to 10                           | ±5%                              | One                   | Four                 | Yes                 | IP22         | 17.8 x 9.5 x 8.9 cm<br>(7" x 3¾" x 3½")  | 90 to 260, 50/60<br>12 VDC† | 250 mA at 230 VAC<br>—             | MY-77122-04<br>MY-77122-06 |
| 13 to 80                            |                                  |                       | Three                |                     |              |                                          | 90 to 260, 50/60<br>12 VDC† | 250 mA at 230 VAC<br>—             | MY-77122-14<br>MY-77122-16 |
| 50 to 300                           |                                  |                       |                      |                     |              |                                          | 90 to 260, 50/60<br>12 VDC† | 250 mA at 230 VAC<br>—             | MY-77122-24<br>MY-77122-26 |
|                                     |                                  |                       |                      |                     |              |                                          |                             |                                    |                            |
| B. C/L digital Microflex pump       |                                  |                       |                      |                     |              |                                          |                             |                                    |                            |
| 0.1 to 300                          | ±0.1%                            | One                   | Three                | Yes                 | IP33         | 20.5 x 14 x 15.7 cm<br>(8¼" x 5½" x 6¾") | 90 to 260                   | 0.34 at 115 VAC,<br>0.2 at 230 VAC | MY-77123-00                |

†Power supply not included with 12 VDC models.

**MY-17050-01 ILAC MRA, NIST-traceable calibration** with data for peristaltic pump drive

**For A models:**

**MY-77120-03 Brackets** for panel mounting. Set of 2

**MY-77200-07 Replacement power supply**, 100 to 240 VAC, for pump 77122-24

**MY-77120-11 Replacement power supply**, 100 to 240 VAC, for pumps 77122-04, -14

**For B model:**

**MY-07523-92 Foot switch**, momentary start/stop; DB25 male; 1.8-m (6-ft) cable

**MY-07523-95 Cable assembly**, DB25 male connector and 7.9-m (25-ft) cable with stripped wire ends

**MY-07523-97 Handheld remote controller**, DB25 male, route tubing through handle for dispensing applications; 1.8-m (6-ft) cable

**MY-77123-80 Replacement single-channel Microflex pump head**

**MY-07525-90 Replacement power supply**, 90 to 260 VAC

| Tube ID (mm) | Tygon® E-Lab    | Silicone (platinum-cured) | Silicone (peroxide-cured) | PharMed® BPT    | Tygon E-LFL (long flex life) | Viton®         | Puri-Flex™     | C-Flex®        | Solvent/hydrocarbon |
|--------------|-----------------|---------------------------|---------------------------|-----------------|------------------------------|----------------|----------------|----------------|---------------------|
| 0.19         | MY-06460-10     | —                         | MY-07625-10               | MY-95809-10     | MY-06449-10                  | —              | MY-96418-10    | MY-95718-10    | MY-95712-10         |
| 0.25         | MY-06460-12     | —                         | MY-07625-12               | MY-95809-12     | MY-06449-12                  | —              | MY-96418-12    | MY-95718-12    | MY-95712-12         |
| 0.51         | MY-06460-18     | MY-95590-18               | MY-07625-18               | MY-95809-18     | MY-06449-18                  | MY-97632-18    | MY-96418-18    | MY-95718-18    | MY-95712-18         |
| 0.89         | MY-06460-26     | MY-95590-26               | MY-07625-26               | MY-95809-26     | MY-06449-26                  | MY-97632-26    | MY-96418-26    | MY-95718-26    | MY-95712-26         |
| 1.14         | MY-06460-30     | MY-95590-30               | MY-07625-30               | MY-95809-30     | MY-06449-30                  | MY-97632-30    | MY-96418-30    | MY-95718-30    | MY-95712-30         |
| 1.42         | MY-06460-34     | MY-95590-34               | MY-07625-34               | MY-95809-34     | MY-06449-34                  | MY-97632-34    | MY-96418-34    | MY-95718-34    | MY-95712-34         |
| 2.06         | MY-06460-42     | MY-95590-42               | MY-07625-42               | MY-95809-42     | MY-06449-42                  | MY-97632-42    | MY-96418-42    | MY-95718-42    | MY-95712-42         |
| 2.79         | MY-06460-48     | MY-95590-48               | MY-07625-48               | MY-95809-48     | MY-06449-48                  | MY-97632-48    | MY-96418-48    | MY-95718-48    | MY-95712-48         |
| Length/pk    | 30.4 m (100 ft) | 15.2 m (50 ft)            | 15.2 m (50 ft)            | 30.4 m (100 ft) | 30.4 m (100 ft)              | 15.2 m (50 ft) | 15.2 m (50 ft) | 15.2 m (50 ft) | 15.2 m (50 ft)      |

#### Flow Rates in mL/min

| rpm        | Microbore pump tubing size (ID) |                |                |               |               |               |              |              |
|------------|---------------------------------|----------------|----------------|---------------|---------------|---------------|--------------|--------------|
|            | 0.19 mm                         | 0.25 mm        | 0.51 mm        | 0.89 mm       | 1.14 mm       | 1.42 mm       | 2.06 mm      | 2.79 mm      |
| 1.7 to 10  | 0.002 to 0.013                  | 0.004 to 0.022 | 0.015 to 0.087 | 0.041 to 0.25 | 0.064 to 0.39 | 0.09 to 0.57  | 0.18 to 1.05 | 0.28 to 1.65 |
| 13 to 80   | 0.017 to 0.10                   | 0.03 to 0.18   | 0.12 to 0.70   | 0.33 to 2.0   | 0.52 to 3.1   | 0.75 to 4.5   | 1.4 to 8.5   | 1.8 to 11.0  |
| 50 to 300  | 0.06 to 0.39                    | 0.11 to 0.66   | 0.43 to 2.6    | 1.2 to 7.5    | 1.9 to 11.4   | 2.8 to 17.1   | 5.3 to 32    | 7.2 to 43    |
| 0.1 to 300 | 0.00013 to 0.39                 | 0.0002 to 0.66 | 0.0009 to 2.6  | 0.003 to 7.5  | 0.004 to 11.4 | 0.006 to 17.1 | 0.011 to 32  | 0.014 to 43  |

## Masterflex® Pump Tubing

### Ensure optimal performance from your Masterflex pump

#### Silicone (platinum-cured)

- Low protein binding, fungus resistant, phthalate and latex-free
- Accommodates a wide temperature range
- USP Class V, EP, FDA, REACH, RoHS, ADF

#### Silicone (peroxide-cured)

- Greater physical compression capability
- Economical, longer tubing life
- USP Class VI, EP, FDA, 3A, REACH, RoHS, ADF

#### BioPharm Silicone (platinum-cured)

- Low absorption; excellent biocompatibility
- Very low extractable; weather, ozone, corona, radiation resistant
- USP Class VI, EP, FDA, 3A, REACH, RoHS, ADF

#### BioPharm Plus Silicone (platinum-cured)

- Ultra-smooth—minimises particle entrapment
- Very low extractable—ideal for lab, biotech, and pharmaceutical
- USP Class VI, EP, FDA, REACH, RoHS, ADF

#### Puri-Flex™

- Heat sealable, weldable, and economical
- Long pump life when compared to silicone or many other TPE tubings
- USP Class VI, FDA, 3A, REACH, RoHS, ADF

#### C-Flex®

- Heat sealable, weldable, economical
- Better resistance to pH fluctuations
- USP Class VI, EP, REACH, RoHS, ADF

#### C-Flex® ULTRA

- Longest life of any C-Flex formulation
- Lowest spallation and reduced residue
- USP Class VI, EP, REACH, RoHS, ADF

#### PharMed® BPT

- >10,000 hours of tubing life; sealable, weldable
- Ideal for tissue and cell culture work
- USP Class VI, EP, FDA, NSF, REACH, RoHS

#### PharmaPure®

- Non-toxic and non-hemolytic biocompatible
- Long life even under pressure with very low spallation
- USP Class VI, EP, FDA, REACH, RoHS

#### GORE® STA-PURE® Series PCS

- Long life—even in pressures up to 100 psi
- Excellent flow stability
- USP Class VI

#### GORE® STA-PURE® Series PFL

- Long life—even in pressures up to 100 psi
- Excellent biocompatibility and flow stability
- USP Class VI

#### Chem-Durance® Bio

- Excellent chemical compatibility and biocompatibility
- Excellent life and durability under pressure
- USP Class VI, FDA, REACH, RoHS

#### Tygon® E-LFL

- Longest life of all Tygon formulations
- Broad chemical resistance
- USP Class VI, EP, FDA, EU Food, REACH, RoHS, ADF

#### Tygon E-Food (B-44-4X)

- Smooth inner surface unaffected by commercial sanitisers
- Non-toxic; will not affect taste or odour
- FDA, EU Food, NSF, 3A, REACH, RoHS, ADF

#### Tygon E-Lab (E-3603)

- Ideal for general transfer use
- Nonaging, nonoxidising, low gas permeability
- USP Class VI, FDA, EU Food, NSF, REACH, RoHS, ADF

#### Tygon Fuel & Lubricant (F-4040-A)

- Formulated to transport hydrocarbons, petroleum products, and distillates
- Low gas permeability
- REACH, RoHS, ADF

#### Tygon Chemical (2001)

- Excellent chemical resistance
- Plasticiser-free with low gas permeability
- FDA 21 CFR 175.300, REACH, RoHS, ADF

#### Norprene® (A-60 G)

- Up to 10,000 hours of tubing life
- Excellent choice for pressure/vacuum applications
- NSF, REACH, RoHS

#### Norprene Food (A 60 F)

- Ideal for high-temperature food applications
- Resists heat, ozone, acids, and alkalis
- FDA, 3A, NSF, REACH, RoHS

#### Viton®

- Perfect for applications where FDA and chemical compatibility is needed
- Resists corrosives, solvents, and oils at elevated temperatures
- FDA, ADF

**MASTERFLEX®**

## Masterflex® Gamma Irradiated Tubing

### Reduce the risk of microbial contamination

- Save time—many sizes of packaged and pretreated tubing available for quick shipment!
- Tubing is double bagged prior to gamma irradiation
- Irradiation at 32.5 kGy offers optimal microbial control for the tubing's critical fluid pathway
- Gamma-sensitive labeling confirms treatment
- Select from popular biocompatible tubing formulations

| Formulation               | Qty/pk        | L/S 16           | L/S 15           | L/S 24           | I/P 73           | I/P 82           |
|---------------------------|---------------|------------------|------------------|------------------|------------------|------------------|
|                           |               | Catalogue number | Catalogue number | Catalogue number | Catalogue number | Catalogue number |
| Silicone (platinum-cured) | 7.6 m (25 ft) | MY-96119-16      | MY-96119-15      | MY-96119-24      | MY-96119-73      | MY-96119-82      |
| PharMed® BPT              | 7.6 m (25 ft) | MY-96122-16      | MY-96122-15      | MY-96122-24      | MY-96122-73      | MY-96122-82      |
| C-Flex®                   | 7.6 m (25 ft) | MY-96123-16      | MY-96123-15      | MY-96123-24      | MY-96123-73      | MY-96123-82      |

|                                         |                | L/S® Precision Pump Tubing     |                                 |                                |                                 |                                 |                                 | L/S® High-Performance Precision Pump Tubing          |                                                      |                                                      |                                                      |
|-----------------------------------------|----------------|--------------------------------|---------------------------------|--------------------------------|---------------------------------|---------------------------------|---------------------------------|------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
|                                         | Qty/pk         | L/S 13<br>0.06 to 36<br>mL/min | L/S 14<br>0.21 to 130<br>mL/min | L/S 16<br>0.8 to 480<br>mL/min | L/S 25<br>1.7 to 1000<br>mL/min | L/S 17<br>2.8 to 1700<br>mL/min | L/S 18<br>3.8 to 2300<br>mL/min | L/S 15<br>1.7 to 1000 mL/min<br>(1.8 to 1100 mL/min) | L/S 24<br>2.8 to 1700 mL/min<br>(3.0 to 1800 mL/min) | L/S 35<br>3.8 to 2300 mL/min<br>(4.3 to 2600 mL/min) | L/S 36<br>4.8 to 2900 mL/min<br>(5.8 to 3400 mL/min) |
| Silicone (platinum-cured)               | 7.6 m (25 ft)  | MY-96410-13                    | MY-96410-14                     | MY-96410-16                    | MY-96410-25                     | MY-96410-17                     | MY-96410-18                     | MY-96410-15                                          | MY-96410-24                                          | MY-96410-35                                          | MY-96410-36                                          |
| Silicone (peroxide-cured)               | 7.6 m (25 ft)  | MY-96400-13                    | MY-96400-14                     | MY-96400-16                    | MY-96400-25                     | MY-96400-17                     | MY-96400-18                     | MY-96400-15                                          | MY-96400-24                                          | MY-96400-35                                          | MY-96400-36                                          |
| BioPharm Silicone (platinum-cured)      | 7.6 m (25 ft)  | MY-96420-13                    | MY-96420-14                     | MY-96420-16                    | MY-96420-25                     | MY-96420-17                     | MY-96420-18                     | MY-96420-15                                          | MY-96420-24                                          | MY-96420-35                                          | MY-96420-36                                          |
| BioPharm Plus Silicone (platinum-cured) | 7.6 m (25 ft)  | MY-96440-13                    | MY-96440-14                     | MY-96440-16                    | MY-96440-25                     | MY-96440-17                     | MY-96440-18                     | MY-96440-15                                          | MY-96440-24                                          | MY-96440-35                                          | MY-96440-36                                          |
| Puri-Flex™                              | 7.6 m (25 ft)  | MY-96419-13                    | MY-96419-14                     | MY-96419-16                    | MY-96419-25                     | MY-96419-17                     | MY-96419-18                     | MY-96419-15                                          | MY-96419-24                                          | MY-96419-35                                          | MY-96419-36                                          |
| C-Flex®                                 | 7.6 m (25 ft)  | MY-06424-13                    | MY-06424-14                     | MY-06424-16                    | MY-06424-25                     | MY-06424-17                     | MY-06424-18                     | MY-06424-15                                          | MY-06424-24                                          | MY-06424-35                                          | MY-06424-36                                          |
| C-Flex® ULTRA                           | 7.6 m (25 ft)  | —                              | MY-06434-14                     | MY-06434-16                    | MY-06434-25                     | MY-06434-17                     | MY-06434-18                     | MY-06434-15                                          | MY-06434-24                                          | MY-06434-35                                          | MY-06434-36                                          |
| PharMed® BPT                            | 7.6 m (25 ft)  | MY-06508-13                    | MY-06508-14                     | MY-06508-16                    | MY-06508-25                     | MY-06508-17                     | MY-06508-18                     | MY-06508-15                                          | MY-06508-24                                          | MY-06508-35                                          | MY-06508-36                                          |
| PharmaPure®                             | 7.6 m (25 ft)  | MY-06435-13                    | MY-06435-14                     | MY-06435-16                    | MY-06435-25                     | MY-06435-17                     | MY-06435-18                     | MY-06435-15                                          | MY-06435-24                                          | —                                                    | —                                                    |
| Gore® STA-PURE® Series PCS              | 35.6 cm (14")  | —                              | MY-96211-14                     | MY-96211-16                    | MY-96211-25                     | MY-96211-17                     | MY-96211-18                     | MY-96211-15                                          | MY-96211-24                                          | MY-96211-35                                          | MY-96211-36                                          |
| Gore® STA-PURE® Series PFL              | 35.6 cm (14")  | —                              | MY-96212-24                     | MY-96212-35                    | MY-96212-36                     | MY-96212-17                     | MY-96212-18                     | MY-96212-15                                          | MY-96212-24                                          | MY-96212-35                                          | MY-96212-36                                          |
| Chem-Durance® Bio                       | 15.2 m (50 ft) | MY-06442-13                    | MY-06442-14                     | MY-06442-16                    | MY-06442-25                     | MY-06442-17                     | MY-06442-18                     | MY-06442-15                                          | MY-06442-24                                          | MY-06442-35                                          | MY-06442-36                                          |
| Tygon® E-LFL                            | 7.6 m (25 ft)  | MY-06440-13                    | MY-06440-14                     | MY-06440-16                    | MY-06440-25                     | MY-06440-17                     | MY-06440-18                     | MY-06440-15                                          | MY-06440-24                                          | MY-06440-35                                          | MY-06440-36                                          |
| Tygon E-Food (B-44-4X)                  | 15.2 m (50 ft) | MY-06418-13                    | MY-06418-14                     | MY-06418-16                    | MY-06418-25                     | MY-06418-17                     | MY-06418-18                     | MY-06418-15                                          | MY-06418-24                                          | MY-06418-35                                          | MY-06418-36                                          |
| Tygon E-Lab (E-3603)                    | 15.2 m (50 ft) | MY-06509-13                    | MY-06509-14                     | MY-06509-16                    | MY-06509-25                     | MY-06509-17                     | MY-06509-18                     | MY-06509-15                                          | MY-06509-24                                          | MY-06509-35                                          | MY-06509-36                                          |
| Tygon Fuel & Lubricant (F-4040-A)       | 15.2 m (50 ft) | MY-06401-13                    | MY-06401-14                     | MY-06401-16                    | MY-06401-25                     | MY-06401-17                     | MY-06401-18                     | MY-06401-15                                          | MY-06401-24                                          | MY-06401-35                                          | MY-06401-36                                          |
| Tygon Chemical (2001)                   | 15.2 m (50 ft) | MY-06475-13                    | MY-06475-14                     | MY-06475-16                    | MY-06475-25                     | MY-06475-17                     | MY-06475-18                     | MY-06475-15                                          | MY-06475-24                                          | —                                                    | —                                                    |
| Norprene® (A-60 G)                      | 15.2 m (50 ft) | MY-06404-13                    | MY-06404-14                     | MY-06404-16                    | MY-06404-25                     | MY-06404-17                     | MY-06404-18                     | MY-06404-15                                          | MY-06404-24                                          | MY-06404-35                                          | MY-06404-36                                          |
| Norprene Food (A 60 F)                  | 15.2 m (50 ft) | MY-06402-13                    | MY-06402-14                     | MY-06402-16                    | MY-06402-25                     | MY-06402-17                     | MY-06402-18                     | MY-06402-15                                          | MY-06402-24                                          | MY-06402-35                                          | MY-06402-36                                          |
| Viton®                                  | 7.6 m (25 ft)  | MY-96412-13                    | MY-96412-14                     | MY-96412-16                    | MY-96412-25                     | MY-96412-17                     | MY-96412-18                     | MY-96412-15                                          | MY-96412-24                                          | MY-96412-35                                          | MY-96412-36                                          |

|                                            | Qty/pk            | I/P® Precision Pump Tubing                  |                                             |                                              | Qty/pk            | I/P® High-Performance Precision Pump Tubing |                                              |                                              | Qty/pk            | B/T® PerfectPosition® Pump Tubing             |                                            |
|--------------------------------------------|-------------------|---------------------------------------------|---------------------------------------------|----------------------------------------------|-------------------|---------------------------------------------|----------------------------------------------|----------------------------------------------|-------------------|-----------------------------------------------|--------------------------------------------|
|                                            |                   | I/P 26<br>0.01 to 4 LPM<br>0.002 to 1.1 GPM | I/P 73<br>0.01 to 8 LPM<br>0.002 to 2.1 GPM | I/P 82<br>0.02 to 13 LPM<br>0.005 to 3.5 GPM |                   | I/P 70<br>0.01 to 8 LPM<br>0.002 to 2.1 GPM | I/P 88<br>0.02 to 17 LPM<br>0.005 to 4.5 GPM | I/P 89<br>0.03 to 19 LPM<br>0.007 to 5.0 GPM |                   | B/T 87<br>0.67 to 18.9 LPM<br>0.17 to 5.0 GPM | B/T 91<br>1.4 to 42 LPM<br>0.4 to 11.1 GPM |
| Silicone<br>(platinum-cured)               | 7.6 m<br>(25 ft)  | MY-96410-26                                 | MY-96410-73                                 | MY-96410-82                                  | 3 m<br>(10 ft)    | MY-96510-70                                 | MY-96510-88                                  | MY-96510-89                                  | 3 m<br>(10 ft)    | MY-96510-87                                   | MY-96510-91                                |
| Silicone<br>(peroxide-cured)               | 7.6 m<br>(25 ft)  | MY-96400-26                                 | MY-96400-73                                 | MY-96400-82                                  | —                 | —                                           | —                                            | —                                            | 3 m<br>(10 ft)    | MY-96400-87                                   | MY-96400-91                                |
| BioPharm Silicone<br>(platinum-cured)      | 7.6 m<br>(25 ft)  | MY-96420-26                                 | MY-96420-73                                 | MY-96420-82                                  | 3 m<br>(10 ft)    | MY-96421-70                                 | MY-96421-88                                  | MY-96421-89                                  | —                 | —                                             | —                                          |
| BioPharm Plus Silicone<br>(platinum-cured) | 7.6 m<br>(25 ft)  | MY-96440-26                                 | MY-96440-73                                 | MY-96440-82                                  | 3 m<br>(10 ft)    | MY-96441-70                                 | MY-96441-88                                  | MY-96441-89                                  | 3 m<br>(10 ft)    | MY-96445-87                                   | MY-96445-91                                |
| Puri-Flex™                                 | 7.6 m<br>(25 ft)  | MY-96419-26                                 | MY-96419-73                                 | MY-96419-82                                  | 3 m<br>(10 ft)    | MY-96419-70                                 | MY-96419-88                                  | MY-96419-89                                  | 3 m<br>(10 ft)    | MY-96419-87                                   | MY-96419-91                                |
| C-Flex®                                    | 7.6 m<br>(25 ft)  | MY-06424-26                                 | MY-06424-73                                 | MY-06424-82                                  | 3 m<br>(10 ft)    | MY-06424-70                                 | MY-06424-88                                  | MY-06424-89                                  | 3 m<br>(10 ft)    | MY-06424-87                                   | MY-06424-91                                |
| PharMed® BPT                               | 7.6 m<br>(25 ft)  | MY-06508-26                                 | MY-06508-73                                 | MY-06508-82                                  | 7.6 m<br>(25 ft)  | MY-06508-70                                 | MY-06508-88                                  | —                                            | 3 m<br>(10 ft)    | MY-06507-87                                   | MY-06507-91                                |
| PharmaPure®                                | 7.6 m<br>(25 ft)  | MY-06435-26                                 | MY-06435-73                                 | MY-06435-82                                  | —                 | —                                           | —                                            | —                                            | 3 m<br>(10 ft)    | MY-06437-87                                   | MY-06437-91                                |
| GORE® STA-PURE®<br>Series PCS              | 0.6 m<br>(2 ft)   | MY-96211-26                                 | MY-96211-73                                 | MY-96211-82                                  | 0.6 m<br>(2 ft)   | —                                           | MY-96211-88                                  | MY-96211-89                                  | —                 | —                                             | —                                          |
| GORE® STA-PURE®<br>Series PFL              | 0.6 m<br>(2 ft)   | MY-96212-26                                 | MY-96212-73                                 | MY-96212-82                                  | 0.6 m<br>(2 ft)   | —                                           | MY-96212-88                                  | MY-96212-89                                  | —                 | —                                             | —                                          |
| Chem-Durance® Bio                          | 15.2 m<br>(50 ft) | MY-06442-26                                 | MY-06442-73                                 | MY-06442-82                                  | 15.2 m<br>(50 ft) | MY-06442-70                                 | MY-06442-88                                  | MY-06442-89                                  | 3 m<br>(10 ft)    | MY-06443-87                                   | MY-06443-91                                |
| Tygon® E-LFL                               | 7.6 m<br>(25 ft)  | MY-06440-26                                 | MY-06440-73                                 | MY-06440-82                                  | 7.6 m<br>(25 ft)  | MY-06440-70                                 | MY-06440-88                                  | MY-06440-89                                  | 3 m<br>(10 ft)    | MY-06440-87                                   | MY-06440-91                                |
| Tygon E-Food<br>(B-44-4X)                  | 15.2 m<br>(50 ft) | MY-06418-26                                 | MY-06418-73                                 | MY-06418-82                                  | 15.2 m<br>(50 ft) | MY-06418-70                                 | MY-06418-88                                  | MY-06418-89                                  | 3 m<br>(10 ft)    | MY-06418-87                                   | MY-06418-91                                |
| Tygon E-Lab<br>(E-3603)                    | 15.2 m<br>(50 ft) | MY-06509-26                                 | MY-06509-73                                 | MY-06509-82                                  | 15.2 m<br>(50 ft) | MY-06509-70                                 | MY-06509-88                                  | MY-06509-89                                  | 15.2 m<br>(50 ft) | —                                             | —                                          |
| Norprene®<br>(A-60 G)                      | 15.2 m<br>(50 ft) | MY-06404-26                                 | MY-06404-73                                 | MY-06404-82                                  | 7.6 m<br>(25 ft)  | MY-06404-70                                 | MY-06404-88                                  | MY-06404-89                                  | 7.6 m<br>(25 ft)  | —                                             | —                                          |
| Norprene Food<br>(A 60 F)                  | 15.2 m<br>(50 ft) | MY-06402-26                                 | MY-06402-73                                 | MY-06402-82                                  | 7.6 m<br>(25 ft)  | —                                           | MY-06402-88                                  | MY-06402-89                                  | 3 m<br>(10 ft)    | MY-06399-87                                   | MY-06399-91                                |

## Masterflex® Pump Accessories

### A. Pulse Dampener

Virtually eliminates pulsation in output flow. Five pairs of fittings and PTFE-pipe thread tape are included. Accepted tubing: all L/S® sizes, I/P® 26, nominal from 1.6 to 9.5 mm ID (1/16" to 3/8" ID). Fittings included (tubing ID x NPT(M) thread): 1/16" x 1/8", 1/8" x 1/8", 3/16" x 1/8", 1/4" x 1/8", and 3/8" x 1/8". Dampener connections: 1/8" NPT(F). Dead volume: 190-mL. Max pressure: 60 psi (4.3 bar). Polyethylene body.



| Description    | Catalogue number   |
|----------------|--------------------|
| Pulse dampener | <b>MY-07596-20</b> |

### B. On/Off Foot Switches

Foot switches allow for remote control of specific functions of your Masterflex drive. All units have a 1.8-m (6-ft) cord. Verify the connector type on your drive before ordering.



| Description                                                                                        | Catalogue number   |
|----------------------------------------------------------------------------------------------------|--------------------|
| DB9 male connector; for 07528-, 07557-, and 07591-series Masterflex drives                         | <b>MY-77595-35</b> |
| DB25 male connector; for 07522-, 07523-, 07551-, 77301-, and 77411-series Masterflex drives        | <b>MY-07523-92</b> |
| 18-pin round water-tight connector; for 07557-, 07591-, and 77411-series Masterflex drives         | <b>MY-07595-43</b> |
| 31-pin round water-tight connector; for 07575-, 77301-, 07594-, and 77420-series Masterflex drives | <b>MY-07575-84</b> |

### C. DB9 Cable Assembly

Use these 9-pin interface cable assemblies with Masterflex L/S® drives (07550- and 07551-series). Cable measures 1.8-m (6-ft) long.



| Description                    | Catalogue number   |
|--------------------------------|--------------------|
| Interface cable with DB9 (M/F) | <b>MY-22050-54</b> |

### D. USB to DB9(M) RS-232 Serial Adapter Cable

Use this cable to connect the Masterflex 07550- and 07551-series Computer Compatible drives to a PC with USB port. Will also connect other peripheral devices equipped with RS-232 serial interface including balances, printers, scanners, and PDA's. Includes 0.6-m (2-ft) cable and driver software. Supports data transfer speeds up to 12 Mbps; compatible with Windows® 2000/XP/Vista/7.



| Description                               | Catalogue number   |
|-------------------------------------------|--------------------|
| USB to DB9(M) RS-232 serial adapter cable | <b>MY-22050-58</b> |

### E. Remote Control Connectors

Combine these connectors with 20 to 22 gauge shielded wire to create your own interface cable. Maximum cable length is 300-m (1000 ft).



| Description                                    | Catalogue number   |
|------------------------------------------------|--------------------|
| DB9 male connector. Wire only the pins needed  | <b>MY-07595-45</b> |
| DB25 male connector. Wire only the pins needed | <b>MY-07523-94</b> |

### F. Handheld Remote Controllers

These handheld controllers are designed to control basic drive operations. Route tubing through the handle for filling and dispensing applications. Handles accept Masterflex tubing sizes L/S® 13, L/S 14, L/S 16, L/S 25, L/S 17, L/S 18, L/S 15, and L/S 24 (up to 11.1 cm nominal OD).



| Description                                                                                                            | Catalogue number   |
|------------------------------------------------------------------------------------------------------------------------|--------------------|
| DB9 male connector; for 07528-series Masterflex L/S drives; control start/stop and direction                           | <b>MY-07528-80</b> |
| DB25 male connector; for 07523-series and 07551-series Masterflex L/S drives; control start/stop, direction, and prime | <b>MY-07523-97</b> |
| Stripped wire ends; control start/stop, direction, and prime                                                           | <b>MY-07595-62</b> |

### G. Remote Control Cable Assemblies

Cable assemblies include terminal connector and 7.9-m (25 ft) cable with stripped wire ends. Verify the connector type on your drive before ordering.

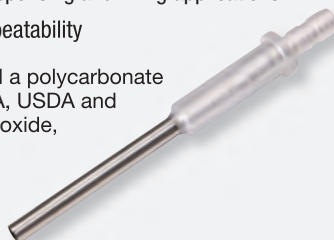


| Description                                                                                        | Catalogue number   |
|----------------------------------------------------------------------------------------------------|--------------------|
| DB9 male connector; for 07528-, 07557-, and 07591-series Masterflex drives                         | <b>MY-07595-47</b> |
| DB25 male connector; for 07522-, 07523-, 07551-, 77301-, and 07594-series Masterflex drives        | <b>MY-07523-95</b> |
| 18-pin round water-tight connector; for 07557-, 07591-, and 77411-series Masterflex drives         | <b>MY-77300-32</b> |
| 31-pin round water-tight connector; for 07575-, 77301-, 07594-, and 77420-series Masterflex drives | <b>MY-07575-80</b> |

## Overlook Filler/Dispensing Nozzles

- Maximise the speed, precision, and accuracy of dispensing and filling applications
- Dimensional tolerances of  $\pm 0.05\%$  for assured repeatability

Nozzles feature a 316L stainless steel needle and a polycarbonate base with hose barb adapter. Materials meet FDA, USDA and USP Class VI requirements. Sterilise by ethylene oxide, autoclaving, or gamma irradiation.



| Tubing ID | Catalogue number   |
|-----------|--------------------|
| 1/32"     | <b>MY-30619-06</b> |
| 1/16"     | <b>MY-30619-01</b> |
| 1/8"      | <b>MY-30619-07</b> |
| 3/16"     | <b>MY-30619-02</b> |
| 1/4"      | <b>MY-30619-08</b> |
| 5/16"     | <b>MY-30619-03</b> |
| 3/8"      | <b>MY-30619-09</b> |
| 1/2"      | <b>MY-30619-04</b> |
| 5/8"      | <b>MY-30619-10</b> |
| 3/4"      | <b>MY-30619-05</b> |

# Ismatec® Reglo Independent Channel Control (ICC) Pumps

## Pump each channel at different speeds and directions!

- Each pump performs the job of up to four separate pumps
- Additional options in channel and roller configurations allow balanced pulsation control at desired flow rate
- USB interface for quick computer connections using redesigned graphical user-interface

These two-, three- and four-channel peristaltic pumps let you individually control the flow and direction of each channel from the intuitive keypad or via your computer—expanding your benchtop pumping capabilities with a single compact unit! Pump delivers continuous pumping or precision dispensing up to 43-mL/min with the capability of bidirectional flow in each channel. Plus, independent channel calibration minimises tube-to-tube differences, resulting in high calibration accuracy.

**ISMATEC®**


Deliver or dispense at multiple flow rates—all from a single space-saving pump.



### Specifications

**Speed:** 0.1 to 100 rpm  
**Accuracy:** ±1%

**Pressure:** up to 1.0 bar (14.5 psi)

ISO9001:2008  
CERTIFIED SUPPLIER



| Channels          | Rollers | Tubing     | Power (VAC, Hz)      | Catalogue number |
|-------------------|---------|------------|----------------------|------------------|
| Benchtop pumps    |         |            |                      |                  |
| Two               | Six     | Three-stop | 100 to 240,<br>50/60 | MY-78001-58      |
|                   | Eight   |            |                      | MY-78001-60      |
|                   | 12      |            |                      | MY-78001-62      |
| Three             | Six     | Three-stop | 100 to 240,<br>50/60 | MY-78001-68      |
|                   | Eight   |            |                      | MY-78001-70      |
|                   | 12      |            |                      | MY-78001-72      |
| Four              | Six     | Three-stop | 100 to 240,<br>50/60 | MY-78001-78      |
|                   | Eight   |            |                      | MY-78001-80      |
|                   | 12      |            |                      | MY-78001-82      |
| Panel-mount pumps |         |            |                      |                  |
| Two               | Eight   | Three-stop | 100 to 240,<br>50/60 | MY-78001-43      |
| Three             | Eight   |            |                      | MY-78001-45      |
| Four              | Eight   |            |                      | MY-78001-47      |
| Four              | 12      |            |                      | MY-78001-49      |

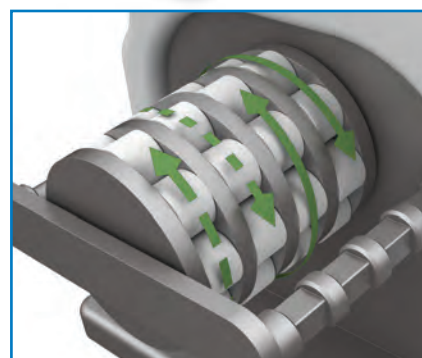
### Accessories

MY-78016-98 Replacement Click'n'go™ tubing cassette, POM-C

MY-78016-96 Click'n'go™ tubing cassette, PVDF

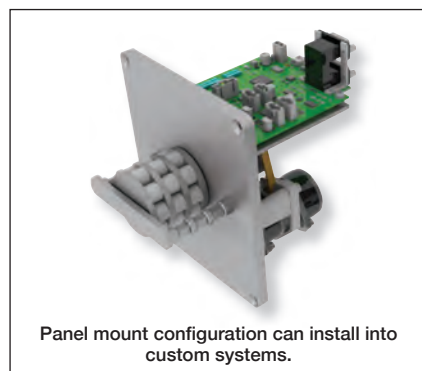
### Flow Ranges (mL/min per channel)

| Tubing ID (mm) | 6 rollers      | 8 rollers      | 12 rollers      |
|----------------|----------------|----------------|-----------------|
| 0.13           | 0.0002 to 0.14 | 0.0002 to 0.11 | 0.0001 to 0.093 |
| 0.51           | 0.0019 to 1.9  | 0.0017 to 1.7  | 0.0016 to 1.6   |
| 0.57           | 0.0024 to 2.4  | 0.0021 to 2.1  | 0.0019 to 1.9   |
| 0.64           | 0.0030 to 3.0  | 0.0026 to 2.6  | 0.0024 to 2.4   |
| 0.95           | 0.0064 to 6.4  | 0.0056 to 5.6  | 0.0050 to 5.0   |
| 1.02           | 0.0073 to 7.3  | 0.0063 to 6.3  | 0.0056 to 5.6   |
| 1.42           | 0.013 to 13    | 0.011 to 11    | 0.0094 to 9.4   |
| 1.75           | 0.019 to 19    | 0.016 to 16    | 0.013 to 13     |
| 1.85           | 0.021 to 21    | 0.017 to 17    | 0.014 to 14     |
| 2.54           | 0.033 to 33    | 0.027 to 27    | 0.019 to 19     |
| 2.79           | 0.036 to 36    | 0.031 to 31    | 0.021 to 21     |
| 3.17           | 0.043 to 43    | 0.035 to 35    | 0.024 to 24     |



### Independent Channel Design

- ✓ Up to four independently controllable channels
- ✓ Flexibility of bidirectional flow in each channel
- ✓ Continuous pumping or precision dispensing
- ✓ Enhanced accuracy through separate channel calibration



Panel mount configuration can install into custom systems.

# Ismatec® IPC Low-Flow High-Accuracy Pumps

## Fewer tubing changes

- Planetary drive for less pulsation and longer tubing life
- Smooth rotation and accurate liquid flow provided by eight SS rollers
- Easy repeatability with four-digit LED
- Simple to use
- Five modes for pumping or dispensing including flow rate, time, batches, drip-free, and volume
- Control via keypad, computer, or remote control
- IP and IP-N analogue pumps available online



Choose from 4, 8, 12, 16 and 24 channel configurations for application flexibility.

**ISMATEC®**



See pages 146–147.

### Specifications

**Accuracy:** better than  $\pm 1\%$

#### Pressure

Click'n'go cassettes: up to 14.5 psi (1 bar)

Pressure lever cassettes: up to 22 psi (1.5 bar)

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4 channel



8 channel



12 channel



16 channel



24 channel



| Flow range per channel<br>(mL/min) | Channels | Rollers | Speed<br>(rpm) | Cassette<br>style                        | Tubing   | Power<br>(VAC, Hz) | Catalogue<br>number |
|------------------------------------|----------|---------|----------------|------------------------------------------|----------|--------------------|---------------------|
| IPC-N models                       |          |         |                |                                          |          |                    |                     |
| 0.0004 to 11                       | Four     | Eight   | 0.11 to 11.25  | Click'n'go                               | Two-stop | 230, 50/60         | MY-78001-06         |
|                                    | Eight    |         |                |                                          |          | 230, 50/60         | MY-78001-16         |
|                                    | 12       |         |                |                                          |          | 230, 50/60         | MY-78001-26         |
|                                    | 16       |         |                |                                          |          | 230, 50/60         | MY-78001-36         |
|                                    | 24       |         |                |                                          |          | 230, 50/60         | MY-78001-46         |
| 0.0004 to 11                       | Eight    | Eight   | 0.11 to 11.25  | Pressure lever<br>(adjustable occlusion) | Two-stop | 230, 50/60         | MY-78000-17         |
|                                    | 16       |         |                |                                          |          | 230, 50/60         | MY-78000-37         |
|                                    | 24       |         |                |                                          |          | 230, 50/60         | MY-78000-47         |
| IPC models                         |          |         |                |                                          |          |                    |                     |
| 0.002 to 44                        | Four     | Eight   | 0.4 to 45      | Click'n'go                               | Two-stop | 230, 50/60         | MY-78001-08         |
|                                    | Eight    |         |                |                                          |          | 230, 50/60         | MY-78001-18         |
|                                    | 12       |         |                |                                          |          | 230, 50/60         | MY-78001-28         |
|                                    | 16       |         |                |                                          |          | 230, 50/60         | MY-78001-38         |
|                                    | 24       |         |                |                                          |          | 230, 50/60         | MY-78001-48         |
| 0.002 to 44                        | Four     | Eight   | 0.4 to 45      | Pressure lever<br>(adjustable occlusion) | Two-stop | 230, 50/60         | MY-78000-09         |
|                                    | Eight    |         |                |                                          |          | 230, 50/60         | MY-78000-19         |
|                                    | 12       |         |                |                                          |          | 230, 50/60         | MY-78000-29         |
|                                    | 16       |         |                |                                          |          | 230, 50/60         | MY-78000-39         |
|                                    | 24       |         |                |                                          |          | 230, 50/60         | MY-78000-49         |

Order extra cartridges/adapters when fast tubing changes are crucial.

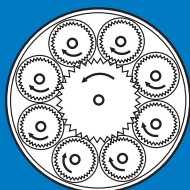
**MY-78001-95 Replacement Click'n'go™ cartridge, POM-C**

**MY-78001-90 Repl. adjust. occlusion cartridge with pressure lever**

**MY-07618-71 Replacement adapters** hold two-stop colour-coded tubing in cartridge. Two adapters required per cartridge. Pack of 12

### Planetary Drive System

With the planetary drive system, each roller is directly driven by the sun wheel. This mechanism prevents axial push-pull friction on the tubing—resulting in longer tubing life, lower pulsation, and high repeatability.



### Pressure Lever Cassettes

These cassettes feature a pressure lever that allows you to set a different tubing pressure for each channel. Depending on the application, tubing material, and diameter, you can adjust to optimal tubing pressure for your requirements.



## Ismatec® ecoline Pumps

### Save space—pumps stack easily

- Durable stainless steel housing and four stabilising feet for easier stacking
- Simple setup and operation—start, stop, and reverse flow with a single switch
- Configurations available to support standard, Masterflex, and stopped tubing

Pumps are controllable remotely via an analog input signal (0 to 5 VDC, 0 to 10 VDC, 0 to 20 mA, or 4 to 20 mA); start/stop drive or reverse direction via contact closure (eight-pin DIN). Input signals from your existing instruments actuate drive functions.

#### A–C. Drive with Single-Channel Pump Head

- Minimise tubing wear and ensure automatic centering with convex roller

#### D. Drive with Masterflex® L/S® Easy-Load® Pump Head

- Change tubing quickly with easily accessible head
- Polysulfone housing and stainless steel rotor

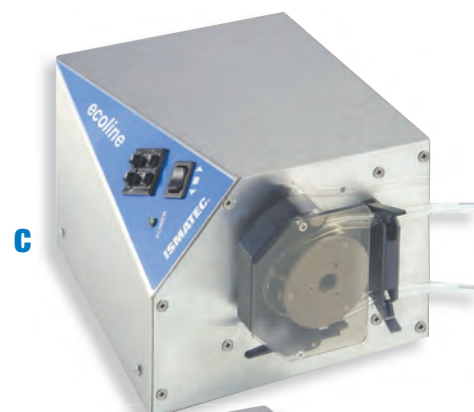
#### E. Drive with Minicartridge Pump Head

- No adjustments needed—spring-loaded cassettes for optimal occlusion at all times Get consistent, low-pulse flow with 6 or 12 rollers

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| Key                                                         | Flow range per channel (mL/min) | Channels | Rollers | Speed          | Tubing                    | Power (VAC, Hz) | Catalogue number |
|-------------------------------------------------------------|---------------------------------|----------|---------|----------------|---------------------------|-----------------|------------------|
| Drive with single-channel pump head                         |                                 |          |         |                |                           |                 |                  |
| A                                                           | 1.7 to 5400                     | One      | Two     | 3.5 to 350 rpm | Pump (pg 147)             | 230, 50/60      | MY-78022-12      |
| B                                                           | 1.6 to 5000                     |          | Three   |                |                           | 230, 50/60      | MY-78022-22      |
| C                                                           | 0.25 to 1300                    |          | Three   |                |                           | 230, 50/60      | MY-78022-32      |
| Drive with Easy-Load pump head                              |                                 |          |         |                |                           |                 |                  |
| D                                                           | 0.23 to 1600                    | One      | Three   | 3.5 to 350 rpm | Masterflex® (pgs 138–139) | 230, 50/60      | MY-78022-52      |
| Drive with Minicartridge pump head and Click'n'go cassettes |                                 |          |         |                |                           |                 |                  |
| E                                                           | 0.01 to 130                     | Eight    | Six     | 3.5 to 350 rpm | Three-stop (pgs 146–147)  | 230, 50/60      | MY-78022-42      |
|                                                             | 0.0075 to 75                    | Four     | 12      |                |                           | 230, 50/60      | MY-78022-62      |



#### Accessories

**MY-07624-64** Optional adjustable occlusion cassette with pressure lever, POM-C

**MY-78016-98** Replacement Click'n'go™ cassette, POM-C

**MY-07610-25** Foot switch, momentary contact

**MY-17050-01** ILAC MRA, NIST-traceable calibration with data for peristaltic pump drive

## Ismatec® Reglo Compact Multichannel Pumps

### No more inconsistent results

- Consistent flow and reduced tubing wear for improved results
- Very small footprint for a stand-alone “smart” pump
- Easy channel changes—Click'n'go™ minicartridges snap in place

**A. Reglo Analog Pumps.** Speed is adjustable in 1% steps (2 to 100%) via two-digit potentiometer. Control speed remotely. Start/stop pump or change direction via contact closure; monitor speed through frequency output.

**B. Reglo Digital Programmable Pumps.** Continuously pump or dispense in any of four modes—flow rate, time, batches, or drip-free. Pumps feature analog input and output controls, as well as an RS-232 interface for control by a computer. Remote control capable.


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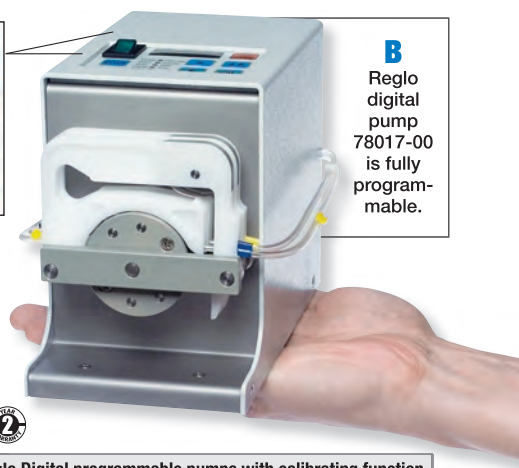

| Channels | Rollers | Speed (rpm) | Tubing                   | A. Reglo Analog pumps           |                 |                    | B. Reglo Digital programmable pumps with calibrating function |                 |                    |
|----------|---------|-------------|--------------------------|---------------------------------|-----------------|--------------------|---------------------------------------------------------------|-----------------|--------------------|
|          |         |             |                          | Flow range per channel (mL/min) | Power (VAC, Hz) | Catalogue number   | Flow range per channel (mL/min)                               | Power (VAC, Hz) | Catalogue number   |
| Two      | Six     | 3.2 to 160  | Three-stop (pgs 146–147) | 0.005 to 68                     | 230, 50/60      | <b>MY-78016-02</b> | 0.003 to 68                                                   | 115/230, 50/60  | <b>MY-78016-42</b> |
|          | Eight   |             |                          | 0.004 to 57                     | 230, 50/60      | <b>MY-78016-12</b> | 0.002 to 57                                                   | 115/230, 50/60  | <b>MY-78017-02</b> |
|          | 12      |             |                          | 0.003 to 38                     | 230, 50/60      | <b>MY-78017-27</b> | 0.002 to 38                                                   | 115/230, 50/60  | <b>MY-78016-47</b> |
| Four     | Six     | 2.0 to 100  | Three-stop (pgs 146–147) | 0.003 to 43                     | 230, 50/60      | <b>MY-78017-32</b> | 0.002 to 43                                                   | 115/230, 50/60  | <b>MY-78017-42</b> |
|          | Eight   |             |                          | 0.003 to 35                     | 230, 50/60      | <b>MY-78016-32</b> | 0.002 to 35                                                   | 115/230, 50/60  | <b>MY-78017-12</b> |
|          | 12      |             |                          | 0.002 to 24                     | 230, 50/60      | <b>MY-78017-37</b> | 0.001 to 24                                                   | 115/230, 50/60  | <b>MY-78017-22</b> |

#### Minicartridges

Optional adjustable-occlusion minicartridge to manually vary pressure on each channel.

**MY-07624-64** Optional adjustable occlusion cassette with pressure lever, POM-C

**MY-07624-96** Replacement Click'n'go cassette, PVDF



**B**  
Reglo digital pump 78017-00 is fully programmable.

# Ismatec® BVP and MCP Quick-Couple Drives

## Create a custom system with interchangeable pump heads

- Mount pump heads quickly with turn-and-lock mounting system
- Select from interchangeable tubing pump heads to achieve flow rates from 0.003 to 3700-mL/min
- Remote control and programming options add versatility
- Process washdown drives are IP65 rated

### BVP Standard and Process Drives

- Prime, flush, or prevent overflow with Max/Slow toggle switch

**BVP Standard Drive** features analog input that remote control of speed, start/stop or flow direction. **BVP Process Washdown Drive** is similar to the BVP Standard model, plus has a digital display for speed. The IP65-rated housing is ideal for dusty, humid, or corrosive environment and in cleanroom areas. The stainless steel housing is sealed for added durability.

### MCP Standard and Process Programmable Digital Drives

- Save time—recall and store up to four programs
- Preprogrammed with tubing sizes and heads for faster setup
- Use the RS-232 interface to automate your lab or operate independently via the keypad—RS-232 in for remote operation; RS-232 out for cascading pumps

**MCP Standard Drive** is programmable to dispense by volume, time, flow rate, batches with variable pause time, stored programs, roller back steps, or number of dispensing cycles. Digital display shows speed, flow rate, dispensing volume, dispensing time, and pause time. **MCP Process Washdown Drive** adds an IP65-rated housing for washdown applications—ideal for dispensing and filling in dusty, humid, or corrosive environment and in cleanroom areas. The stainless steel housing is sealed for added durability.



BVP Standard drive 78002-10



MCP Standard programmable digital drive 78002-00 can store four programs downloaded from your computer.



BVP Process washdown drive 78003-10



### REQUIRED System Components

- 1) Quick-couple drive . . . 144
- 2) Pump head . . . 144-145
- 3) Tubing . . . . . 146-147

### Specifications

Duty cycle: continuous

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| Model                       | BVP Standard drive                                                                                                                           | BVP Process washdown drive                                                                                                                               | MCP Standard drive                                                                                                                                                                          | MCP Process washdown drive                                                                                                                                                                   |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Catalogue number            | MY-78002-12                                                                                                                                  | MY-78003-10                                                                                                                                              | MY-78002-02                                                                                                                                                                                 | MY-78003-08                                                                                                                                                                                  |
| Programmable                | No                                                                                                                                           | No                                                                                                                                                       | Yes                                                                                                                                                                                         | Yes                                                                                                                                                                                          |
| Speed                       | 2.4 to 240 rpm (indicated as 1 to 99.9%)                                                                                                     | 1 to 240 rpm                                                                                                                                             | 1 to 240 rpm                                                                                                                                                                                | 1 to 240 rpm                                                                                                                                                                                 |
| Resolution                  | 0.1% via 3-digit potentiometer                                                                                                               | 0.1 rpm                                                                                                                                                  | 0.1 rpm                                                                                                                                                                                     | 0.1 rpm                                                                                                                                                                                      |
| Flow range/channel          | 0.003 to 3700 mL/min                                                                                                                         | 0.001 to 3700 mL/min                                                                                                                                     | 0.001 to 3700 mL/min                                                                                                                                                                        | 0.001 to 3700 mL/min                                                                                                                                                                         |
| Flow rate settings          | —                                                                                                                                            | —                                                                                                                                                        | μL/min, mL/min, L/min                                                                                                                                                                       | μL/min, mL/min, L/min                                                                                                                                                                        |
| Remote control capabilities | Speed control: 0 to 5 VDC, 0 to 10 VDC, 0 to 20 mA, 4 to 20 mA<br>Speed output: 0 to 10 VDC or 0 to 12 kHz<br>Start/stop, rotation direction | Speed control: 0 to 5 VDC, 0 to 10 VDC, 0 to 20 mA, 4 to 20 mA<br>Speed output: 0 to 10 VDC or 0 to 7.2 kHz<br>Start/stop, rotation direction, autostart | RS-232 (1 in, 1 out), 1 analogue<br>Speed control: 0 to 5 VDC, 0 to 10 VDC, 0 to 20 mA, 4 to 20 mA<br>Speed output: 0 to 10 VDC or 0 to 12 kHz<br>Start/stop, rotation direction, autostart | RS-232 (1 in, 1 out), 1 analogue<br>Speed control: 0 to 5 VDC, 0 to 10 VDC, 0 to 20 mA, 4 to 20 mA<br>Speed output: 0 to 10 VDC or 0 to 7.2 kHz<br>Start/stop, rotation direction, autostart |
| IP rating                   | IP30                                                                                                                                         | IP65                                                                                                                                                     | IP30                                                                                                                                                                                        | IP65                                                                                                                                                                                         |
| Power                       | 230 VAC, 50/60 Hz                                                                                                                            | 230 VAC, 50/60 Hz                                                                                                                                        | 230 VAC, 50/60 Hz                                                                                                                                                                           | 115 to 230 VAC, 50/60 Hz                                                                                                                                                                     |

## Pump Heads for Quick-Couple Drives

### A. Washdown Pump Heads

**Dismantle for easy cleaning**—ideal for hygienic applications. Convex rollers and concave tube-bed maintain the integrity of material being pumped. Coated aluminum with stainless steel rollers.

| Flow range (mL/min) | Channels | Rollers | Tubing          | Wall thickness | Max pressure     | Catalogue number |
|---------------------|----------|---------|-----------------|----------------|------------------|------------------|
| 0.49 to 3700        | One      | Two     | Pump (page 147) | 1.6 mm         | 1.5 bar (22 psi) | MY-78002-11      |
| 3.6 to 3100         | One      | Two     |                 | 2.4 mm         | 2.5 bar (36 psi) | MY-78002-13      |
| 0.45 to 3400        | One      | Three   |                 | 1.6 mm         | 1.5 bar (22 psi) | MY-78002-16      |
| 3.3 to 2900         | One      | Three   |                 | 2.4 mm         | 2.5 bar (36 psi) | MY-78002-14      |



## B. Minicartridge Pump Heads

**Minimise pulsation with more rollers.** Get up to 24 channels with 78002-50 and 07623-00 below. Get up to 16 channels of the smoothest flow with 78002-90 and 07623-30 below.

| Flow range (mL/min) | Channels         | Rollers | Tubing                        | Max pressure                       | Catalogue number   |
|---------------------|------------------|---------|-------------------------------|------------------------------------|--------------------|
| 0.002 to 100        | Eight (up to 24) | Six     | Three-stop<br>(pages 146–147) | 1.0 bar<br>(14.5 psi) <sup>†</sup> | <b>MY-78002-50</b> |
| 0.001 to 57         | Four (up to 16)  | 12      |                               |                                    | <b>MY-78002-90</b> |

<sup>†</sup>22 psi (1.5 bar) achievable with pressure lever cassettes.

**MY-07623-00 Eight-channel extension pump head.** Use up to two with 78002-50 to get 24 channels

**MY-07623-30 Four-channel extension pump head.** Use up to three with 78002-90 to get 16 channels

## C. Standard Cartridge Pump Heads

**Ideal pump head for multichannel applications at higher flow rates.** Click'n'go™ cartridges allow quick and easy tubing changes.

| Flow range (mL/min) | Channels | Rollers | Tubing                      | Max pressure                       | Catalogue number   |
|---------------------|----------|---------|-----------------------------|------------------------------------|--------------------|
| 0.002 to 230        | Eight    | Eight   | Two-stop<br>(pages 146–147) | 1.0 bar<br>(14.5 psi) <sup>†</sup> | <b>MY-78002-34</b> |
|                     | 12       |         |                             |                                    | <b>MY-78002-36</b> |

<sup>†</sup>22 psi (1.5 bar) achievable with pressure lever cassettes.

## D. Tubing Bed Pump Heads

**Ideal for delicate substances which require gentle occlusion setting.** Spring-loaded tubing bed adjusts occlusion for optimum pressure and tubing.

| Flow range (mL/min) | Channels | Rollers | Tubing             | Max pressure        | Catalogue number   |
|---------------------|----------|---------|--------------------|---------------------|--------------------|
| 1.1 to 1110         | Two      | Six     | Pump<br>(page 147) | 1.5 bar<br>(22 psi) | <b>MY-07611-10</b> |
| 0.09 to 530         | Three    |         |                    |                     | <b>MY-07611-15</b> |

## E. Quick-Change Pump Heads

**Hinged tube-bed allows rapid tube change-over.** Easily accessible pump head. Self-centering tube-track results in longer tubing-life. Stainless steel rollers.

| Flow range (mL/min) | Channels | Rollers | Tubing             | Max pressure        | Catalogue number   |
|---------------------|----------|---------|--------------------|---------------------|--------------------|
| 0.08 to 880         | One      | Three   | Pump<br>(page 147) | 1.5 bar<br>(22 psi) | <b>MY-78002-22</b> |
| 0.44 to 2800        |          |         |                    |                     | <b>MY-78002-20</b> |

## F. High-Viscosity Pump Head

**Ideal for media with high viscosity or with a certain level of solid content.** Accepts tubing with 1.6-mm and 2.4-mm wall thickness (for elevated differential pressures). Stainless steel rollers. Pressure on tubing adjustable via rollers.

| Flow range (mL/min) | Channels | Rollers | Tubing             | Max pressure | Catalogue number   |
|---------------------|----------|---------|--------------------|--------------|--------------------|
| 0.41 to 3600        | One      | Three   | Pump<br>(page 147) | 36 psi       | <b>MY-78002-60</b> |

## G. Masterflex® Easy-Load® Pump Heads

**Allows for fast and easy tubing changes.** Accepts Masterflex L/S® pump tubing (see pages 138–139). Requires mounting plate below.

| Flow range (mL/min) | Channels        | Rollers | Tubing                                         | Max pressure | Catalogue number   |
|---------------------|-----------------|---------|------------------------------------------------|--------------|--------------------|
| 3.4 to 672          | One (up to two) | Three   | L/S 15, L/S 24                                 | 20 psi       | <b>MY-07514-12</b> |
| 0.12 to 912         |                 |         | L/S 13, L/S 14, L/S 16, L/S 17, L/S 18, L/S 25 |              | <b>MY-07514-10</b> |

**MY-78020-90 Mounting plate** for Masterflex Easy-Load pump head

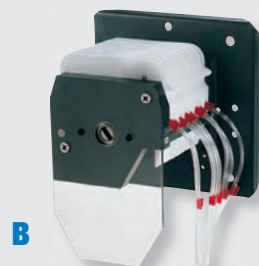
**MY-07013-05 Mounting hardware** to stack two Easy-Load pump heads

## H. PTFE-Diaphragm Pump Heads

**Pump almost any fluid** with these chemically inert PTFE-diaphragm pump heads. Pump heads withstand temperatures up to 66°C (150°F). Requires adapter 78002-70 to mount to Quick-Couple drives.

| Flow range (mL/min) | Ball check | Rollers     | Tubing             | Max pressure | Catalogue number   |
|---------------------|------------|-------------|--------------------|--------------|--------------------|
| 0.5 to 60           | Glass      | 1/8" NPT(M) | Pump<br>(page 147) | 75 psi       | <b>MY-07090-62</b> |
| 4 to 480            | PTFE       | 1/4" NPT(M) |                    | 50 psi       | <b>MY-07090-42</b> |

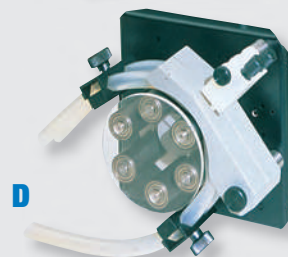
**MY-78002-70 Adapter** for mounting PTFE Diaphragm pump heads onto 78002- and 78003-series drives



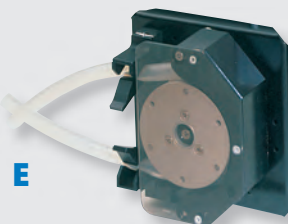
B



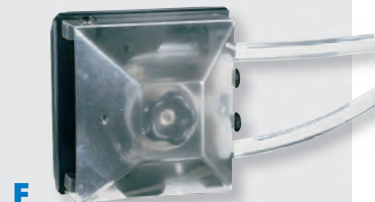
C



D



E



F



G



H

# Two- and Three-Stop Colour-Coded Tubing for Ismatec® Pumps

## Don't confuse your tubing— colour-coded stops for easy identification

**STEP 1:** Choose a tubing material & select the catalogue number prefix.

Make sure to choose two-stop or three-stop tubing based on what your pump uses!

**STEP 3:** Complete your system with extension tubing.

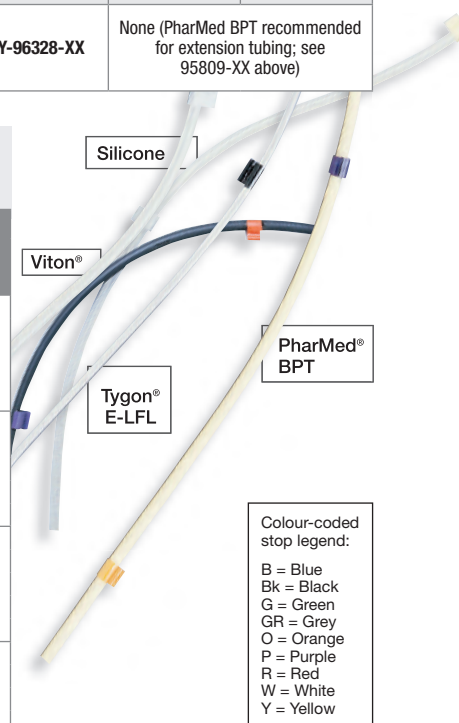
Connect stopped and extension tubing with barbed fittings or splicing.

| Material                  | Description                                                                                                                                                                                                                | Suffixes for the available sizes                                                                       | No. of tubes per pack (length) | Two-Stop Tubing | Three-Stop Tubing | Extension Tubing                                                        |                 |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------|-----------------|-------------------|-------------------------------------------------------------------------|-----------------|
|                           |                                                                                                                                                                                                                            |                                                                                                        |                                | Cat. no. prefix | Cat. no. prefix   | Length/pk                                                               | Cat. no. prefix |
| Tygon® E-LFL              | <ul style="list-style-type: none"> <li>Complies with USP Class VI, EP, FDA, EU food, REACH, RoHS, ADF</li> <li>Longest life of all Tygon formulations</li> <li>Broad chemical resistance</li> </ul>                        | 10, 12, 14, 18, 22, 24, 26, 28, 30, 31, 33, 34, 36, 37, 40, 42, 43, 47, 48                             | 12 tubes (40.6 cm)             | MY-96449-XX     | MY-96450-XX       | 100 ft                                                                  | MY-06449-XX     |
| Tygon® E-Food             | <ul style="list-style-type: none"> <li>Complies with FDA, EU Food, NSF, 2A, REACH, RoHS, ADF</li> <li>Smooth inner surface; non-toxic</li> </ul>                                                                           | 10, 12, 18, 26, 30, 34, 42, 48                                                                         | 12 tubes (40.6 cm)             | MY-96457-XX     | MY-96458-XX       | 100 ft                                                                  | MY-06457-XX     |
| Tygon® E-Lab              | <ul style="list-style-type: none"> <li>Complies with USP Class VI, FDA, EU Food, NSF, REACH, RoHS, ADF</li> <li>Ideal for general transfer applications</li> <li>Non-ageing/non-oxidizing, low gas permeability</li> </ul> | 09, 10, 12, 14, 16, 18, 19, 22, 24, 26, 27, 28, 29, 30, 31, 32, 34, 36, 38, 39, 40, 42, 44, 46, 48, 49 | 12 tubes (40.6 cm)             | MY-96460-XX     | MY-96461-XX       | 100 ft                                                                  | MY-06460-XX     |
| Silicone (peroxide cured) | <ul style="list-style-type: none"> <li>Complies with USP Class VI, EP, FDA, 3A, REACH, RoHS, ADF</li> <li>Greater physical compression capability</li> <li>Economical, long tubing life</li> </ul>                         | 10, 12, 14, 18, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48                                 | Six tubes (40.6 cm)            | MY-07616-XX     | MY-07624-XX       | 50 ft                                                                   | MY-07625-XX     |
| Silicone (platinum cured) | <ul style="list-style-type: none"> <li>Complies with USP Class V, EP, FDA, REACH, RoHS, ADF</li> <li>Low protein binding, fungus-resistant, phthalate and latex free</li> </ul>                                            | 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48                                                 | Six tubes (40.6 cm)            | MY-95602-XX     | MY-95603-XX       | 50 ft                                                                   | MY-95612-XX     |
| Viton®                    | <ul style="list-style-type: none"> <li>Complies with FDA, ADF</li> <li>Resists corrosives, solvents, and oils at elevated temperatures</li> </ul>                                                                          | 18, 22, 24, 26, 30, 34, 38, 42, 46, 48                                                                 | 12 tubes (30.5 cm)             | MY-97628-XX     | MY-97629-XX       | 50 ft                                                                   | MY-97632-XX     |
| PharMed® BPT              | <ul style="list-style-type: none"> <li>Complies with USP Class VI, EP, FDA, NSF, REACH, RoHS</li> <li>Over 10,000 hours of tubing life, sealable, weldable</li> <li>Ideal for tissue and cell culture work</li> </ul>      | 10, 12, 14, 18, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48                                 | 12 tubes (40.6 cm)             | MY-95723-XX     | MY-95714-XX       | 100 ft                                                                  | MY-95809-XX     |
| Autoclavable PharMed® BPT |                                                                                                                                                                                                                            | 12, 32, 48                                                                                             | Six tubes (40.6 cm)            | MY-95692-XX     | MY-95693-XX       |                                                                         |                 |
| PVC Solvent/hydrocarbon   | <ul style="list-style-type: none"> <li>Resistant to ozone and UV light</li> <li>Resists to cracking, swelling, hardening</li> <li>Ideal for fuels, distillates, lubricants, and glycol-based coolants</li> </ul>           | 10, 12, 14, 18, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48                                 | 12 tubes (40.6 cm)             | MY-95606-XX     | MY-95605-XX       | 50 ft                                                                   | MY-95712-XX     |
| Chem-Durance® Bio         | <ul style="list-style-type: none"> <li>Suitable for acetone, MEK, and other highly aggressive chemicals</li> <li>Long life and outstanding fatigue resistance</li> </ul>                                                   | 14, 18, 24, 26, 30, 34, 36, 42, 48                                                                     | Six tubes (40.6 cm)            | MY-96327-XX     | MY-96328-XX       | None (PharMed BPT recommended for extension tubing; see 95809-XX above) |                 |

### STEP 2:

Find your desired flow rate and select the catalogue number suffix. See page 147 for more.

| Flow rate <sup>1</sup> (mL/min) with Fixed-Speed Pumps (order online) |                    |                    |                    |                    |                    |                    |                    | Cat. no. suffix | ID (mm) | Colour-coded stops |
|-----------------------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-----------------|---------|--------------------|
| 78012-00, 78012-05                                                    | 78012-20, 78012-25 | 78012-30, 78012-35 | 78012-40, 78012-45 | 78012-50, 78012-55 | 78012-60, 78012-65 | 78012-80, 78012-85 | 78012-90, 78012-95 |                 |         |                    |
| 0.047                                                                 | 0.14               | 0.047              | 0.093              | 0.056              | 0.11               | 0.17               | 0.056              | -10             | 0.19    | O/R                |
| 0.081                                                                 | 0.24               | 0.08               | 0.16               | 0.1                | 0.19               | 0.29               | 0.1                | -12             | 0.25    | O/B                |
| 0.19                                                                  | 0.56               | 0.19               | 0.37               | 0.22               | 0.43               | 0.65               | 0.22               | -14             | 0.38    | O/G                |
| 0.25                                                                  | 0.75               | 0.25               | 0.5                | 0.29               | 0.58               | 0.86               | 0.29               | -16             | 0.44    | G/Y                |
| 0.34                                                                  | 1                  | 0.34               | 0.67               | 0.38               | 0.77               | 1.2                | 0.38               | -18             | 0.51    | O/Y                |
| 0.52                                                                  | 1.6                | 0.52               | 1                  | 0.6                | 1.2                | 1.8                | 0.6                | -22             | 0.64    | O/W                |
| 0.73                                                                  | 2.2                | 0.73               | 1.5                | 0.84               | 1.7                | 2.5                | 0.84               | -24             | 0.76    | Bk/Bk              |
| 1                                                                     | 2.9                | 1                  | 2                  | 1.1                | 2.3                | 3.4                | 1.1                | -26             | 0.89    | O/O                |
| 1.3                                                                   | 3.8                | 1.3                | 2.5                | 1.5                | 2.9                | 4.4                | 1.5                | -28             | 1.02    | W/W                |
| 1.6                                                                   | 4.7                | 1.6                | 3.1                | 1.8                | 3.6                | 5.4                | 1.8                | -30             | 1.14    | R/R                |
| 2                                                                     | 5.9                | 2                  | 3.9                | 2.3                | 4.6                | 6.9                | 2.3                | -32             | 1.3     | GR/GR              |
| 2.3                                                                   | 6.8                | 2.3                | 4.6                | 2.7                | 5.4                | 8                  | 2.7                | -34             | 1.42    | Y/Y                |
| 2.6                                                                   | 7.7                | 2.6                | 5.1                | 3                  | 6                  | 9.1                | 3                  | -36             | 1.52    | Y/B                |
| 2.9                                                                   | 8.8                | 2.9                | 5.8                | 3.5                | 6.9                | 10                 | 3.5                | -38             | 1.65    | B/B                |
| 3.5                                                                   | 10                 | 3.5                | 7                  | 4.2                | 8.4                | 13                 | 4.2                | -40             | 1.85    | G/G                |
| 4.1                                                                   | 12                 | 4.1                | 8.2                | 5                  | 10                 | 15                 | 5                  | -42             | 2.06    | P/P                |
| 4.8                                                                   | 14                 | 4.8                | 9.5                | 5.8                | 12                 | 17                 | 5.8                | -44             | 2.29    | P/Bk               |
| 5.5                                                                   | 16                 | 5.5                | 11                 | 6.6                | 13                 | 20                 | 6.6                | -46             | 2.54    | P/O                |
| 6.2                                                                   | 19                 | 6.2                | 12                 | 7.4                | 15                 | 22                 | 7.4                | -48             | 2.79    | P/W                |



**STEP 2 (continued):** Find your desired flow rate and select the catalogue number suffix.

| Flow ranges <sup>†</sup> (mL/min) with indicated pump system (page 143 and online) |                    |                    |            |            |                    |                    |                    |                    |                    |                    |                    |                    | Cat. no. suffix | ID (mm) | Colour-coded stops |
|------------------------------------------------------------------------------------|--------------------|--------------------|------------|------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-----------------|---------|--------------------|
| 78016-00, 78016-02                                                                 | 78016-10, 78016-12 | 78016-30, 78016-32 | 78016-42   | 78017-02   | 78017-10, 78017-12 | 78017-20, 78017-22 | 78017-25, 78017-27 | 78017-30, 78017-32 | 78017-35, 78017-37 | 78017-45, 78017-47 | 78022-40, 78022-42 | 78022-60, 78022-62 |                 |         |                    |
| 0.009-0.45                                                                         | 0.008-0.37         | 0.005-0.23         | 0.005-0.45 | 0.004-0.37 | 0.003-0.23         | 0.003-0.21         | 0.007-0.34         | 0.006-0.28         | 0.005-0.21         | 0.004-0.34         | 0.01-0.98          | 0.0075-0.75        | -10             | 0.19    | O/R                |
| 0.015-0.76                                                                         | 0.013-0.65         | 0.009-0.41         | 0.008-0.76 | 0.007-0.65 | 0.005-0.41         | 0.004-0.38         | 0.012-0.61         | 0.01-0.48          | 0.008-0.38         | 0.007-0.61         | 0.017-1.7          | 0.013-1.3          | -12             | 0.25    | O/B                |
| 0.035-1.7                                                                          | 0.03-1.5           | 0.019-0.94         | 0.017-1.7  | 0.015-1.5  | 0.01-0.94          | 0.009-0.88         | 0.028-1.4          | 0.022-1.1          | 0.018-0.88         | 0.014-1.4          | 0.038-3.8          | 0.031-3.1          | -14             | 0.38    | O/G                |
| 0.046-2.3                                                                          | 0.04-2             | 0.025-1.3          | 0.023-2.3  | 0.02-2     | 0.013-1.3          | 0.012-1.2          | 0.038-1.9          | 0.029-1.4          | 0.024-1.2          | 0.019-1.9          | 0.05-5             | 0.041-4.1          | -16             | 0.44    | G/Y                |
| 0.061-3.1                                                                          | 0.054-2.7          | 0.034-1.7          | 0.031-3.1  | 0.027-2.7  | 0.017-1.7          | 0.016-1.6          | 0.05-2.5           | 0.038-1.9          | 0.031-1.6          | 0.025-2.5          | 0.067-6.7          | 0.055-5.5          | -18             | 0.51    | O/Y                |
| 0.096-4.8                                                                          | 0.084-4.2          | 0.052-2.6          | 0.048-4.8  | 0.042-4.2  | 0.026-2.6          | 0.024-2.4          | 0.077-3.9          | 0.06-3             | 0.048-2.4          | 0.039-3.9          | 0.1-10             | 0.085-8.5          | -22             | 0.64    | O/W                |
| 0.13-6.7                                                                           | 0.12-5.8           | 0.073-3.6          | 0.067-6.7  | 0.058-5.8  | 0.036-3.6          | 0.33-3.3           | 0.11-5.3           | 0.084-4.2          | 0.067-3.3          | 0.053-5.3          | 0.15-15            | 0.12-12            | -24             | 0.76    | Bk/Bk              |
| 0.18-9                                                                             | 0.16-7.9           | 0.098-4.9          | 0.09-9     | 0.079-7.9  | 0.049-4.9          | 0.044-4.4          | 0.14-7.1           | 0.11-5.7           | 0.088-4.4          | 0.071-7.1          | 0.2-20             | 0.15-15            | -26             | 0.89    | O/O                |
| 0.23-12                                                                            | 0.2-10             | 0.13-6.3           | 0.12-12    | 0.1-10     | 0.063-6.3          | 0.056-5.6          | 0.18-9             | 0.15-7.3           | 0.11-5.6           | 0.09-9             | 0.26-26            | 0.2-20             | -28             | 1.02    | W/W                |
| 0.29-14                                                                            | 0.25-12            | 0.16-7.8           | 0.14-14    | 0.12-12    | 0.078-7.8          | 0.067-6.7          | 0.22-11            | 0.18-9             | 0.13-6.7           | 0.11-11            | 0.32-32            | 0.24-24            | -30             | 1.14    | R/R                |
| 0.37-18                                                                            | 0.31-16            | 0.2-10             | 0.18-18    | 0.16-16    | 0.1-10             | 0.083-8.3          | 0.26-13            | 0.23-11            | 0.17-8.3           | 0.13-13            | 0.4-40             | 0.29-29            | -32             | 1.3     | GR/GR              |
| 0.43-21                                                                            | 0.37-18            | 0.23-11            | 0.21-21    | 0.18-18    | 0.11-11            | 0.094-9.4          | 0.3-15             | 0.27-13            | 0.19-9.4           | 0.15-15            | 0.47-47            | 0.33-33            | -34             | 1.42    | Y/Y                |
| 0.48-24                                                                            | 0.41-20            | 0.26-13            | 0.24-24    | 0.2-20     | 0.13-13            | 0.1-10             | 0.33-17            | 0.3-15             | 0.21-10            | 0.17-17            | 0.53-53            | 0.36-36            | -36             | 1.52    | Y/B                |
| 0.56-28                                                                            | 0.47-23            | 0.29-15            | 0.28-28    | 0.23-23    | 0.15-15            | 0.12-12            | 0.37-19            | 0.35-17            | 0.23-12            | 0.19-19            | 0.61-61            | 0.41-41            | -38             | 1.65    | B/B                |
| 0.67-34                                                                            | 0.56-28            | 0.35-17            | 0.34-34    | 0.28-28    | 0.17-17            | 0.13-13            | 0.43-21            | 0.42-21            | 0.27-13            | 0.21-21            | 0.73-73            | 0.47-47            | -40             | 1.85    | G/G                |
| 0.79-40                                                                            | 0.66-33            | 0.41-20            | 0.4-40     | 0.33-33    | 0.2-20             | 0.15-15            | 0.49-24            | 0.5-25             | 0.3-15             | 0.24-24            | 0.87-87            | 0.53-53            | -42             | 2.06    | P/P                |
| 0.92-46                                                                            | 0.76-38            | 0.48-24            | 0.46-46    | 0.38-38    | 0.24-24            | 0.17-17            | 0.55-27            | 0.58-29            | 0.34-17            | 0.27-27            | 1-100              | 0.6-60             | -44             | 2.29    | P/Bk               |
| 1.1-53                                                                             | 0.88-44            | 0.55-27            | 0.53-53    | 0.44-44    | 0.27-27            | 0.19-19            | 0.62-31            | 0.66-33            | 0.39-19            | 0.31-31            | 1.2-120            | 0.68-68            | -46             | 2.54    | P/O                |
| 1.2-59                                                                             | 0.99-50            | 0.62-31            | 0.59-59    | 0.5-50     | 0.31-31            | 0.21-21            | 0.69-34            | 0.74-37            | 0.43-21            | 0.34-34            | 1.3-130            | 0.75-75            | -48             | 2.79    | P/W                |

## How to order two- and three-stop tubing

**Step 1:** Choose a tubing material and select the corresponding catalogue number prefix from the table on page 146. Make sure to choose two-stop or three-stop tubing based on what your pump uses!

**Step 2:** Find your pump system and desired flow rate and select the catalogue number suffix from the tables on pages 146–147.

**Step 3:** Complete your system with extension tubing from page 146. You can connect stopped tubing and extension tubing with hose barb fittings or by splicing together.

| Flow ranges <sup>†</sup> (mL/min) with indicated pump (pages 142, 145, and online) |            |            |            |            |            | Cat. no. suffix | ID (mm) | Colour-coded stops |
|------------------------------------------------------------------------------------|------------|------------|------------|------------|------------|-----------------|---------|--------------------|
| IPC-N models                                                                       | IPC models | 78002-34   | 78002-36   | 78002-50   | 78002-90   |                 |         |                    |
| 0.0007-0.066                                                                       | 0.003-0.26 | 0.004-0.94 | 0.004-0.94 | 0.003-0.67 | 0.003-0.51 | -10             | 0.19    | O/R                |
| 0.001-0.1                                                                          | 0.005-0.41 | 0.008-1.8  | 0.008-1.8  | 0.005-1.1  | 0.004-0.91 | -12             | 0.25    | O/B                |
| 0.003-0.22                                                                         | 0.009-0.86 | 0.019-4.5  | 0.019-4.5  | 0.011-2.6  | 0.009-2.1  | -14             | 0.38    | O/G                |
| 0.003-0.28                                                                         | 0.011-1.1  | 0.025-6.1  | 0.025-6.1  | 0.014-3.5  | 0.012-2.8  | -16             | 0.44    | G/Y                |
| 0.004-0.38                                                                         | 0.015-1.5  | 0.034-8.2  | 0.034-8.2  | 0.019-4.6  | 0.016-3.8  | -18             | 0.51    | O/Y                |
| 0.006-0.58                                                                         | 0.023-2.3  | 0.053-13   | 0.053-13   | 0.03-7.2   | 0.024-5.8  | -22             | 0.64    | O/W                |
| 0.009-0.81                                                                         | 0.032-3.2  | 0.074-18   | 0.074-18   | 0.042-10   | 0.033-8    | -24             | 0.76    | Bk/Bk              |
| 0.011-1.1                                                                          | 0.044-4.4  | 0.1-24     | 0.1-24     | 0.057-14   | 0.044-11   | -26             | 0.89    | O/O                |
| 0.014-1.4                                                                          | 0.057-5.7  | 0.13-31    | 0.13-31    | 0.073-18   | 0.056-13   | -28             | 1.02    | W/W                |
| 0.017-1.7                                                                          | 0.07-7     | 0.16-38    | 0.16-38    | 0.09-22    | 0.067-16   | -30             | 1.14    | R/R                |
| 0.022-2.2                                                                          | 0.089-8.9  | 0.2-47     | 0.2-47     | 0.11-27    | 0.083-20   | -32             | 1.3     | GR/GR              |
| 0.026-2.6                                                                          | 0.11-11    | 0.23-55    | 0.23-55    | 0.13-32    | 0.094-23   | -34             | 1.42    | Y/Y                |
| 0.03-3                                                                             | 0.12-12    | 0.26-62    | 0.26-62    | 0.15-36    | 0.1-25     | -36             | 1.52    | Y/B                |
| 0.035-3.5                                                                          | 0.14-14    | 0.3-71     | 0.3-71     | 0.17-42    | 0.12-28    | -38             | 1.65    | B/B                |
| 0.43-4.3                                                                           | 0.17-17    | 0.36-86    | 0.36-86    | 0.21-50    | 0.13-32    | -40             | 1.85    | G/G                |
| 0.52-5.2                                                                           | 0.21-21    | 0.43-100   | 0.43-100   | 0.25-59    | 0.15-37    | -42             | 2.06    | P/P                |
| 0.062-6.2                                                                          | 0.25-25    | 0.51-120   | 0.51-120   | 0.29-69    | 0.17-41    | -44             | 2.29    | P/Bk               |
| 0.075-7.5                                                                          | 0.3-30     | 0.63-150   | 0.62-150   | 0.33-79    | 0.19-46    | -46             | 2.54    | P/O                |
| 0.088-8.8                                                                          | 0.35-35    | 0.74-180   | 0.74-180   | 0.37-89    | 0.21-52    | -48             | 2.79    | P/W                |

<sup>†</sup>Flow ranges (mL/min) are nominal. Actual flow depends on factors such as occlusion, fluid viscosity, temperature, and pressure.

## Pump Tubing for Ismatec® Pumps

| ID (mm) | OD (mm) | Wall (mm) | Tygon® E-Lab |             | Silicone (peroxide cured) |             | Silicone (platinum cured) |             | Norprene® |             | Viton® |             |
|---------|---------|-----------|--------------|-------------|---------------------------|-------------|---------------------------|-------------|-----------|-------------|--------|-------------|
|         |         |           | Ft/pk        | Cat. no.    | Ft/pk                     | Cat. no.    | Ft/pk                     | Cat. no.    | Ft/pk     | Cat. no.    | Ft/pk  | Cat. no.    |
| 0.8     | 4.0     | 1.6       | 50           | MY-06509-13 | 25                        | MY-96400-13 | 25                        | MY-95802-01 | 50        | MY-06404-13 | 25     | MY-96412-13 |
| 1.6     | 4.8     | 1.6       | 50           | MY-06407-72 | 25                        | MY-96400-14 | 25                        | MY-96410-14 | 50        | MY-06410-01 | 25     | MY-96412-14 |
| 3.2     | 6.4     | 1.6       | 50           | MY-06407-76 | 25                        | MY-06411-67 | 25                        | MY-95802-05 | 50        | MY-06410-02 | —      | —           |
| 4.8     | 8.0     | 1.6       | 50           | MY-06407-78 | 25                        | MY-06411-03 | 25                        | MY-95802-09 | 50        | MY-06410-04 | 25     | MY-96412-25 |
| 6.4     | 9.5     | 1.6       | 50           | MY-06407-80 | 25                        | MY-06411-71 | 25                        | MY-95802-12 | 50        | MY-06410-05 | 25     | MY-96412-18 |
| 6.4     | 11.1    | 2.4       | 50           | MY-06509-24 | 25                        | MY-06411-72 | 25                        | MY-95802-13 | 50        | MY-06410-06 | 25     | MY-96412-24 |
| 8.0     | 11.2    | 1.6       | 50           | MY-06407-83 | 25                        | MY-06411-75 | 25                        | MY-95802-15 | 50        | MY-06410-08 | 25     | MY-06434-05 |
| 9.5     | 12.7    | 1.6       | 50           | MY-06407-86 | 25                        | MY-06411-12 | 25                        | MY-95802-18 | 50        | MY-06410-10 | 25     | MY-06434-06 |
| 11.1    | 14.3    | 1.6       | 50           | MY-06407-87 | —                         | —           | —                         | —           | 50        | MY-06410-13 | —      | —           |
| 12.8    | 16.0    | 1.6       | 25           | MY-06407-90 | 25                        | MY-06411-80 | 25                        | MY-95802-21 | 50        | MY-06410-14 | 25     | MY-06434-07 |
| 16.0    | 22.4    | 3.2       | —            | —           | 25                        | MY-06411-58 | 25                        | MY-95802-25 | 50        | MY-06410-17 | 25     | MY-06434-08 |

| ID (mm) | Flow ranges (mL/min) with indicated pump system <sup>††</sup> (pages 143, 145, and online) |               |               |               |                       |                       |                       |                       |                       |
|---------|--------------------------------------------------------------------------------------------|---------------|---------------|---------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|         | 78025-00, -05                                                                              | 78022-10, -12 | 78022-20, -22 | 78022-30, -32 | 07611-10 <sup>‡</sup> | 07611-15 <sup>‡</sup> | 78002-20 <sup>‡</sup> | 78002-22 <sup>‡</sup> | 78002-60 <sup>‡</sup> |
| 0.8     | —                                                                                          | —             | —             | 0.25 to 25    | —                     | 0.22 to 22            | —                     | 0.072 to 17           | —                     |
| 1.6     | —                                                                                          | 1.7 to 170    | 1.6 to 160    | 0.9 to 90     | —                     | 0.63 to 63            | 0.44 to 100           | 0.026 to 62           | 0.41 to 99            |
| 3.2     | —                                                                                          | 6.6 to 660    | 5.9 to 590    | 3.5 to 350    | 2.6 to 260            | 2.4 to 240            | 1.7 to 400            | 1.0 to 240            | 1.5 to 370            |
| 4.8     | —                                                                                          | 15.0 to 1500  | 13.0 to 1300  | 7.7 to 770    | 5.5 to 550            | 5.3 to 530            | 3.6 to 860            | 2.2 to 530            | 3.4 to 830            |
| 6.4     | 37 to 3700                                                                                 | 25.0 to 2500  | 23.0 to 2300  | 13.0 to 1300  | 8.9 to 890            | —                     | 6.0 to 1400           | 3.7 to 880            | 6.2 to 1500           |
| 8.0     | —                                                                                          | 37.0 to 3700  | 34.0 to 3400  | —             | 11.1 to 1110          | —                     | 8.8 to 2100           | —                     | 9.5 to 2300           |
| 9.5     | 80 to 8000                                                                                 | 48.0 to 4800  | 44.0 to 4400  | —             | —                     | —                     | 12 to 2800            | —                     | 13 to 3000            |
| 11.1    | —                                                                                          | 54.0 to 5400  | 50.0 to 5000  | —             | —                     | —                     | —                     | —                     | 15 to 3600            |
| 12.8    | 100 to 10000                                                                               | —             | —             | —             | —                     | —                     | —                     | —                     | —                     |
| 16.0    | 130 to 13000                                                                               | —             | —             | —             | —                     | —                     | —                     | —                     | —                     |

<sup>††</sup>Note: Lower flow rates achievable with digital drives.

<sup>‡</sup>Flow rate with Digital Drive 78002-10; lower flow rate achievable with Programmable Drive 78002-00.

## Ismatec® Gear Pumps

### High-performance drives with numerous features

- Magnetically driven pump heads provide smooth, accurate, pulseless fluid delivery
- Remote control and programming options add versatility
- Adapter plate to accept either A or B Mount gear pump heads is included

#### Reglo-Z/ZS Analogue Drives

- Two-digit potentiometer provides a 100:1 turndown
- Four analogue signal plus relay contact permit remote control from a variety of instruments
- **Reglo-Z Analogue:** compact enough to fit in the palm of your hand
- **Reglo-ZS Analogue:** modular design has drive and controller separated by a 2-m long cable

#### Reglo-Z/ZS Digital Drives

- Program pump for dispensing fluids by flow rate, volume, time or number of dispensing cycles
- Drip-free dispensing—program pump to draw back fluid (only compatible with certain heads)
- RS-232 interface allows control via a PC
- **Reglo-Z Digital:** compact yet lets you operate in multiple modes
- **Reglo-ZS Digital:** modular design has drive and controller separated by a 2-m long cable

#### BVP-Z Standard Drive

- Economical yet versatile drive lets you adjust speed in 0.1% steps
- Press the MAX button for extra rpm and quick pump priming
- Electronic feedback system compensates for changes in load to maintain  $\pm 1\%$  speed accuracy

#### MCP-Z Series Drives

- Dispense by time, volume, flow rate, batches with variable pause time, stored programs, roller backsteps, or number of dispensing cycles
- Use the RS-232 interface to automate your lab or operate independently via the keypad
- Digital display shows speed, flow rate, dispensing volume, dispensing time, and pause time
- Store and recall up to four programs
- **MCP-Z Process:** IP65-rated stainless steel housing is ideal for dusty, humid, or corrosive environments and cleanroom areas



BVP-Z Standard

Reglo-Z Digital



MCP-Z Process

Reglo-ZS Analogue



#### REQUIRED System Components

- 1) Drive ..... below
- 2) Pump head ..... 150
- 3) Tubing ..... online

#### Specifications

Duty cycle: continuous

**ISMATEC®**

ISO9001:2008  
CERTIFIED SUPPLIER



| Model                       | Reglo-Z Analogue                                                                                          | Reglo-ZS Analogue | Reglo-Z Digital                                              | Reglo-ZS Digital | BVP-Z Standard                                                                                                           | MCP-Z Standard                                                                                                                                                          | MCP-Z Process |
|-----------------------------|-----------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Catalogue number            | MY-78008-05                                                                                               | MY-78008-25       | MY-78008-15                                                  | MY-78008-35      | MY-78004-17                                                                                                              | MY-78004-37                                                                                                                                                             | MY-78004-47   |
| Programmable                | No                                                                                                        |                   | Yes                                                          |                  | No                                                                                                                       | Yes                                                                                                                                                                     |               |
| Speed                       | 50 to 5000 rpm (indicated as 1 to 99%)                                                                    |                   | 50 to 5000 rpm                                               |                  | 60 to 6000 rpm (indicated as 1 to 99.9%)                                                                                 | 60 to 6000 rpm                                                                                                                                                          |               |
| Resolution                  | 0.1% via 2-digit potentiometer                                                                            |                   | 1 rpm                                                        |                  | 0.1% via 3-digit potentiometer                                                                                           | 0.1 rpm                                                                                                                                                                 |               |
| Flow rate settings          | —                                                                                                         |                   | mL/min (flow), mL (dispensing)                               |                  | —                                                                                                                        | μL/min, mL/min, L/min                                                                                                                                                   |               |
| Remote control capabilities | Speed control: 0 to 5 VDC, 0 to 10 VDC, 0 to 20 mA, 4 to 20 mA<br>Speed output: 0 to 10 kHz<br>Start/stop |                   | RS-232<br>Speed output: 0 to 12 kHz<br>Start/stop, autostart |                  | Speed control: 0 to 5 VDC, 0 to 10 VDC, 0 to 20 mA, 4 to 20 mA<br>Speed output: 0 to 10 VDC or 0 to 12 kHz<br>Start/stop | RS-232 (1 in, 1 out), 1 analogue<br>Speed control: 0 to 5 VDC, 0 to 10 VDC, 0 to 20 mA, 4 to 20 mA<br>Speed output: 0 to 10 VDC or 0 to 12 kHz<br>Start/stop, autostart |               |
| IP rating                   | IP30                                                                                                      |                   | IP30                                                         |                  | IP30                                                                                                                     | IP30                                                                                                                                                                    | IP65          |
| Power                       | 230 VAC, 50 Hz                                                                                            |                   | 230 VAC, 50 Hz                                               |                  | 230 VAC, 50 Hz                                                                                                           | 230 VAC, 50 Hz                                                                                                                                                          |               |

#### Accessories

**MY-78016-91 Foot switch** for Reglo-Z/ZS Analogue drives. On/off control, 1.8-m cable  
**MY-78016-93 Foot switch** for Reglo-Z/ZS Digital drives. On/off control, 1.8-m cable

**MY-78016-91 Foot switch** for BVP-Z drive. Start/stop operation, 1.8-m cable  
**MY-78003-07 Foot switch** for MCP-Z drives. Start/stop operation, 1.8-m cable



# Ismatec® Piston Pumps

## No valves to clog or wear out

- Get high pressure, accurate dispensing, and great repeatability—all with one pump system
- Only one moving part—the piston
- Remotely control speed, start/stop flow, or change fluid direction
- Differential pressure up to 6.9 bar (100 psi)
- Applications include dispensing of organic solvents, acids/bases, and highly corrosive acids including HF

### Reglo-CPF Analogue Drive

- Two-digit potentiometer provides a 100:1 turndown
- Four analogue signal plus relay contact permit remote control from a variety of instruments
- Compatible with RH pump heads for flow rates up to 180-mL/min

### Reglo-CPF Digital Drive

- Program pump for dispensing fluids by flow rate, volume, time, or number of dispensing cycles
- Drip-free dispensing—program pump to draw back fluid
- RS-232 interface allows control via a PC
- Compatible with RH pump heads for flow rates up to 180-mL/min

### MCP-CPF Process Drive

- IP65 rated, stainless steel housing—ideal for pumping in dusty, humid, or corrosive environment and in cleanroom areas
- Dispense by time, volume, flow rate, batches with variable pause time, stored programs, roller backsteps, or number of dispensing cycles
- Use the RS-232 interface to automate your lab or operate independently via the keypad
- Digital display shows speed, flow rate, dispensing volume, dispensing time, and pause time
- Store and recall up to four programs
- Compatible with the full range of RH and Q series of pump heads for higher overall flow rates to 2300-mL/min



Reglo-CPF Analogue

Reglo-CPF Digital



MCP-CPF Process



#### REQUIRED System Components

- 1) Drive ..... below
- 2) Pump head ..... 150
- 3) Tubing ..... below

#### Specifications

Duty cycle: continuous

**ISMATEC®**

ISO9001:2008  
CERTIFIED SUPPLIER



| Model                       | Reglo-CPF Analogue                                                                                                           | Reglo-CPF Digital                                                                                                                                 | MCP-CPF Process                                                                                                                                                                             |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Catalogue number            | MY-78020-47                                                                                                                  | MY-78020-52                                                                                                                                       | MY-78020-30                                                                                                                                                                                 |
| Programmable                | No                                                                                                                           | Yes                                                                                                                                               | Yes                                                                                                                                                                                         |
| Speed                       | 18 to 1800 rpm (indicated as 1 to 99%)                                                                                       | 40 to 1800 rpm                                                                                                                                    | 10.0 to 1800 rpm                                                                                                                                                                            |
| Resolution                  | 1% via 2-step potentiometer                                                                                                  | 0.1 rpm                                                                                                                                           | 0.1 rpm                                                                                                                                                                                     |
| Flow rate settings          | —                                                                                                                            | μL/min, mL/min                                                                                                                                    | μL/min, mL/min, L/min                                                                                                                                                                       |
| Remote control capabilities | Speed control: 0 to 5 VDC, 0 to 10 VDC, 0 to 20 mA, 4 to 20 mA<br>Speed output: 0 to 9 kHz<br>Start/stop, rotation direction | RS-232<br>Speed control: 0 to 5 VDC, 0 to 10 VDC, 0 to 20 mA, 4 to 20 mA<br>Speed output: 0 to 9 kHz<br>Start/stop, autostart, rotation direction | RS-232 (1 in, 1 out), 1 analogue<br>Speed control: 0 to 5 VDC, 0 to 10 VDC, 0 to 20 mA, 4 to 20 mA<br>Speed output: 0 to 10 VDC or 0 to 12 kHz<br>Start/stop, autostart, rotation direction |
| IP rating                   | IP30                                                                                                                         | IP30                                                                                                                                              | IP65                                                                                                                                                                                        |
| Power                       | 230 VAC, 50 Hz                                                                                                               | 230 VAC, 50 Hz                                                                                                                                    | 115 to 230 VAC, 50/60 Hz                                                                                                                                                                    |

#### Accessories

**MY-78016-91 Foot switch** for Reglo-CPF Analogue drive. On/off control, 1.8-m cable

**MY-78016-93 Foot switch** for Reglo-CPF Digital drive. On/off control, 1.8-m cable

**MY-78003-07 Foot switch** for MCP-CPF Process drive. Start/stop operation, 1.8-m cable

#### Tubing

| Description                            | Length | Catalogue number |
|----------------------------------------|--------|------------------|
| PTFE tubing coil, 4.48 mm ID x 6 mm OD | 7.5 m  | MY-06605-54      |
| Lab tubing coil, 3.2 mm ID             | 15 m   | MY-06507-05      |
| Lab tubing coil, 6.4 mm ID             | 15 m   | MY-06507-11      |
| Lab tubing coil, 9.5 mm ID             | 15 m   | MY-06507-15      |



## Micropump® B-Mount Gear Pump Heads for Ismatec® Drives on page 148

- Compact and easy to service
- Hermetically sealed and magnetically coupled pump heads offer leak- and contamination-free pumping

**Suction Shoe Pump Heads** provide enhanced pumping performance at elevated differential pressures and are well-suited for higher temperature ranges. Not recommended for applications requiring suction lift.

**Cavity-Style Pump Heads** produce smooth, precise flow with low friction and offer excellent chemical resistance. Recommended for applications requiring a modest suction lift.

### Specifications

**Max viscosity:**  
0.2 to 1500 cp

**Suction lift:**  
Suction shoe: 30.5 cm H<sub>2</sub>O at 3450 rpm  
Cavity shoe: 61 cm H<sub>2</sub>O at 1750 rpm

**Duty cycle:** continuous



| mL/rev                  | Flow rate† (mL/min) |        | Max differential pressure | Max system pressure | Gear material | Seals  | Housing | Temperature range | Internal bypass | Port size   | Catalogue number |
|-------------------------|---------------------|--------|---------------------------|---------------------|---------------|--------|---------|-------------------|-----------------|-------------|------------------|
|                         | Min                 | Max    |                           |                     |               |        |         |                   |                 |             |                  |
| Suction shoe pump heads |                     |        |                           |                     |               |        |         |                   |                 |             |                  |
| 0.017                   | 0.85                | 102‡   | 20.3 psi                  | 304.6 psi           | Graphite      | PTFE   | 316 SS  | -46 to 177°C      | —               | 1/8" NPT(F) | MY-74004-23      |
| 0.042                   | 2.1                 | 252‡   | 40.6 psi                  | 304.6 psi           | Graphite      | PTFE   | 316 SS  | -46 to 177°C      | —               | 1/8" NPT(F) | MY-74004-29      |
| 0.084                   | 4.2                 | 504‡   | 40.6 psi                  | 304.6 psi           | Graphite      | PTFE   | 316 SS  | -46 to 177°C      | —               | 1/8" NPT(F) | MY-74004-31      |
| 0.092                   | 4.6                 | 552‡   | 75.4 psi                  | 304.6 psi           | PPS           | PTFE   | 316 SS  | -46 to 177°C      | —               | 1/8" NPT(F) | MY-74004-33      |
| 0.084                   | 4.2                 | 504‡   | 75.4 psi                  | 304.6 psi           | PEEK          | PTFE   | 316 SS  | -46 to 177°C      | —               | 1/8" NPT(F) | MY-74004-73      |
| 0.58                    | 29                  | 3190†† | 50.8 psi                  | 304.6 psi           | PPS           | Viton® | 316 SS  | -29 to 177°C      | Yes             | 1/8" NPT(F) | MY-74004-19      |
| 0.58                    | 29                  | 3190†† | 50.8 psi                  | 304.6 psi           | PEEK          | Viton  | 316 SS  | -29 to 177°C      | —               | 1/8" NPT(F) | MY-74004-77      |
| 1.17                    | 58.5                | 6435†† | 50.8 psi                  | 304.6 psi           | PPS           | Viton  | 316 SS  | -29 to 177°C      | Yes             | 1/8" NPT(F) | MY-74004-21      |
| Cavity style pump heads |                     |        |                           |                     |               |        |         |                   |                 |             |                  |
| 0.64                    | 32                  | 3520†† | 50.8 psi                  | 304.6 psi           | PTFE          | PTFE   | 316 SS  | -50 to 130°F      | Yes             | 1/8" NPT(F) | MY-74004-01      |
| 0.64                    | 32                  | 3520†† | 50.8 psi                  | 304.6 psi           | PTFE          | PTFE   | 316 SS  | -50 to 130°F      | —               | 1/8" NPT(F) | MY-74004-09      |
| 0.64                    | 32                  | 3520†† | 81.2 psi                  | 304.6 psi           | PEEK          | PTFE   | 316 SS  | -50 to 130°F      | —               | 1/8" NPT(F) | MY-74004-75      |
| 0.91                    | 45.5                | 5005†† | 50.8 psi                  | 304.6 psi           | PTFE          | PTFE   | 316 SS  | -50 to 130°F      | —               | 1/8" NPT(F) | MY-74004-11      |
| 0.64                    | 32                  | 3520†† | 75.4 psi                  | 304.6 psi           | PPS           | PTFE   | 316 SS  | -50 to 130°F      | Yes             | 1/8" NPT(F) | MY-74004-05      |
| 0.64                    | 32                  | 3520†† | 75.4 psi                  | 319.1 psi           | PPS           | PTFE   | 316 SS  | -50 to 130°F      | —               | 1/8" NPT(F) | MY-74004-17      |

†Flow rates without differential pressure.

‡Based on max rpm of 6000.

††Based on max rpm of 5500.

## RH-Series Piston Pump Heads for All Ismatec® Drives on page 149

- Small displacement provides excellent precision
- Fully adjustable from zero to maximum

| Type                           | RH0                      |                                                    | RH1                      |                                                    |                                                    |
|--------------------------------|--------------------------|----------------------------------------------------|--------------------------|----------------------------------------------------|----------------------------------------------------|
| Catalogue number               | MY-07117-59*             | MY-07117-55                                        | MY-07117-69*             | MY-07117-65                                        | MY-07117-63                                        |
| Stroke volume                  | 5 to 50 µL               |                                                    | 10 to 100 µL             |                                                    |                                                    |
| Flow range, Reglo-CPF Analogue | 0.09 to 90 mL/min        |                                                    | 0.18 to 180 mL/min       |                                                    |                                                    |
| Flow range, Reglo-CPF Digital  | 0.2 to 90 mL/min         |                                                    | 0.4 to 180 mL/min        |                                                    |                                                    |
| Flow range, MCP-CPF Process    | 0.050 to 90 mL/min       |                                                    | 0.1 to 180 mL/min        |                                                    |                                                    |
| Piston                         | Ceramic                  | Ceramic                                            | Ceramic                  | Ceramic                                            | Ceramic                                            |
| Cylinder case                  | PVDF                     | PVDF                                               | PVDF                     | PVDF                                               | ETFE                                               |
| Cylinder liner                 | Ceramic                  | Ceramic                                            | Ceramic                  | Ceramic                                            | Ceramic                                            |
| Lip seals                      | Rulon® AR                | Rulon AR                                           | Rulon AR                 | Rulon AR                                           | Rulon AR                                           |
| Gland washers                  | PTFE                     | PTFE                                               | PTFE                     | PTFE                                               | PTFE                                               |
| Max temperature                | 100°C                    | 100°C                                              | 100°C                    | 100°C                                              | 100°C                                              |
| Max differential pressure      | 100 psi                  | 100 psi                                            | 100 psi                  | 100 psi                                            | 100 psi                                            |
| Flow ports                     | 1/4" NPT(F) <sup>†</sup> | Two compression fittings for 6 to 6.4 mm OD tubing | 1/4" NPT(F) <sup>†</sup> | Two compression fittings for 6 to 6.4 mm OD tubing | Two compression fittings for 6 to 6.4 mm OD tubing |

\*For low flow rates, 1/4"-28 (female) adapters are recommended.

<sup>†</sup>Tubing adapters are sold separately online.



Go online for our comprehensive selection of Piston Pump Heads for Ismatec Drives.

## Q-Series Piston Pump Heads for Ismatec® MCP-CPF Process Drive on page 149

| Type                      | Q0                       | Q1                       |                          | Q2                       |                          | Q3                                     |
|---------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|----------------------------------------|
| Catalogue number          | MY-07104-50              | MY-07104-56              | MY-07104-54              | MY-07104-66              | MY-07104-64              | MY-07104-53                            |
| Stroke volume             | 3.2 to 80 µL             | 12.8 to 320 µL           |                          | 28.8 to 720 µL           |                          | 51.2 to 1280 µL                        |
| Flow range                | 0.04 to 144 mL/min       | 0.04 to 144 mL/min       |                          | 0.29 to 1300 mL/min      |                          | 0.51 to 2300 mL/min                    |
| Piston                    | 316 SS                   | Ceramic                  | Ceramic                  | Ceramic                  | Ceramic                  | Ceramic                                |
| Cylinder case             | 316 SS                   | 316 SS                   | PVDF                     | 316 SS                   | PVDF                     | PVDF                                   |
| Cylinder liner            | Carbon                   | Ceramic                  | Ceramic                  | Ceramic                  | Ceramic                  | Ceramic                                |
| Lip seals                 | Rulon® J                 | Rulon AR                 | Rulon AR                 | Rulon AR                 | Rulon AR                 | Rulon AR                               |
| Gland washers             | PTFE                     | PTFE                     | PTFE                     | PTFE                     | PTFE                     | PTFE                                   |
| Cylinder head seal        | PTFE                     | PTFE                     | None                     | PTFE                     | None                     | None                                   |
| Max temperature           | 100°C                    | 177°C                    | 100°C                    | 177°C                    | 100°C                    | 100°C                                  |
| Max differential pressure | 100 psi                  | 100 psi                  | 60 psi                   | 100 psi                  | 60 psi                   | 24.6 psi <1600 rpm, 7.3 psi >1600 rpm) |
| Flow ports                | 1/4" NPT(F) <sup>†</sup> | 1/4" NPT(F) <sup>†</sup> | 1/4" NPT(F) <sup>†</sup> | 1/4" NPT(F) <sup>†</sup> | 1/4" NPT(F) <sup>†</sup> | 1/4" NPT(F) <sup>†</sup>               |

<sup>†</sup>Tubing adapters are sold separately online.



## Magnetic Drive Centrifugal Pumps with Enclosed Motors

### Handles thrust loads easily

- Quiet, energy efficient, and compact
- Internal cooling circuit reduces bearing temperature
- Enclosed motor for use in harsh environments

These moderate-flow polypropylene centrifugal pumps are ideal for process, recirculation, or transfer applications where continuous flow and excellent chemical resistance are needed.

High-head polypropylene pumps with a closed impeller design to significantly increase output pressure are available online.



#### Specifications

##### Wetted parts:

Front/rear casings and impeller/magnet capsule: glass fiber-reinforced polypropylene  
Spindle and thrust ring: alumina ceramic 99.5%  
Casing O-ring: Viton®  
Bearing: PTFE

**Duty cycle:** continuous

| Max flow |      | Max head |      | Max pressure |      | Max temp |     | Connections |        | Motor |      | Power |      | Dimensions (L x W x H) |                        | Catalogue number |
|----------|------|----------|------|--------------|------|----------|-----|-------------|--------|-------|------|-------|------|------------------------|------------------------|------------------|
| LPM      | GPM  | m        | ft   | bar          | psi  | °C       | °F  | Hose barb   | NPT(M) | hp    | rpm  | VAC   | Amps | cm                     | in.                    |                  |
| 11.7     | 3.1  | 2.1      | 6.8  | 0.3          | 4.3  | 80       | 176 | ½           | —      | 1/125 | 3400 | 220   | 0.20 | 10.5 x 7.6 x 9.5       | 4 1/8 x 3 x 3 3/4      | MY-72010-15      |
| 18.9     | 5    | 3.4      | 11.1 | 0.5          | 7.2  | 80       | 176 | ½           | —      | 1/75  | 3400 | 220   | 0.15 | 18.1 x 9.5 x 11.4      | 7 1/8 x 3 3/4 x 4 1/2  | MY-72010-25      |
| 37.9     | 10   | 5.4      | 17.7 | 0.8          | 11.6 | 80       | 176 | —           | ¾      | 1/16  | 3400 | 220   | 0.55 | 25.7 x 12.1 x 10.5     | 10 1/8 x 4 3/4 x 4 1/8 | MY-72010-45      |
| 134.8    | 35.6 | 11.9     | 39   | 1.8          | 25.6 | 80       | 176 | —           | 1      | 1/3   | 3400 | 220   | 1.70 | 32.4 x 15.9 x 17.8     | 12 3/4 x 6 1/4 x 7     | MY-72010-85      |

## Plastic Hand-Operated Drum Pumps

### No need for electricity

#### A. Siphon/Lift Pail Pump

- Suitable for pumping petroleum or water
- Cut inlet tube to fit your container
- Maximum suction lift is 3.0-m (10 ft)

#### B. Threaded Hand Pump

- Pump detergents, water, dilute alkalies, and waxes from drums or pails—¾" NPT(M) intake threads permit semi-permanent attachment
- Optional adapters securely attach pump to a variety of containers—order online

#### C. Water and Chemical Siphons

- Portable and self-priming
- Water siphon (70607-00) includes 152 cm (60") outlet hose—cut to fit your application
- Chemical siphons (70607-10, -12) are suitable for acid and alkali transfer

#### D. HDPE Hand Pump

- Pump acids, alcohols, and alkalies from carboys, drums, or tanks
- Self-cleaning ball valve ensures free flow and long pump life
- Order optional LDPE adapter to securely mount pump on drums (order online)

#### E. Heavy-Duty Acid-Transfer Pumps

- Withstand temperatures up to 49°C (120°F)
- Self-priming, will not siphon



A



B



C



D



06433-20



06433-30

| Key | Strokes/<br>L (gal.) | Wetted materials                                          | Container<br>adapter   | Inlet tube  |             | Outlet hose |             | Dimensions |              | Catalogue<br>number |
|-----|----------------------|-----------------------------------------------------------|------------------------|-------------|-------------|-------------|-------------|------------|--------------|---------------------|
|     |                      |                                                           |                        | L cm (in.)  | OD cm (in.) | L cm (in.)  | OD cm (in.) | L cm (in.) | Dia in. (cm) |                     |
| A   | 9 (32)               | PVC barrel/hose, urethane-buna valves/seals               | —                      | 182.9 (72") | 1.3 (½")    | 91.4 (36")  | 1.3 (½")    | 22.9 (9")  | 3.2 (1¼")    | MY-06437-00         |
| B   | 7 (26)               | PVC body/hose, Hypalon® valves/seals, LDPE tube           | ¾" NPT(M)              | 45.7 (18")  | 1.4 (⅝")    | 22.9 (9")   | 1.4 (⅝")    | 30.5 (12") | 3.2 (1¼")    | MY-06433-60         |
| C   | 9 (32)               | PVC barrel, shaft, and hose                               | —                      | —           | —           | 152.4 (60") | 1.4 (⅝")    | 22.9 (9")  | 3.2 (1¼")    | MY-70607-00         |
|     | 7 (26)               | PVC barrel/hose, LDPE tube                                | —                      | 45.7 (18")  | 1.4 (⅝")    | 121.9 (48") | 1.4 (⅝")    | 30.5 (12") | 3.2 (1¼")    | MY-70607-10         |
|     | 4 (12)               | PVC barrel/hose, PVC tube                                 | 50.8 mm (2") IPS       | —           | —           | 121.9 (48") | 1.4 (⅝")    | 91.4 (36") | 3.2 (1¼")    | MY-70607-12         |
| D   | 2 (6)                | All HDPE parts                                            | None                   | 86.4 (34")  | 3.8 (1½")   | 35.6 (14")  | 3.8 (1½")   | 96.5 (38") | 4.6 (1⅞")    | MY-06425-05         |
| E   | 2 (6)                | Solvent-welded PVC body and shaft, Viton valves and seals | —                      | —           | —           | 91.4 (36")  | 3.2 (1¼")   | 96.5 (38") | 3.8 (1½")    | MY-06433-10         |
|     |                      |                                                           | 50.8 mm (2") IPS       |             |             |             |             |            |              | MY-06433-20         |
|     |                      |                                                           | Buttress Dell drum (M) |             |             |             |             |            |              | MY-06433-30         |

## Cole-Parmer® Digital Gear Drive with Remote Control

### Easier setup and operation

- Graphical LCD shows four operating modes: continuous run, timed dispense, copy dispense, volume dispense
- Get pulse-free flow in combination with Micropump® A-mount head
- Maximise bench space by stacking up to three drives
- Control speed, direction, and more remotely via DB25 female connector on drive

Analogue remote control options and simple programming make this drive ideal for automated process applications. Drive features set time interval between dispense cycles for hands-free dispensing. Batch count function counts down and displays batches completed. Cumulative volume function enables you to track total volume dispensed over multiple cycles.

**What's included:** 1.8-m (6-ft) line cord with IEC 320/CEE22 socket. These models are shipped with country specific plug/cord set; please specify ultimate destination when ordering.



| Motor rpm  | Motor hp | Motor type | Power VAC | Power Hz | Catalogue number |
|------------|----------|------------|-----------|----------|------------------|
| 36 to 3600 | 1/10     | PMDC       | 90 to 260 | 50/60    | MY-75211-70      |

**MY-07523-92 Foot switch** for momentary start/stop; 1.8-m (6-ft) cable

**MY-07523-94 Male connector, DB25.** Use to create your own cable



## Micropump® A-Mount Gear Pump Heads

### Achieve leak-free, pulseless flow

Magnetic drive and static seals keep fluids securely inside and potential contaminants out. Chemical resistant materials handle a wide range of materials (due to wetted parts). Pump heads are compact and easy to service. Modular design offers flexibility—configure to your specific needs.

**Cavity-style pump heads** are designed to handle higher inlet pressures. Models without internal bypass are reversible.

**Suction shoe pump heads** provide higher delivery pressures and temperatures, plus increased efficiency.

#### Specifications

|                                      |                                                               |
|--------------------------------------|---------------------------------------------------------------|
| <b>Port size:</b> 1/8" NPT(F)        | <b>Suction lift</b>                                           |
| <b>Max viscosity:</b> 0.2 to 1500 cp | Cavity style: 60.9 mm (24") of H <sub>2</sub> O (at 1750 rpm) |
| <b>Duty cycle:</b> continuous        | Suction shoe: 30.4 mm (12") of H <sub>2</sub> O (at 3450 rpm) |



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| Flow rates (mL/min)*                                                                       |           |             | Max pressure psi (bar) |            | Wetted parts |             |        | Temperature range           | Internal bypass | Max speed (rpm) | Catalogue number | Service kits     |
|--------------------------------------------------------------------------------------------|-----------|-------------|------------------------|------------|--------------|-------------|--------|-----------------------------|-----------------|-----------------|------------------|------------------|
| mL/rev                                                                                     | at 50 rpm | at 5000 rpm | Diff                   | System     | Body         | Gears       | Seals  |                             |                 |                 |                  | Catalogue number |
| Cavity-style pump heads reversible units without internal bypass                           |           |             |                        |            |              |             |        |                             |                 |                 |                  |                  |
| 0.32                                                                                       | 16        | 1600        | 80 (5.5)               | 300 (20.7) | 316 SS       | PEEK        | PTFE   | -43 to 77°C (-45 to 170°F)  | No              | 10,000          | MY-73011-00      | —                |
|                                                                                            |           |             | 80 (5.5)               |            |              | PEEK        |        | -43 to 77°C (-45 to 170°F)  | Yes             |                 | MY-73011-02      | —                |
|                                                                                            |           |             | 50 (3.4)               |            |              | PTFE        |        | -43 to 54°C (-45 to 130°F)  | Yes             |                 | MY-07002-17      | MY-07144-38      |
| 0.64                                                                                       | 32        | 3200        | 50 (3.4)               | 300 (20.7) | 316 SS       | PTFE        | PTFE   | -43 to 54°C (-45 to 130°F)  | Yes             | 10,000          | MY-07002-23      | —                |
|                                                                                            |           |             | 75 (5.2)               |            |              | PPS         |        | -43 to 77°C (-45 to 170°F)  |                 |                 | MY-07001-80      | MY-07001-89      |
| 0.64                                                                                       | 32        | 3200        | 80 (5.5)               | 300 (20.7) | 316 SS       | PEEK        | PTFE   | -43 to 77°C (-45 to 170°F)  | No              | 10,000          | MY-73011-08      | MY-73011-84      |
|                                                                                            |           |             |                        |            |              | Yes         |        | MY-73011-10                 | MY-73011-84     |                 |                  |                  |
| 0.91                                                                                       | 45.5      | 4550        | 80 (5.5)               | 300 (20.7) | 316 SS       | PEEK        | PTFE   | -43 to 77°C (-45 to 170°F)  | No              | 10,000          | MY-73011-16      | —                |
|                                                                                            |           |             | 80 (5.5)               |            |              | PEEK        |        | -43 to 77°C (-45 to 170°F)  | Yes             |                 | MY-73011-18      | —                |
|                                                                                            |           |             | 50 (3.4)               |            |              | PTFE        |        | -43 to 54°C (-45 to 130°F)  | Yes             |                 | MY-07001-40      | MY-07001-41      |
| Suction shoe pump heads provide higher delivery pressures and superior priming performance |           |             |                        |            |              |             |        |                             |                 |                 |                  |                  |
| 0.017                                                                                      | 0.85      | 85          | 20 (1.4)               | 300 (20.7) | 316 SS       | Graphite    | PTFE   | -46 to 120°C (-50 to 250°F) | No              | 8000            | MY-07002-25      | MY-07144-42      |
| 0.042                                                                                      | 2.10      | 210         | 40 (2.8)               |            |              |             |        | -46 to 120°C (-50 to 250°F) | No              | 8000            | MY-07002-26      | —                |
| 0.084                                                                                      | 4.20      | 420         | 40 (2.8)               |            |              |             |        | -46 to 120°C (-50 to 250°F) | No              | 8000            | MY-07002-27      | MY-07144-47      |
| 0.092                                                                                      | 4.60      | 460         | 75 (5.2)               | 300 (20.7) | 316 SS       | PEEK        | PTFE   | -46 to 120°C (-50 to 250°F) | No              | 8000            | MY-73003-14      | —                |
|                                                                                            |           |             | PPS                    |            |              | MY-07002-14 |        | MY-07144-34                 |                 |                 |                  |                  |
| 0.261                                                                                      | 13.05     | 1305        | 125 (8.6)              | 300 (20.7) | 316 SS       | PEEK        | Viton® | -46 to 120°C (-50 to 250°F) | Yes             | 10,000          | MY-73004-00      | —                |
| 0.58                                                                                       | 29.00     | 2900        | 125 (8.6)              | 300 (20.7) | 316 SS       | PEEK        | Viton  | -46 to 120°C (-50 to 250°F) | Yes             | 10,000          | MY-73004-02      | —                |
|                                                                                            |           |             | 75 (5.2)               |            |              | PPS         |        | MY-07003-02                 |                 |                 | MY-07003-12      |                  |
| 1.17                                                                                       | 58.50     | 5850        | 60 (4.1)               | 300 (20.7) | 316 SS       | PEEK        | Viton  | -46 to 120°C (-50 to 250°F) | Yes             | 10,000          | MY-73004-04      | —                |
|                                                                                            |           |             | PPS                    |            |              | MY-07003-04 |        | MY-07003-14                 |                 |                 |                  |                  |

**Note:** Viscosity for all models is dependent on the parameters (flow rate, pressures, etc.) of the specific application.

<sup>†</sup>Flow rates based on water at room temperature and no back pressure.

## Cole-Parmer® Console Gear Pump Drive

### Optimise performance of long-term applications

- 100:1 speed-control range controls flow over a wide range for applications from filtration to chemical injection
- Easy operation with single-turn speed control
- Separate on-off switch maintains speed (flow) setting when drive is powered down
- Maximum continuous-duty flow rates possible with Micropump® A-mount pump heads (order on page 152)

**What's included:** 1.8-m (6-ft) modular line power cord with IEC 320/CEE 22 connector (specify country to obtain correct plug).



Micropump pump head is not included; order on page 152

**Cole-Parmer®**



#### Specifications

**Duty cycle:** continuous **Max system temperature:** 40°C (104°F)  
**Remote control:** (input/output) none

| Motor      |      |      | Power |       | Catalogue number |
|------------|------|------|-------|-------|------------------|
| rpm        | hp   | Type | VAC   | Hz    |                  |
| 50 to 5000 | 0.07 | PMDC | 230   | 50/60 | MY-75211-17      |

## Cole-Parmer® Digital Gear Pump Benchtop Systems

### Ready to use—drive and pump head included!

- Operate in continuous run, volume dispense, copy dispense, and timed dispense modes with precise, pulseless batching and dispensing
- IP33-rated enclosure protects the internal components from dust and splashes



**Cole-Parmer®**

#### Specifications

**Remote control:** (input) on/off by contact closure (0 to 10 VDC, 0 to 20 mA, 4 to 20 mA), and remote output

**Speed:** 36 to 3600 rpm  
**Power:** 90 to 260 VAC

| Flow rates (mL/min) <sup>†</sup> |           |             | Max pressure |              | Wetted parts |          |        | Catalogue number |
|----------------------------------|-----------|-------------|--------------|--------------|--------------|----------|--------|------------------|
| mL/rev                           | at 50 rpm | at 3600 rpm | Diff (psi)   | System (psi) | Body         | Gears    | Seals  |                  |
| 0.017                            | 0.85      | 85          | 20           | 300          | 316 SS       | Graphite | PTFE   | MY-74014-72      |
| 0.092                            | 4.60      | 331.2       | 75           | 300          | 316 SS       | PPS      | PTFE   | MY-74014-02      |
| 0.092                            | 4.60      | 331.2       | 75           | 300          | 316 SS       | PPS      | PPS    | MY-74014-62      |
| 0.261                            | 13.05     | 939.6       | 125          | 300          | 316 SS       | PEEK     | Viton® | MY-74014-52      |
| 0.316                            | 15.8      | 1137.6      | 75           | 300          | 316 SS       | PPS      | PTFE   | MY-74014-12      |
| 0.64                             | 32.0      | 2304        | 75           | 300          | 316 SS       | PPS      | PTFE   | MY-74014-22      |
| 0.91                             | 45.5      | 3276        | 75           | 300          | 316 SS       | PPS      | PTFE   | MY-74014-32      |
| 1.17                             | 58.5      | 4212        | 60           | 300          | 316 SS       | PPS      | Viton  | MY-74014-42      |

<sup>†</sup>Flow rates in mL/min are based on water at room temperature and no back pressure.

**MY-07523-92 Foot switch**, momentary start/stop, 1.8-m cable

**MY-06385-10 Male pipe adapter (straight)**; PP, ¼" OD x 3/8" NPT(M). Pk of 10

**MY-40610-36 Male pipe adapter**; natural PP, ½" NPT(M) x ¼" ID. Pk of 10

## Ismatec® Compact Programmable Digital Gear Pump Drive

### Dispense by flow rate, volume, time, or number of cycles

- No more dripping—program pump to draw back fluid
- Control via a PC with RS-232 interface
- Get more accuracy with calibration of flow rate and dispensing volume

Multiple operating modes allow for continuous pumping by drive speed, flow rate, volume, time, volume within a preset time or interval dispensing by volume with a pause or time with a pause. The number of dispensing cycles may also be programmed into the pump. Pump is capable of reversible rotation (only with cavity style pump heads that indicate reversible operation). Both dispensing and pause time settings are from 0.1 seconds up to 999 hours.

**ISMATEC®**

#### Specifications

**Duty cycle:** continuous from 50 to 5000 rpm

**Remote control (input):**

Speed control: RS-232 (DB9F) for pump control via PC. 9600 baud, 8 bit, 1 stopbit, no parity  
 On/off control: TTL contact (for footswitch)  
 0 to 10 kHz for speed monitoring

**Remote control (output):**

RS-232 (DB9M) for cascading of up to 8 pumps

**Max temperature:** 40°C (104°F)

**IP rating:** IP30



Programmable digital drive 78008-15 with Micropump pump head 07002-14 (head not included; order on page 152)

| Motor rpm  | Power                      | Catalogue number |
|------------|----------------------------|------------------|
| 50 to 5000 | 90 to 264 VAC, 47 to 63 Hz | MY-78008-15      |

**MY-70738-94 Replacement adapter plate**; for mounting Micropump A-mount pump heads (except models 07003-02, 07003-04, 73004-00, 73004-02, 73004-04)

**What's included:** an adapter plate for mounting Micropump® A pump heads (except 07003-02, 07003-04, 73004-00, 73004-02, and 73004-04), a driving magnet and a 1.8-m (6-ft) power cord with plug.

